3-Tier-Project-By Mayur Chaudhari

3-Tier Web Application on AWS



ARCHITECTURE AT A GLANCE

1 WEB TIER
Nginx | public subnets | External Load Balancer

2 APP TIER
Node.js :4000 | private subnets | Internal Load Balancer

3 DATABASE TIER
Aurora MySQL | private subnets | Writer + Standby

WHAT'S INSIDE

1. Network: VPC, subnets, gateways, routes
2. Security groups (guest lists)
3. Database: Aurora MySQL + subnet group
4. IAM role and S3 code bucket
5. App server, Node.js, PM2 and AMI
6. Internal load balancer + launch template
7. Web server, Nginx and AMI
8. External load balancer + Auto Scaling
9. Test the app, then clean up safely
10. Production guide + go-live checklist

BUILT WITH

VPC | EC2 | ALB | Auto Scaling | RDS Aurora | S3 | IAM | NAT Gateway | Session Manager | Nginx | Node.js | PM2

Highly Available | **Highly Secure** | **Highly Scalable** | **Fault Tolerant**

PROJECT BY

Mayur Chaudhari

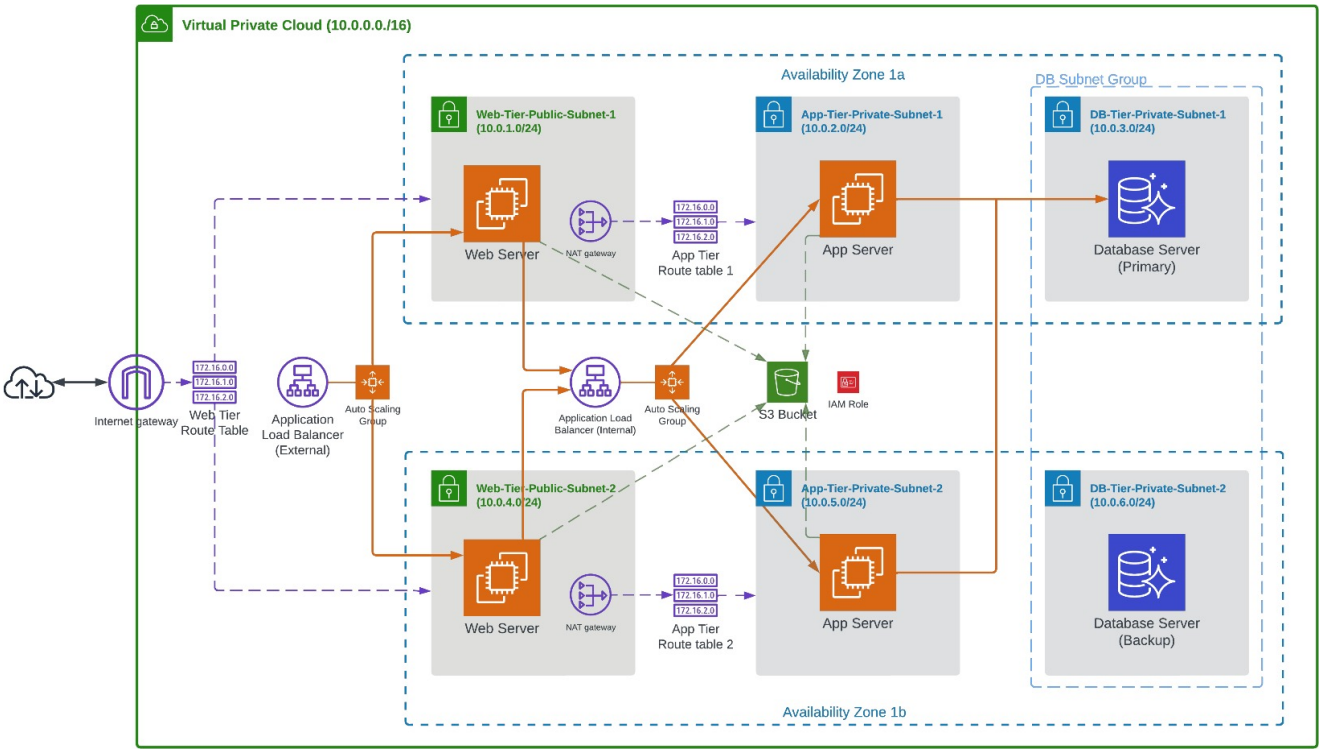
AWS Region: Asia Pacific (Mumbai) ap-south-1 | Beginner friendly | Production notes included

1. What is this project? (30-second version)

You build a small website on AWS that shows **transactions** (amount + description) from a database. You can add and delete items. The website is built so that it keeps working even if one server or one whole data-centre fails, and so that only the right parts of it are open to the internet.

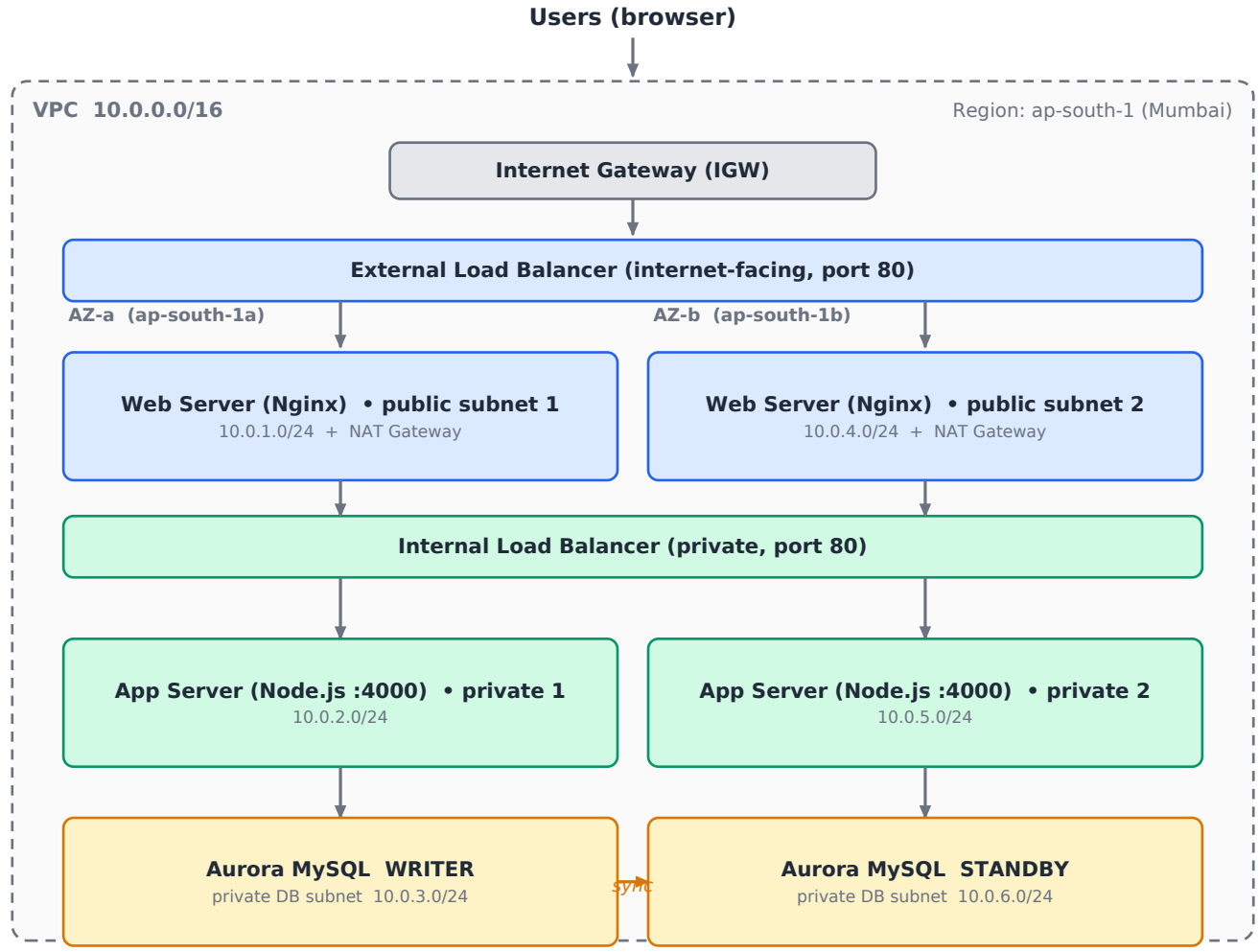
> **Think of a restaurant**
> **Web tier** = the waiter who talks to customers (the only person customers meet).
> **App tier** = the kitchen that does the real work (customers never enter).
> **Database tier** = the store-room with all ingredients (only the kitchen can enter).
> Each room has its own door and its own security guard. That is the whole idea of 3-tier.

Architecture diagram



Architecture of the project: custom VPC 10.0.0.0/16, two Availability Zones, web tier in public subnets, app and database tiers in private subnets, load balancers, NAT gateways, S3 and IAM role.

The same thing, simplified



Blue = web tier (public). Green = app tier (private). Orange = database tier (private). Every tier exists twice, once in each Availability Zone (AZ = a separate data-centre in the same city).

The four promises of this design

Promise	Meaning in simple words	How it is done here
Highly Available	Website stays up if one part breaks	Everything duplicated in 2 Availability Zones (1a, 1b)
Highly Secure	Strangers cannot reach the inside	Private subnets + security groups + no public IP on app/DB
Highly Scalable	More visitors = more servers automatically	Auto Scaling Groups behind load balancers
Fault Tolerant	A failure is fixed by itself	Load balancer health checks + ASG replaces bad servers; DB standby

2. AWS words made easy

You will meet 14 AWS names. Here is each one in plain language:

Name	What it really is	Everyday example
VPC	Your own private network inside AWS	Your own gated colony
Subnet	A smaller section of the VPC, lives in one AZ	A street inside the colony

Name	What it really is	Everyday example
Public subnet	Subnet that can talk to the internet directly	Street facing the main road
Private subnet	Subnet with no direct internet access	Inner street, no main-road gate
Internet Gateway (IGW)	Door between the VPC and the internet	Colony main gate
NAT Gateway	Lets private servers go OUT (updates, downloads) but blocks anyone coming IN	Peephole: you can look out, nobody walks in
Route Table	Signboard telling traffic which way to go	Road signs
Security Group (SG)	Firewall around a server or load balancer	Guard with a guest list
EC2	A virtual server (computer)	One rented computer
AMI	A saved copy (photo) of a fully set-up server	Cloned computer image
Launch Template	Recipe for new servers: which AMI, size, SG, role	Order form
Target Group	List of servers a load balancer can send traffic to	Team list
Load Balancer (ALB)	Spreads visitors across servers; skips sick ones	Traffic police
Auto Scaling Group	Keeps N servers alive; adds/removes by load	Manager who hires/fires staff
RDS / Aurora	Managed database (backups, patching, failover done by AWS)	Hired storekeeper
S3	Cloud file storage; here it holds the app code	Cloud pen-drive
IAM Role	Permission badge a server wears	Employee ID card
SSM Session Manager	Open a terminal into a server from the AWS console, no SSH key, no public IP	Remote video-call into the server

3. Step 1-2: Build the network

Everything lives inside one VPC. Nothing works until the network is right, so this is the most important part.

3.1 VPC and the 6 subnets

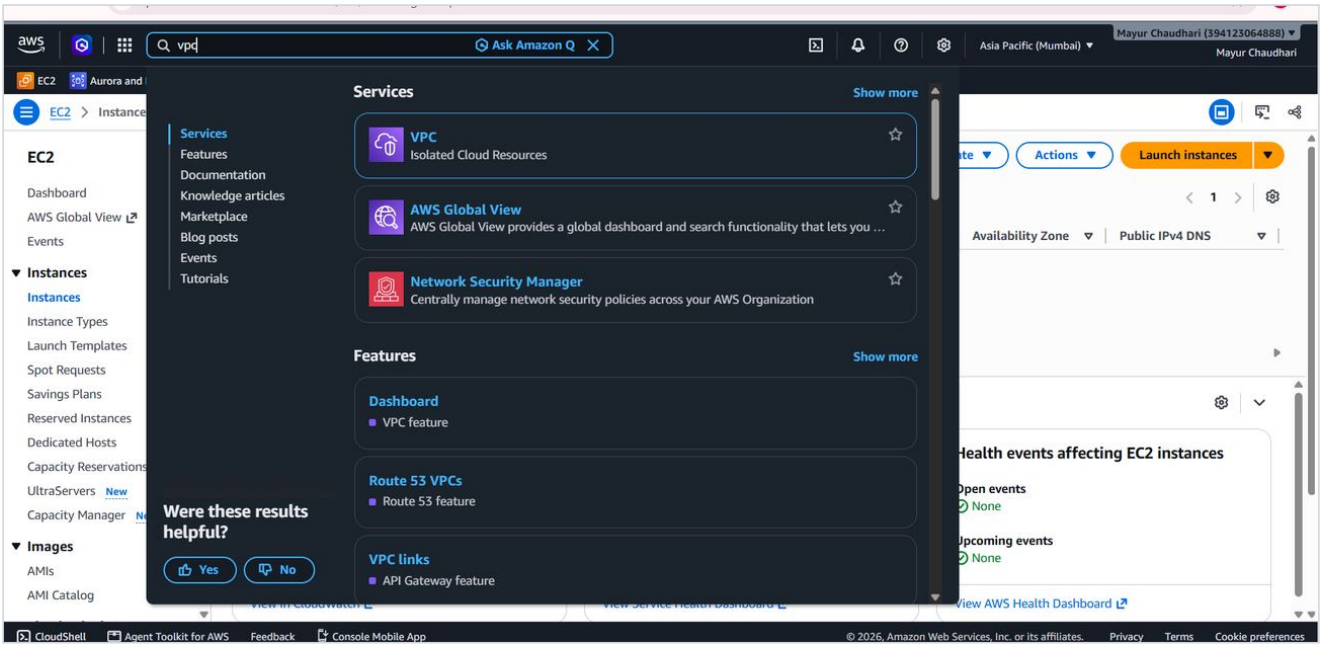
VPC CIDR = **10.0.0.0/16** (that is 65,536 private IP addresses). It is cut into six /24 subnets (256 addresses each), three per AZ:

Subnet name	AZ	CIDR	Type	Holds
Web-Tier-Subnet-1-Public	1a	10.0.1.0/24	Public	Web server, NAT-GW-1
App-Tier-Subnet-1-Private	1a	10.0.2.0/24	Private	App server
DB-Tier-Subnet-1-Private	1a	10.0.3.0/24	Private	DB writer
Web-Tier-Subnet-2-Public	1b	10.0.4.0/24	Public	Web server, NAT-GW-2
App-Tier-Subnet-2-Private	1b	10.0.5.0/24	Private	App server
DB-Tier-Subnet-2-Private	1b	10.0.6.0/24	Private	DB standby

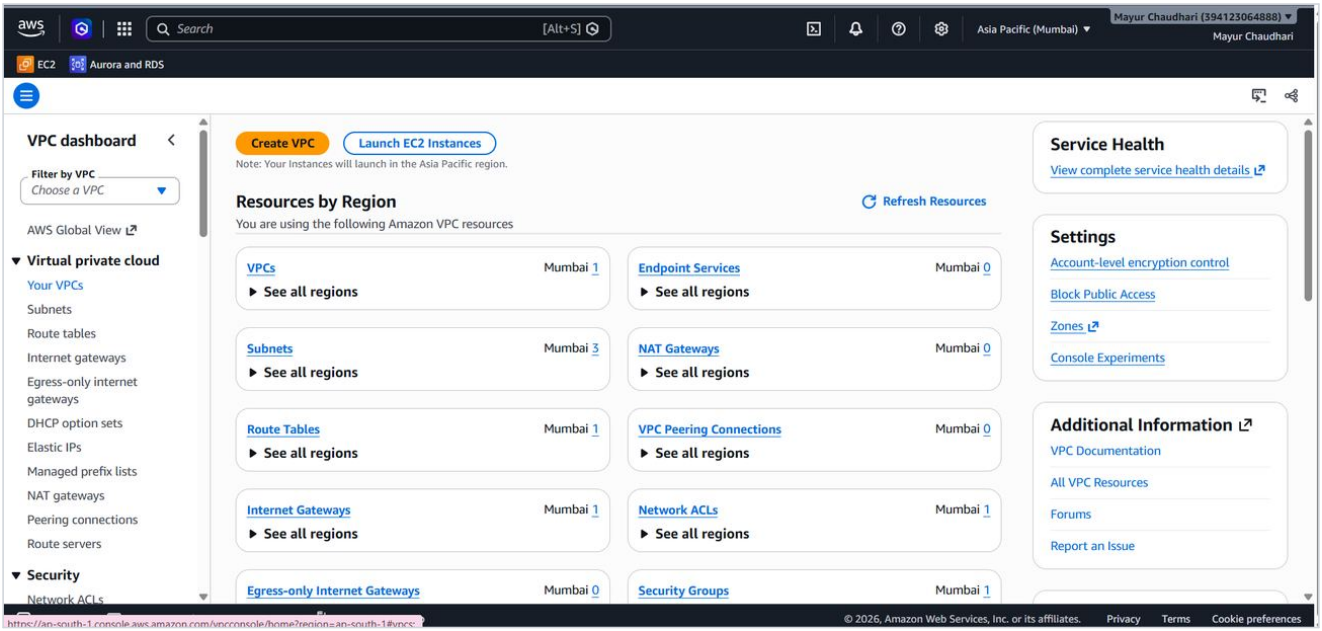
Extra setting: on both **public** subnets turn on *auto-assign public IPv4*, so servers there can be reached from the internet.

Step 1 - Create the VPC and subnets

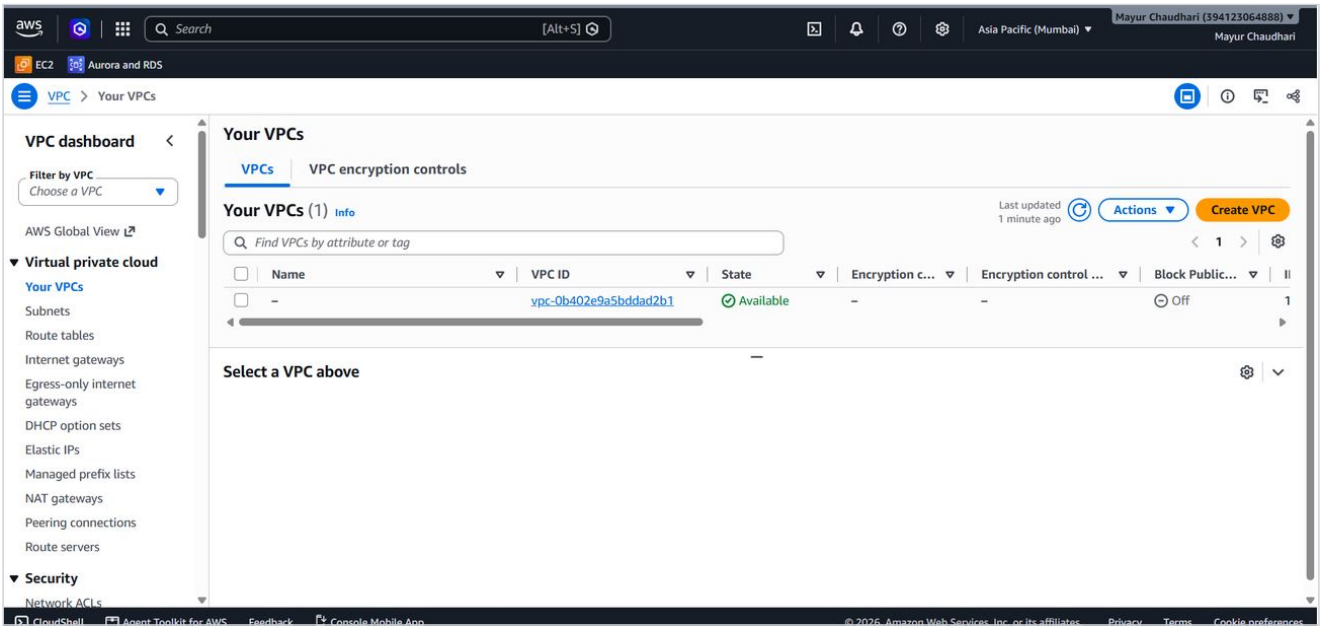
Step 1.1 Search for VPC in the AWS console.



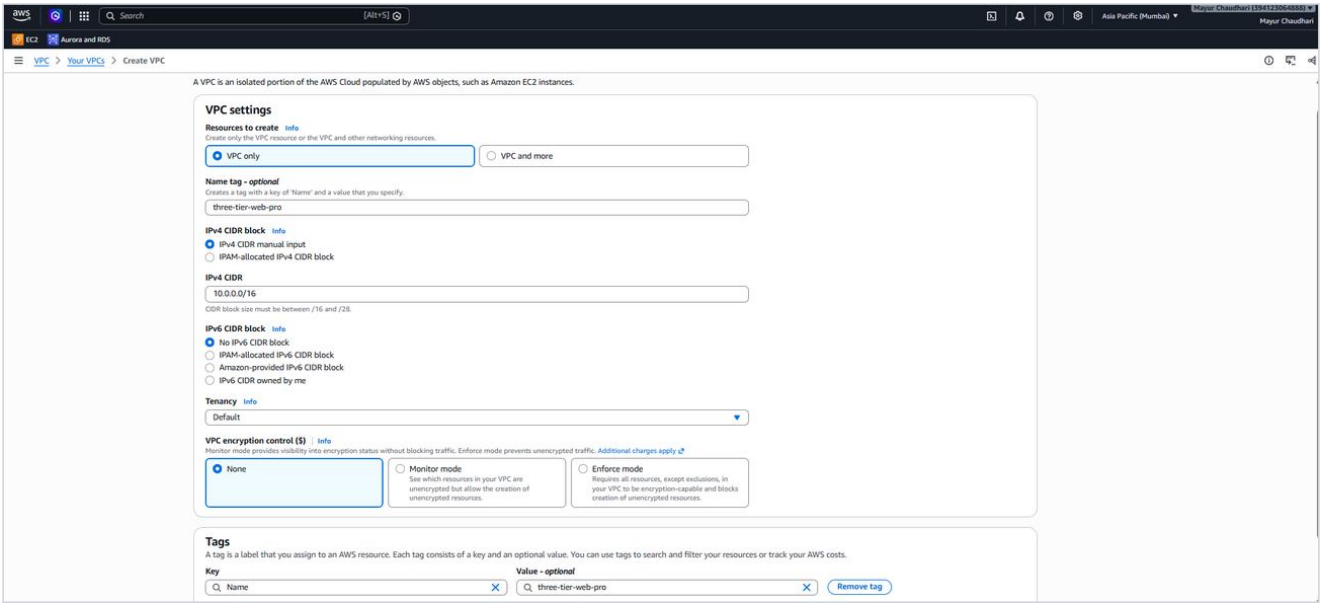
Step 1.2 VPC dashboard.



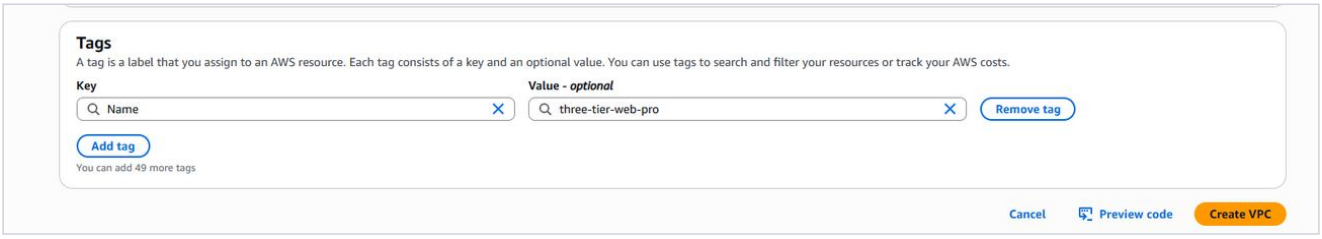
Step 1.3 Your VPCs list.



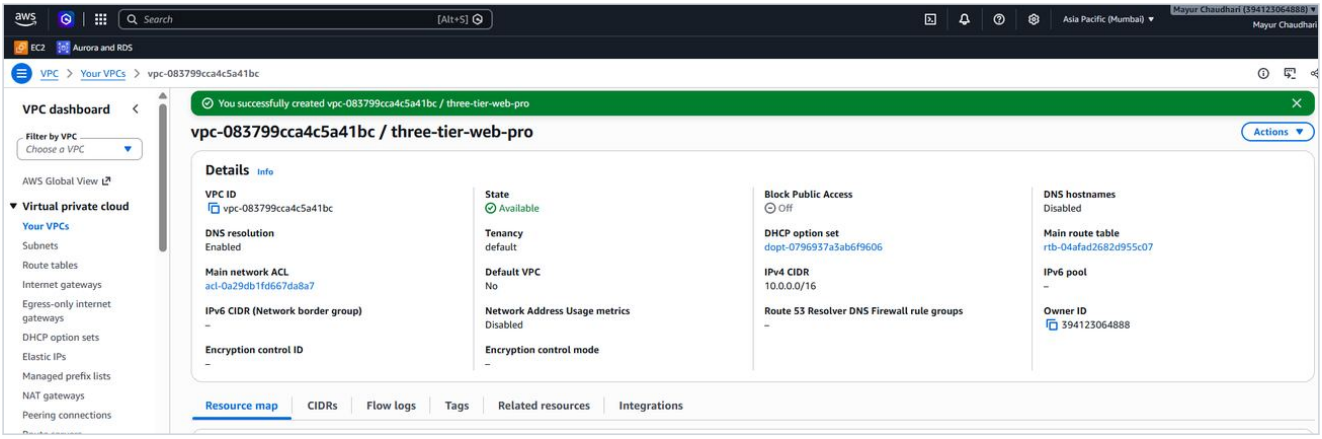
Step 1.4 Create VPC: name and CIDR block 10.0.0.0/16.



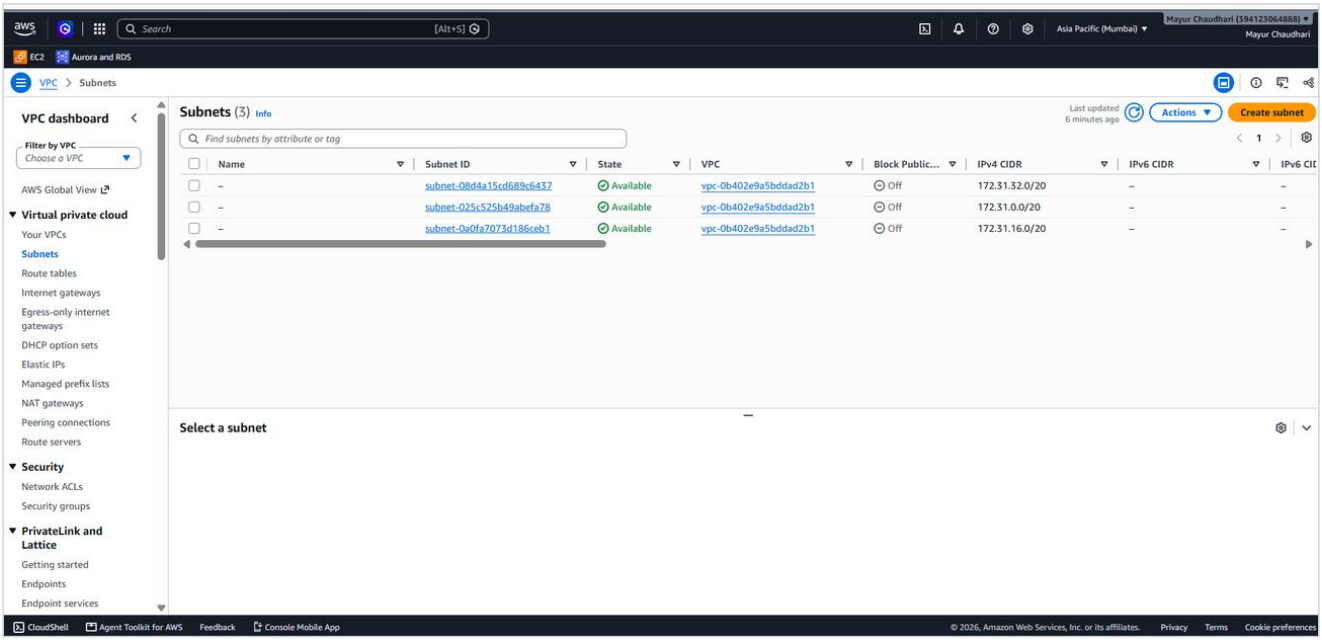
Step 1.5



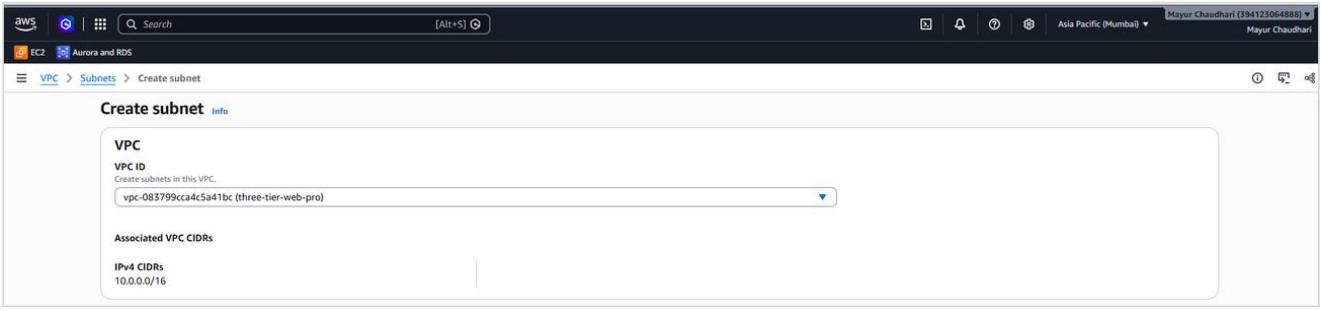
Step 1.6 VPC created.



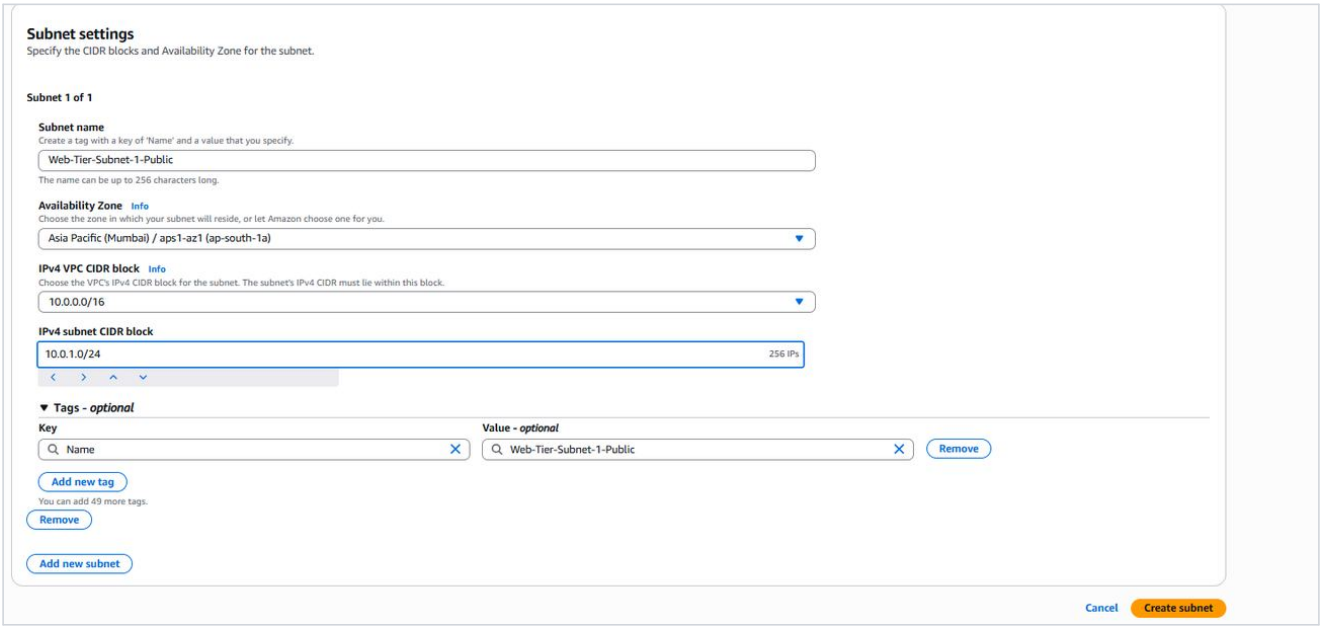
Step 1.7 Subnets page.



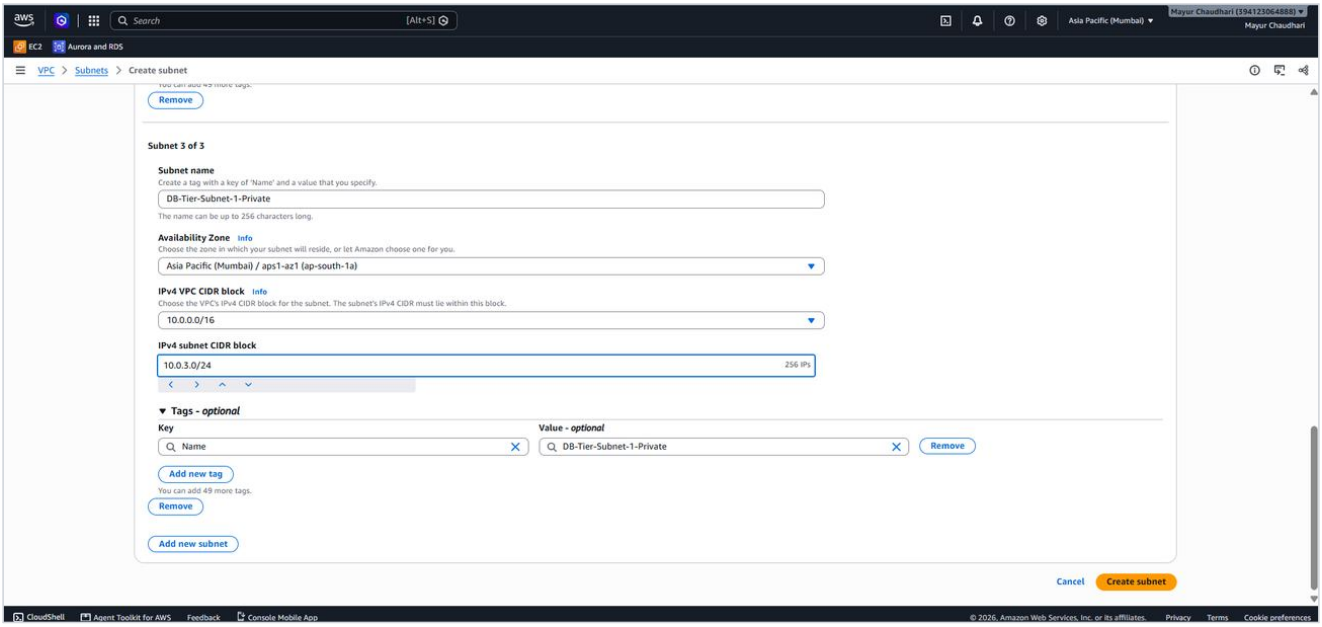
Step 1.8 Create subnet: choose your VPC.



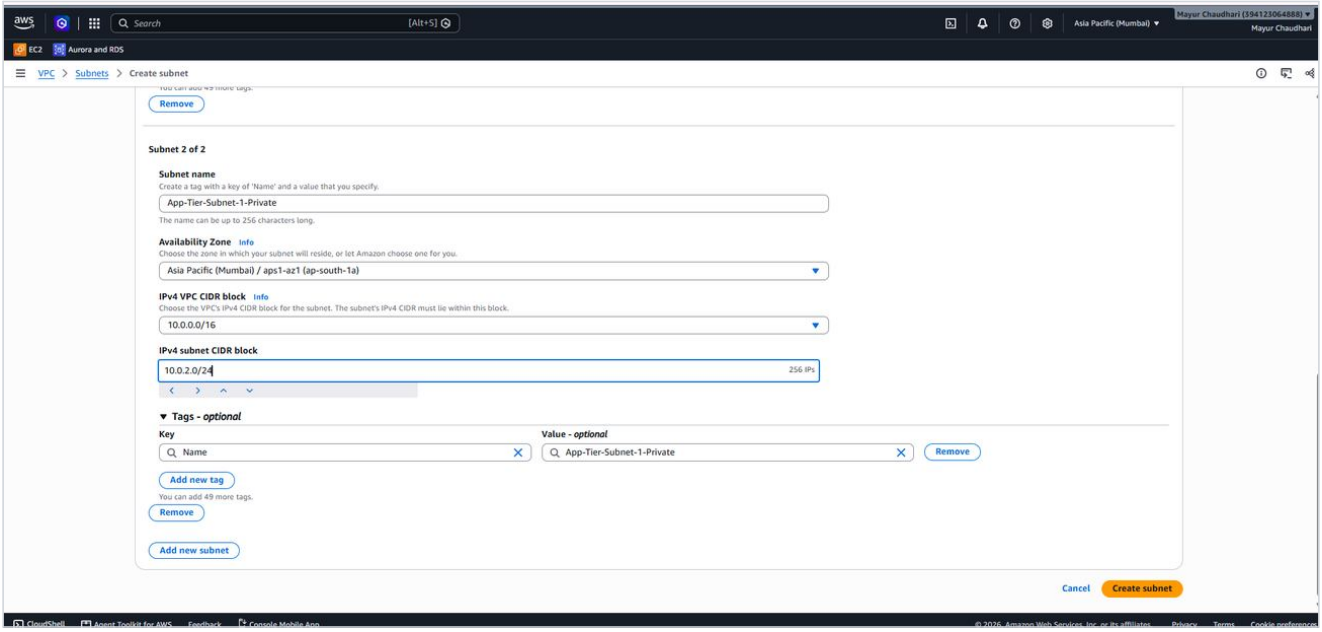
Step 1.9



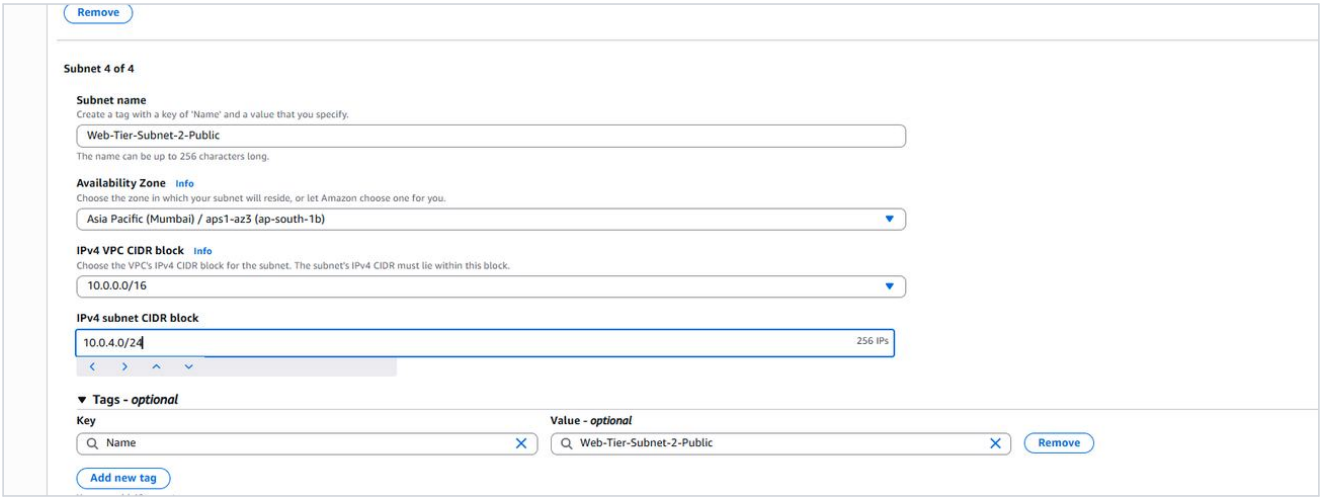
Step 1.10



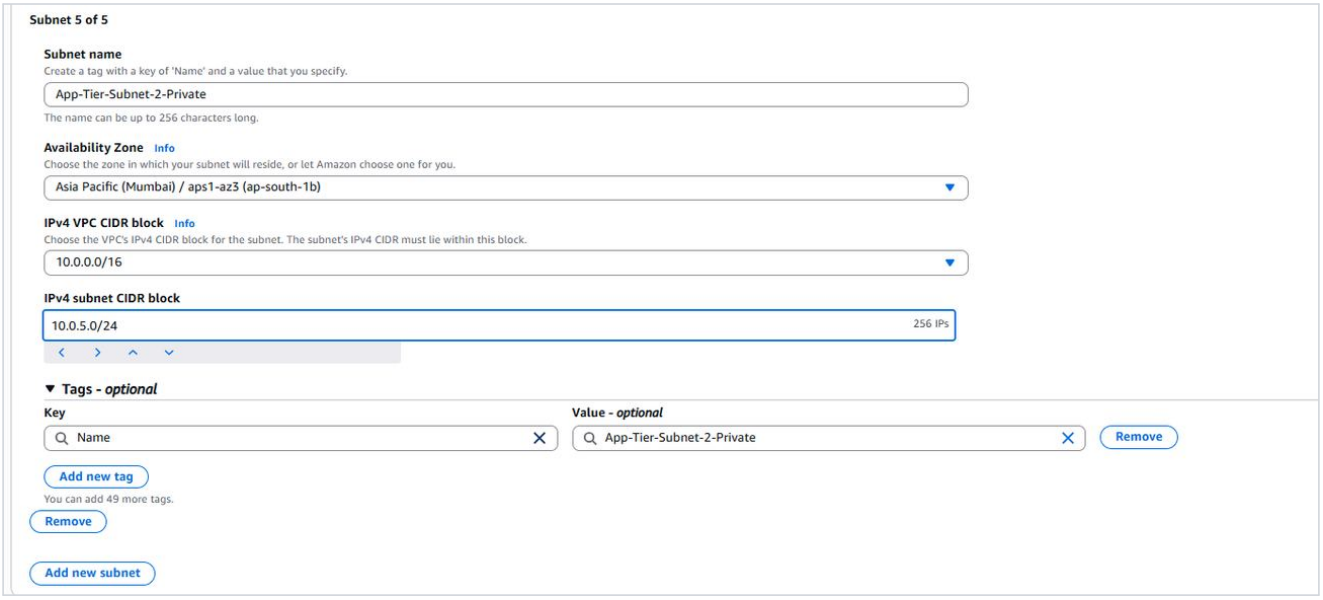
Step 1.11



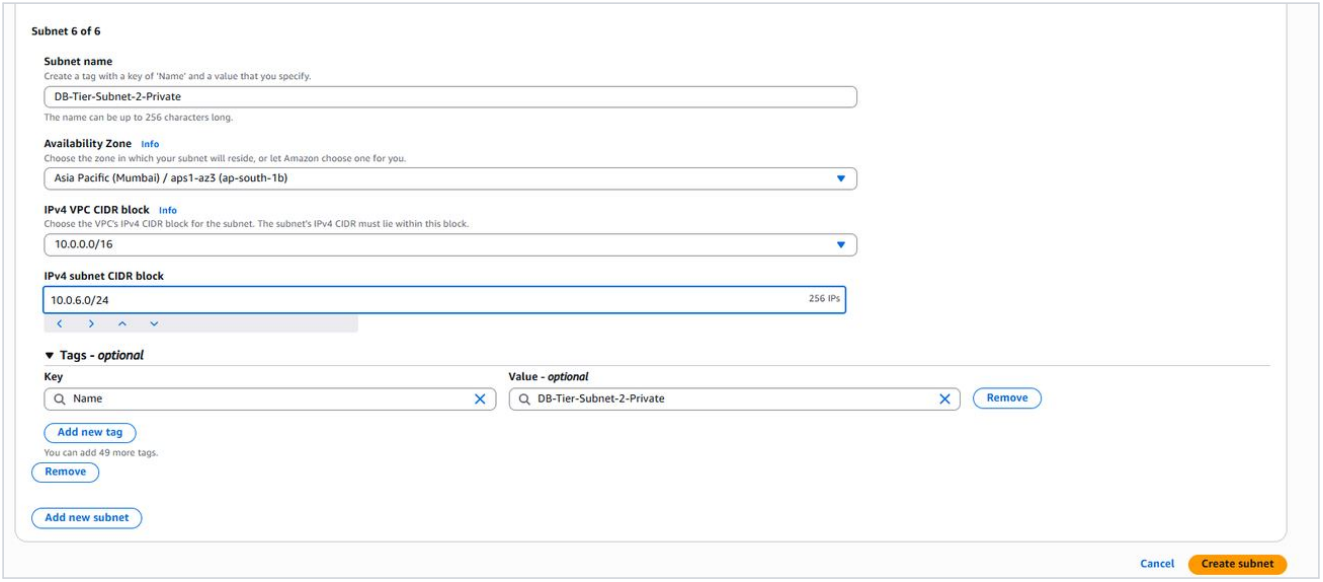
Step 1.12



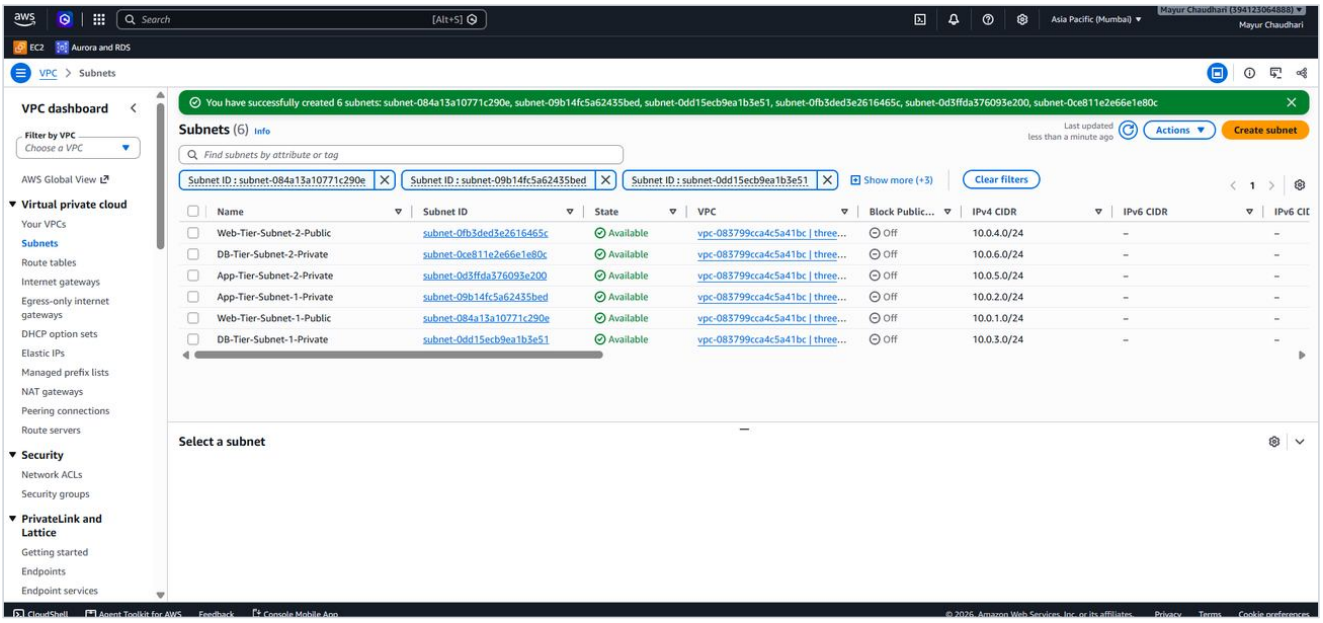
Step 1.13



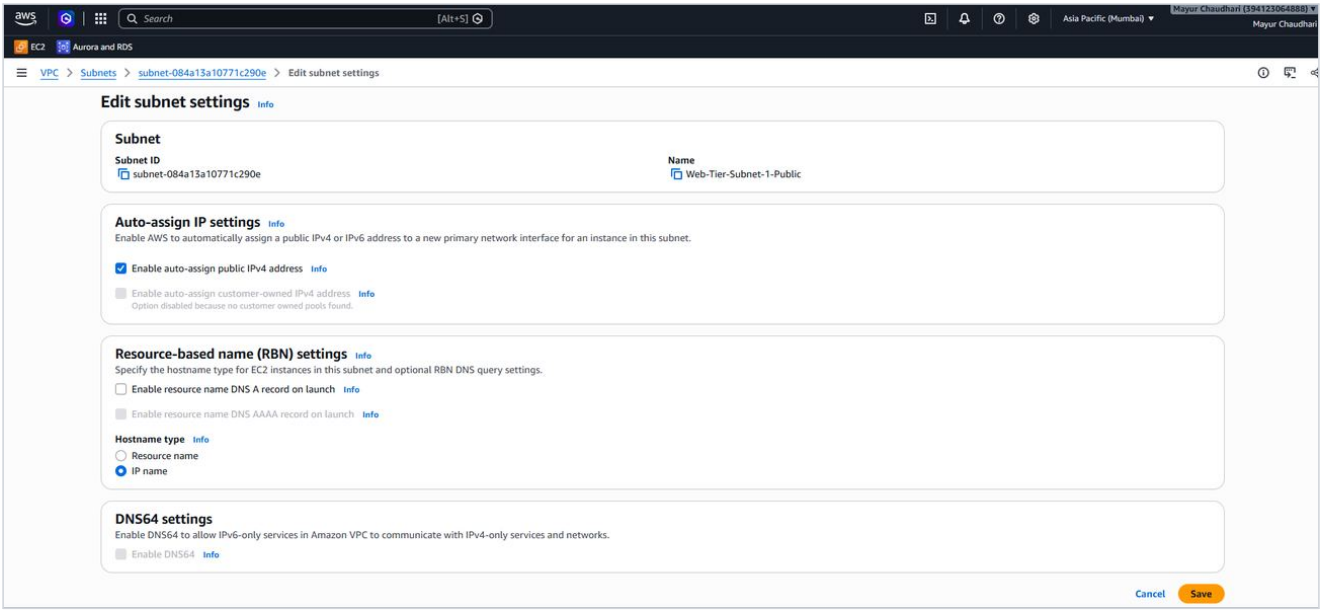
Step 1.14



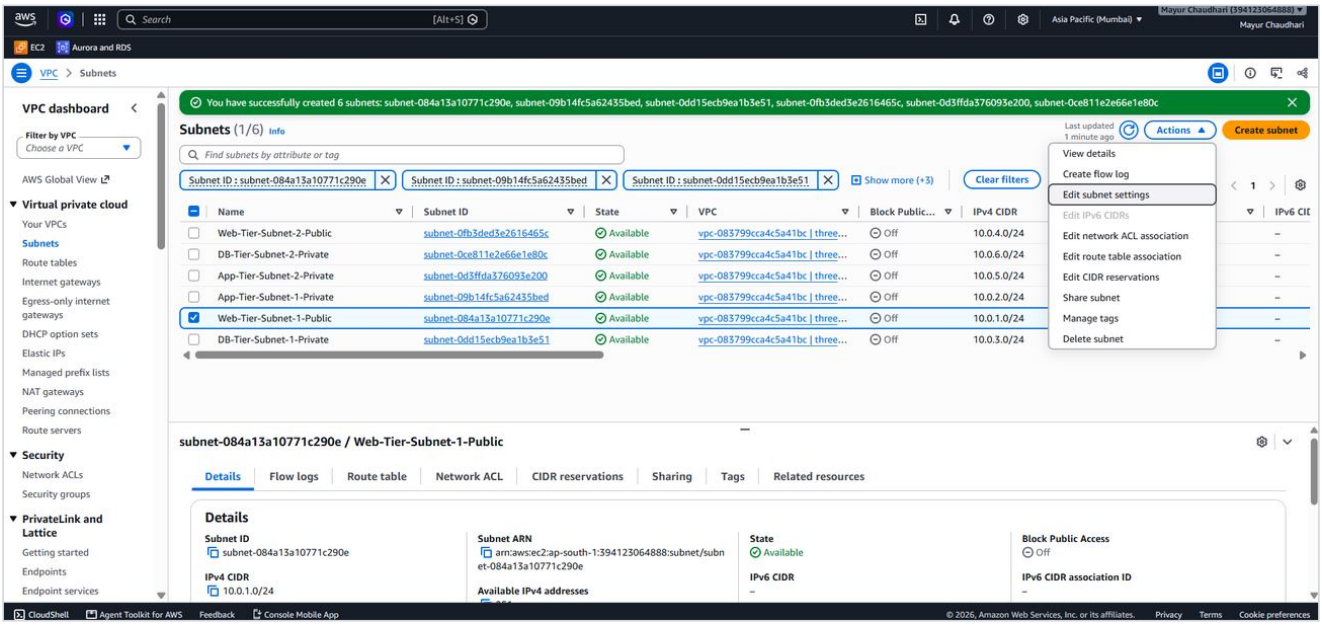
Step 1.15 All six subnets created.



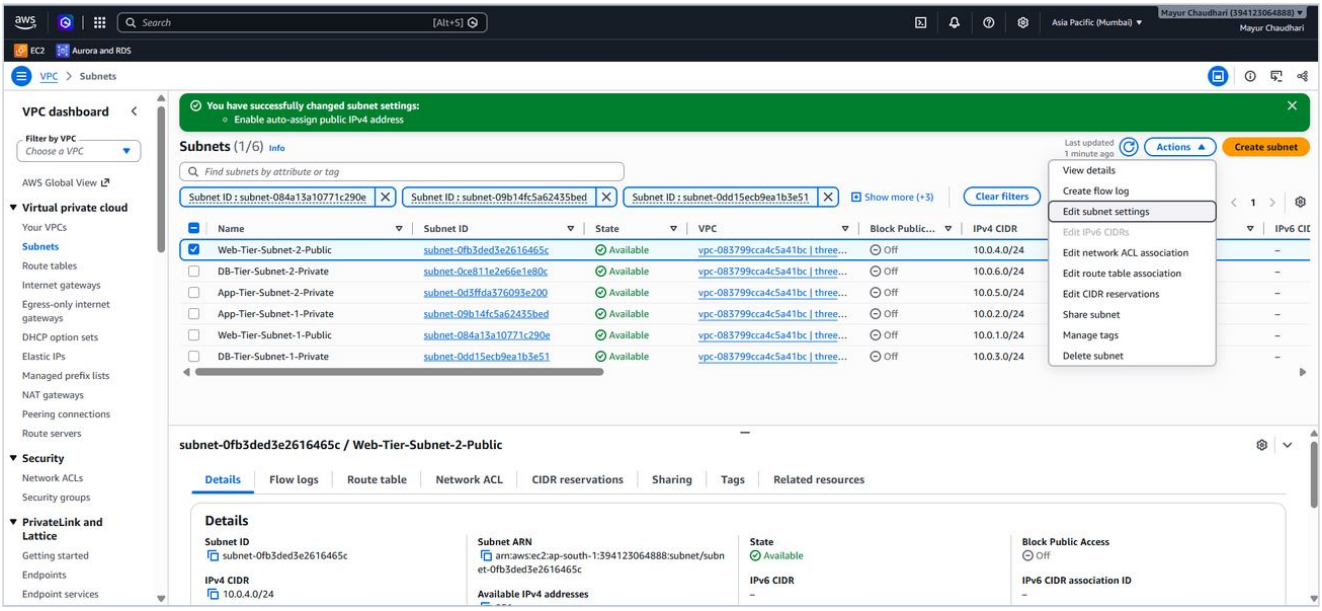
Step 1.16 Edit subnet settings: enable auto-assign public IPv4 on the public subnets.



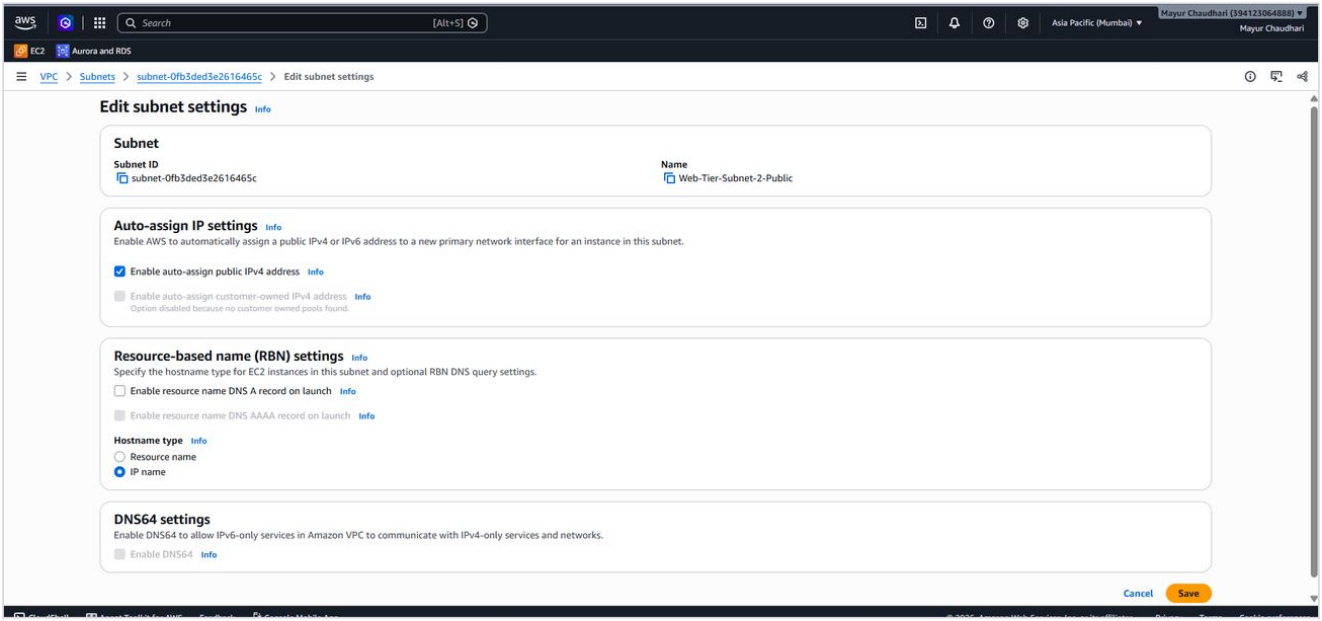
Step 1.17



Step 1.18



Step 1.19



3.2 Gateways and route tables

A subnet is "public" or "private" only because of its **route table**. Three route tables are made:

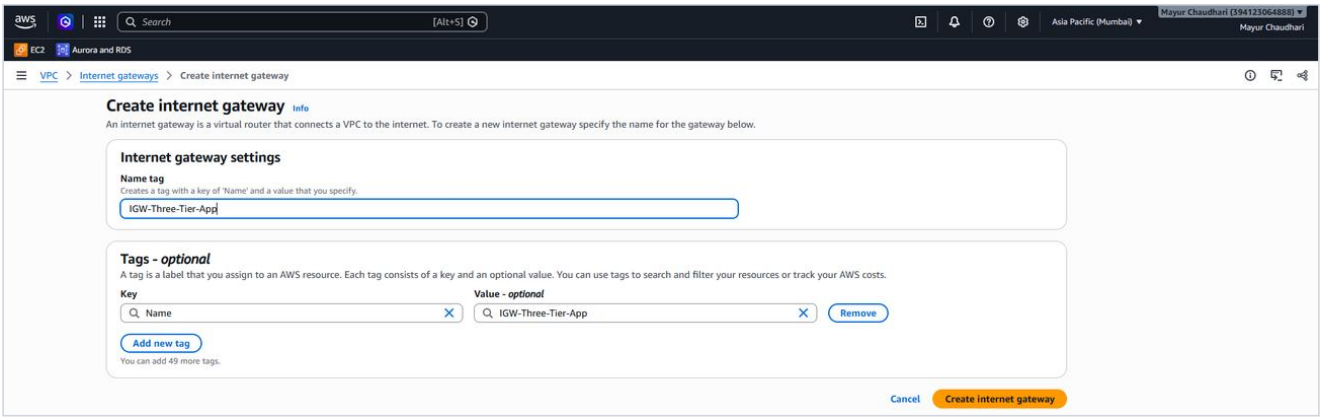
Route table	Attached to	Rule: 0.0.0.0/0 (anything else) goes to...	Result
Web-Tier-RT	Both public web subnets	Internet Gateway	Public: internet in and out
App-Tier-RT-1	App subnet 1 (AZ-a)	NAT-GW-1	Private: outbound only
App-Tier-RT-2	App subnet 2 (AZ-b)	NAT-GW-2	Private: outbound only

Why 2 NAT Gateways?
One per AZ. If AZ-a dies, AZ-b's app servers still reach the internet. Why are they needed at all? The app server is in a private subnet, but it must download MySQL client, Node.js and npm packages once, and talk to S3. NAT gives it outbound-only internet.

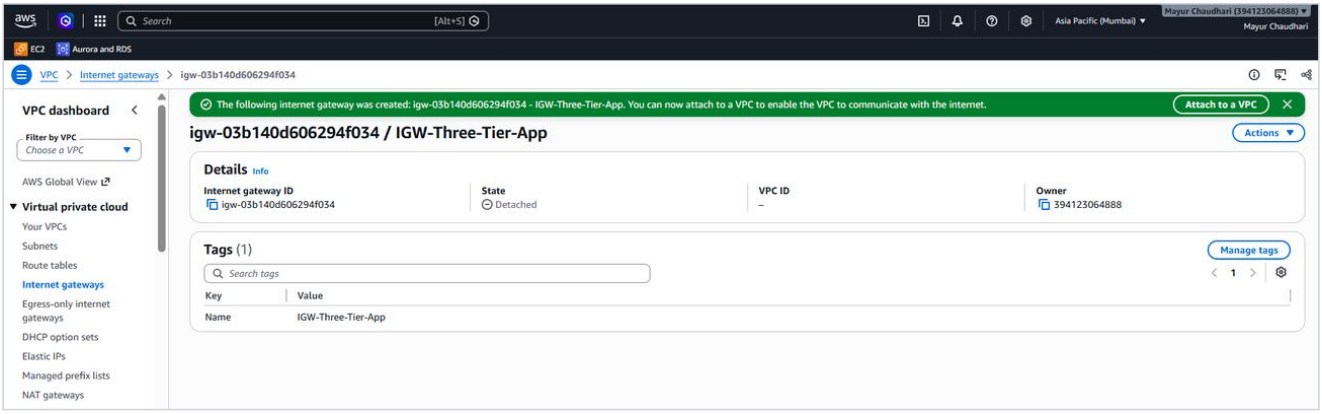
Tip from the original doc: open **VPC > Resource map** after this step and hover over each item to check the lines match the architecture picture.

Step 2 - Internet Gateway, NAT Gateways and route tables

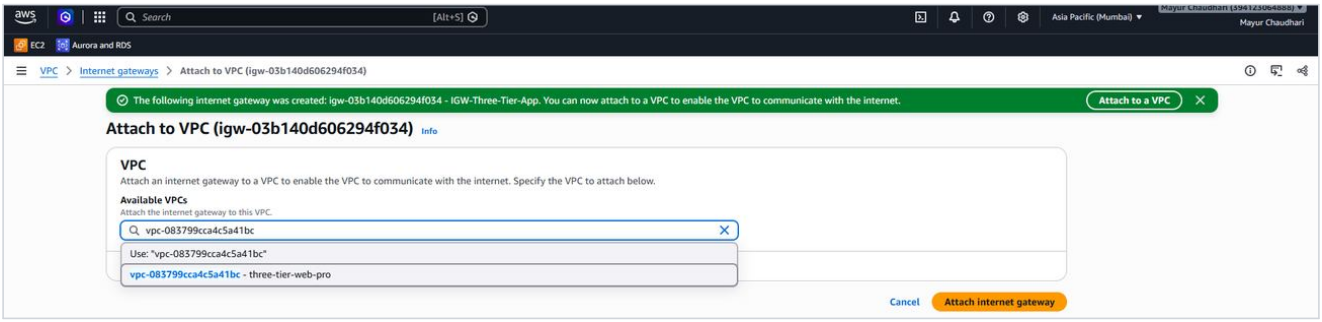
Step 2.1 Create Internet Gateway.



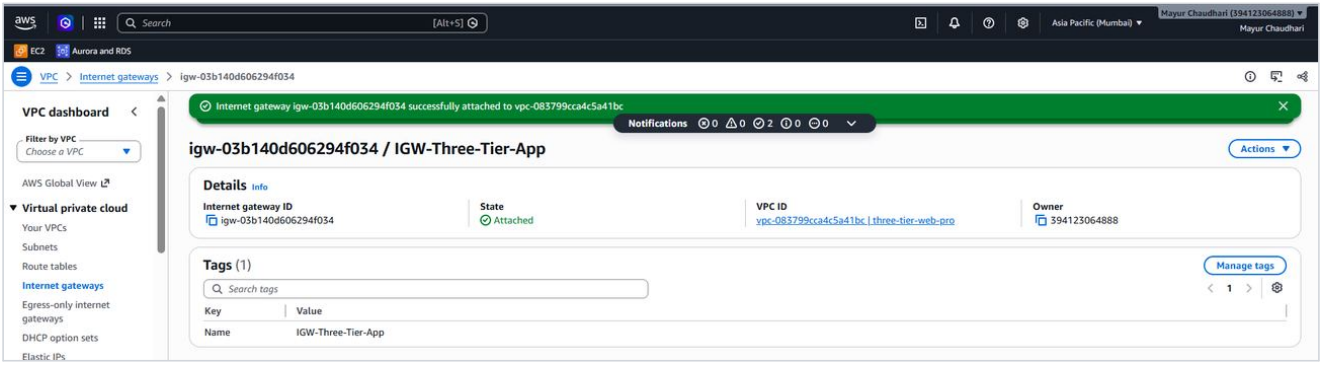
Step 2.2 Internet Gateway created.



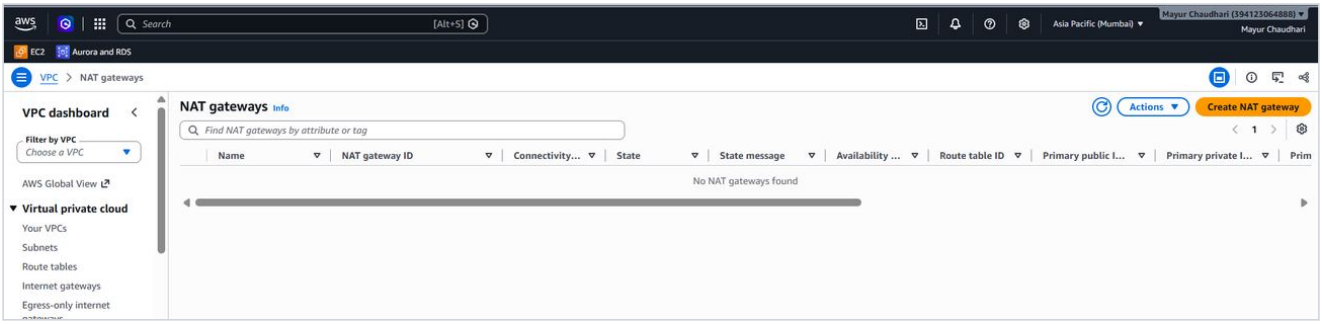
Step 2.3 Attach the Internet Gateway to your VPC.



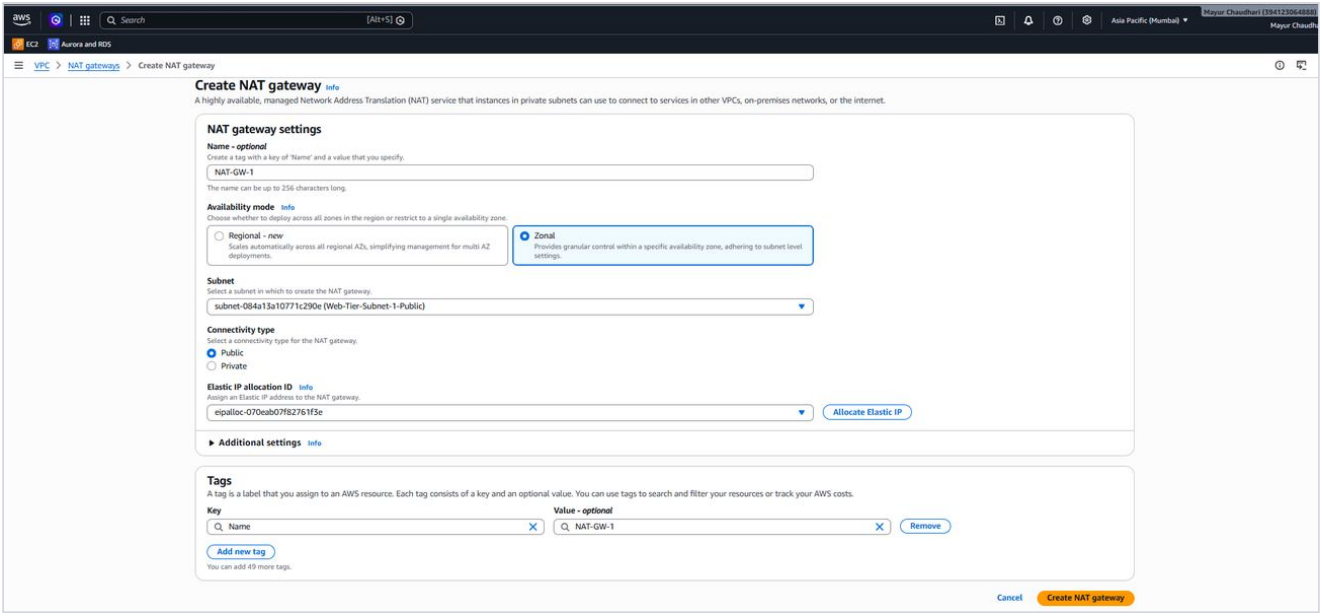
Step 2.4 Internet Gateway attached.



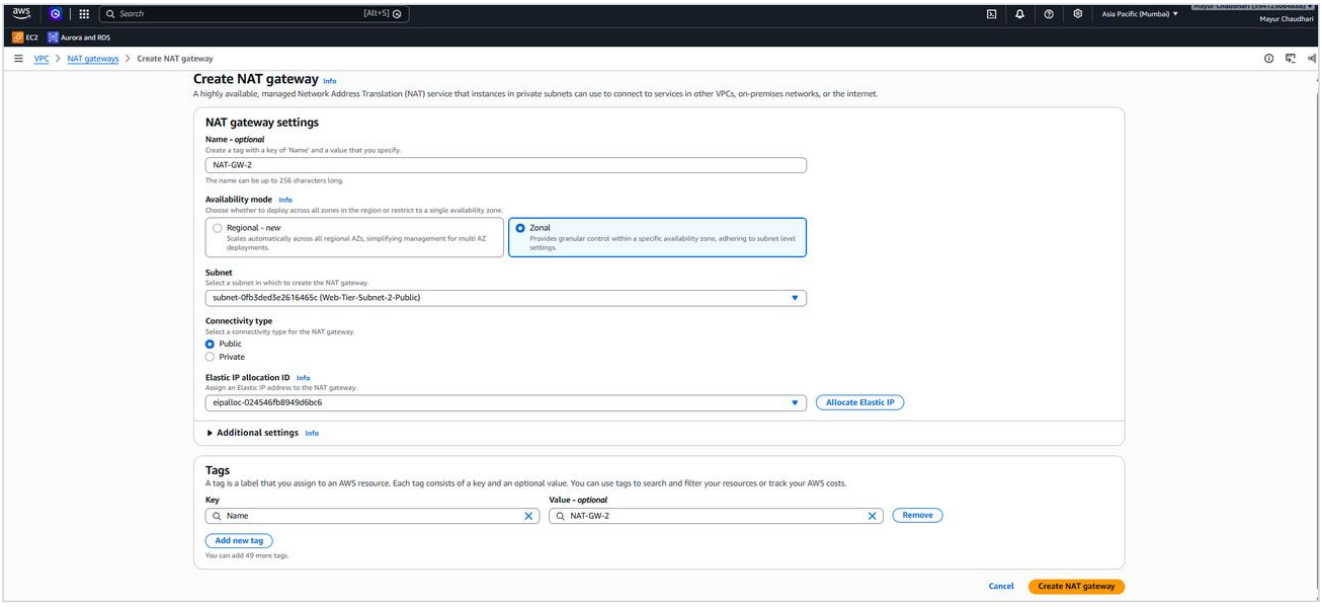
Step 2.5 NAT Gateways page.



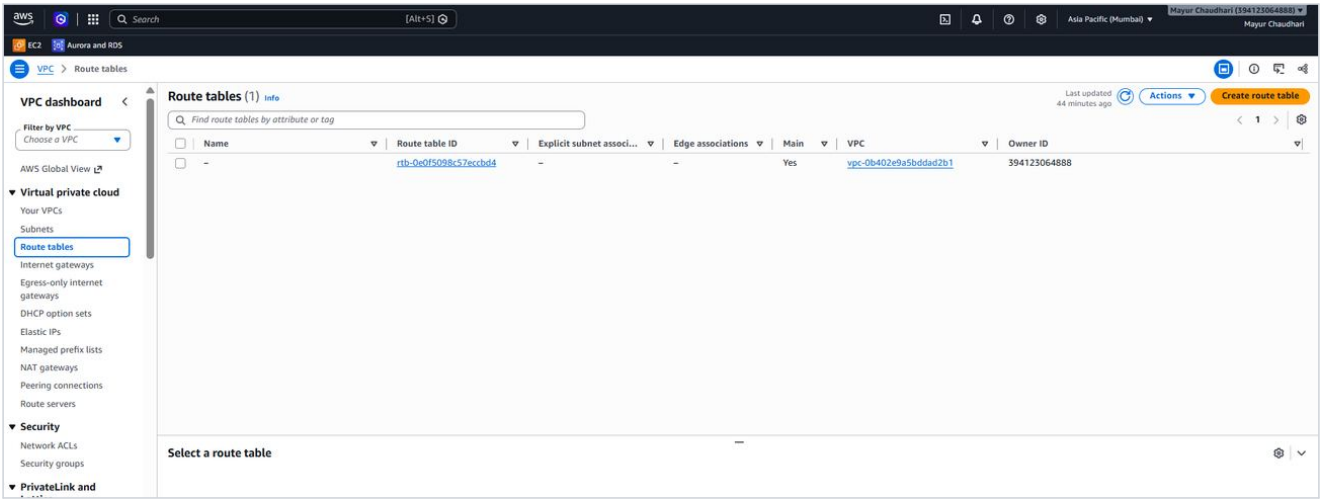
Step 2.6 Create NAT Gateway 1 (public subnet, AZ-a).



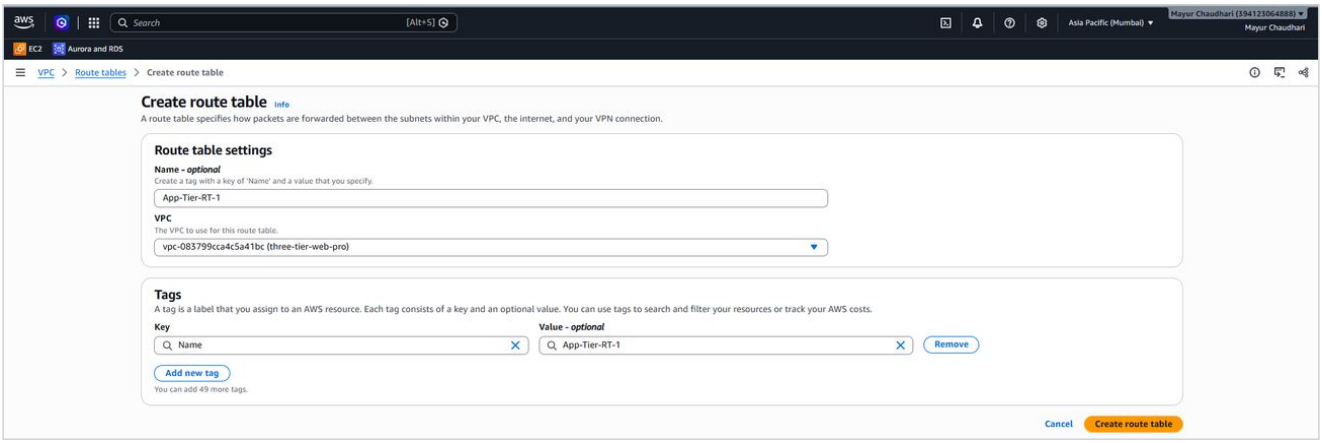
Step 2.7 Create NAT Gateway 2 (public subnet, AZ-b).



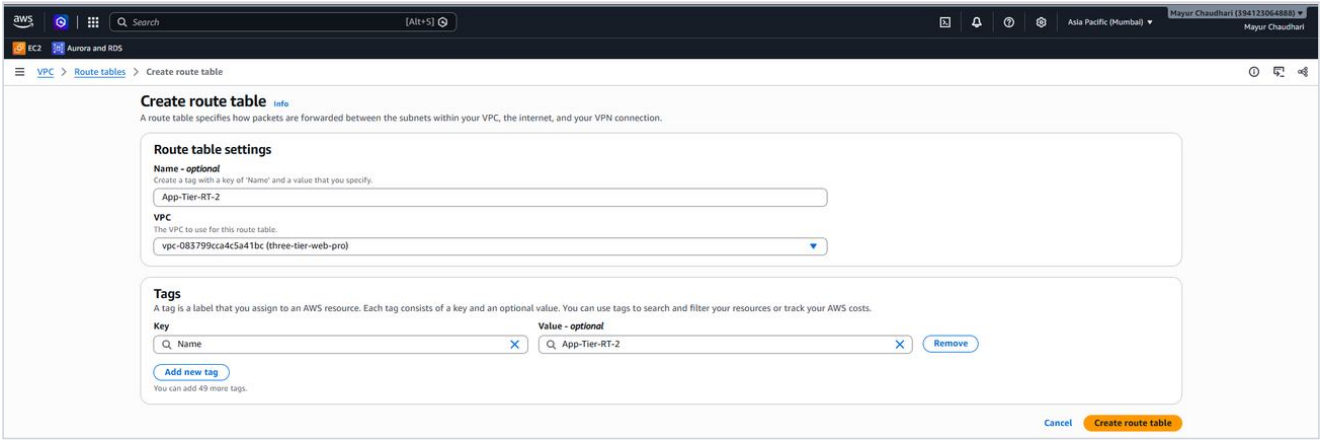
Step 2.8 Route tables page.



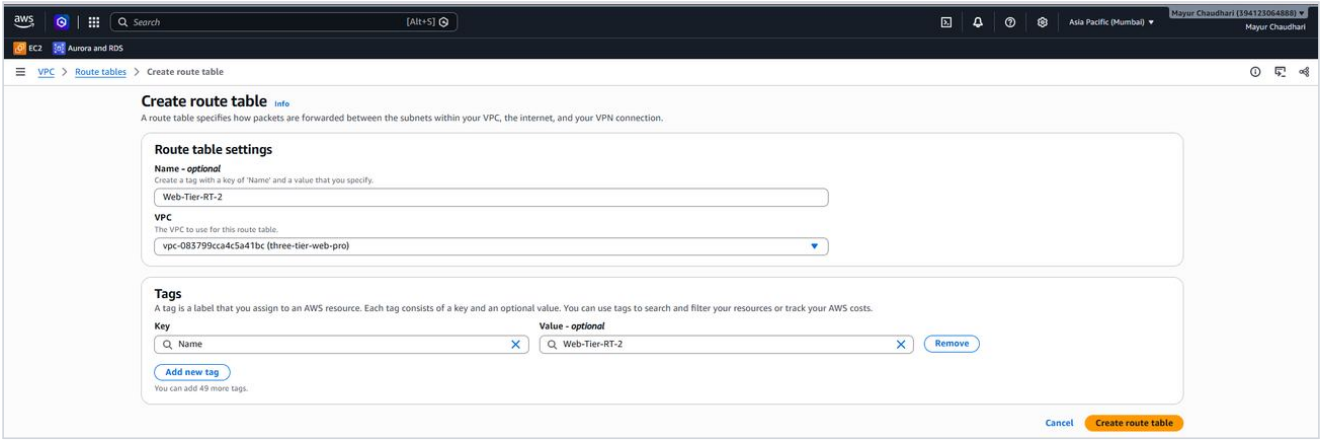
Step 2.9 Create route table.



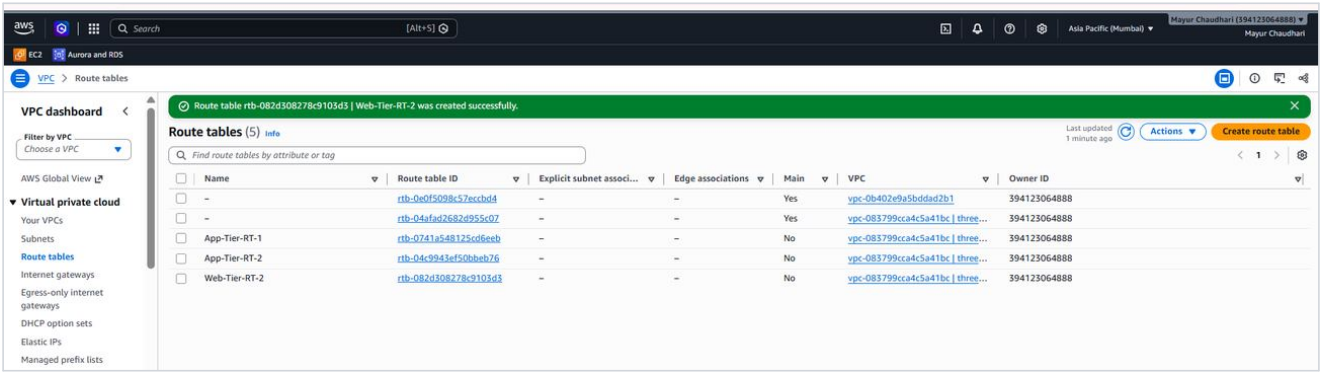
Step 2.10 Create route table.



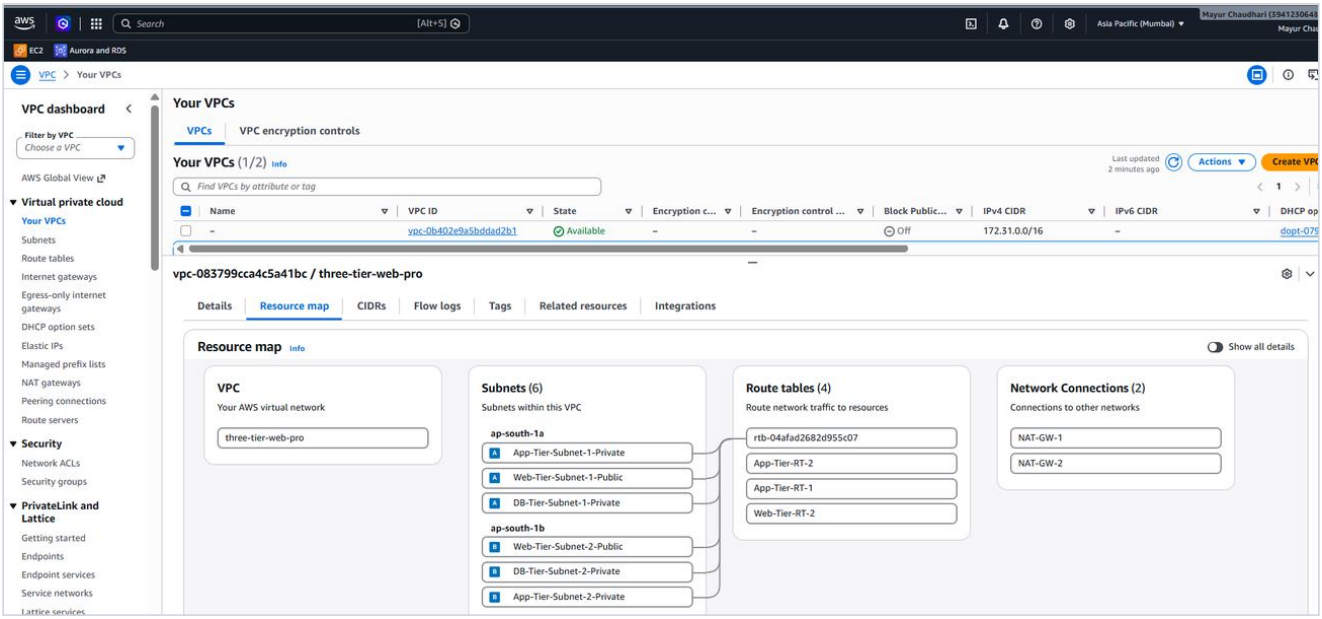
Step 2.11 Create route table.



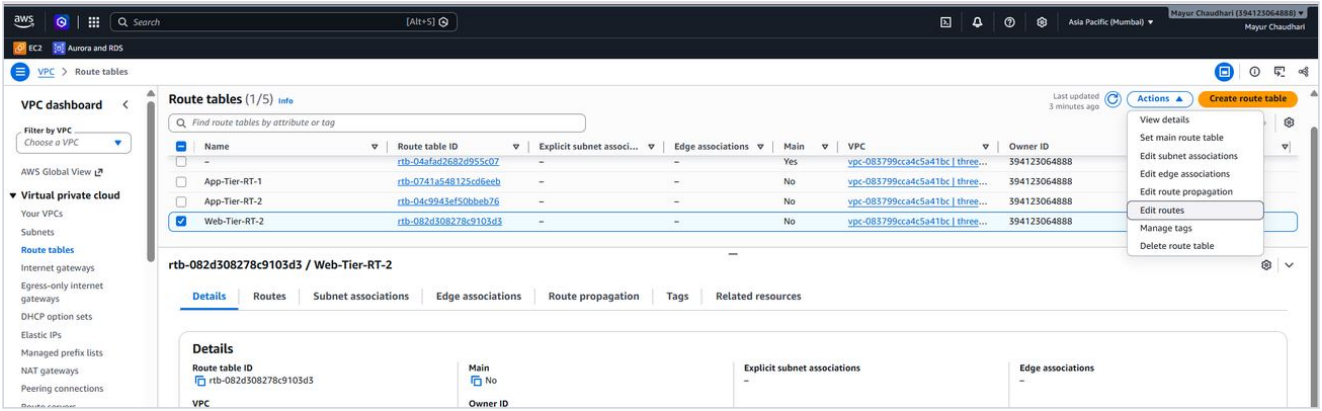
Step 2.12 Route tables created.



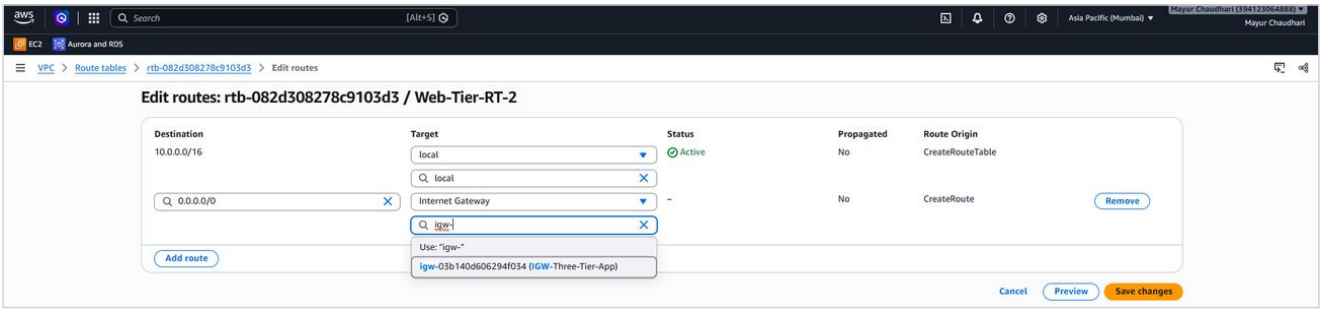
Step 2.13



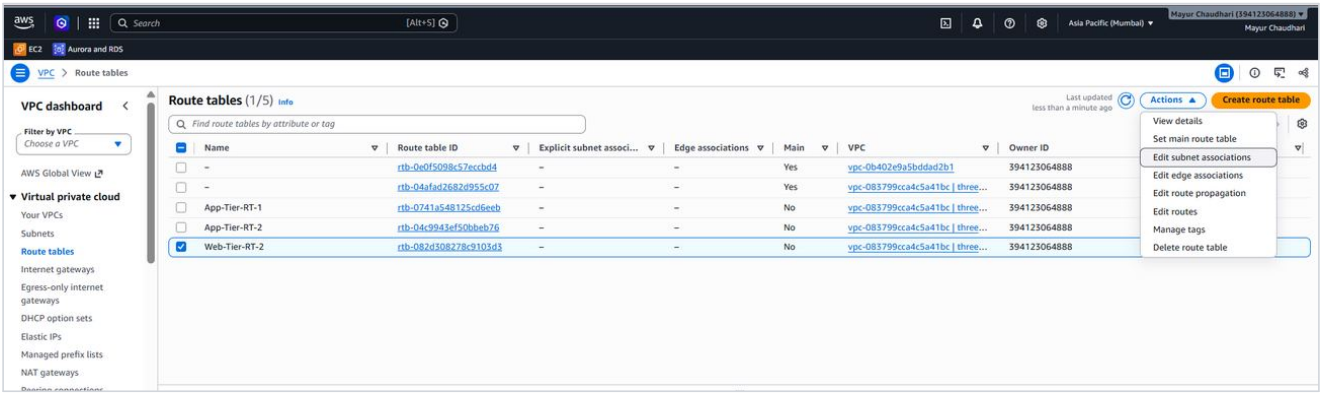
Step 2.14



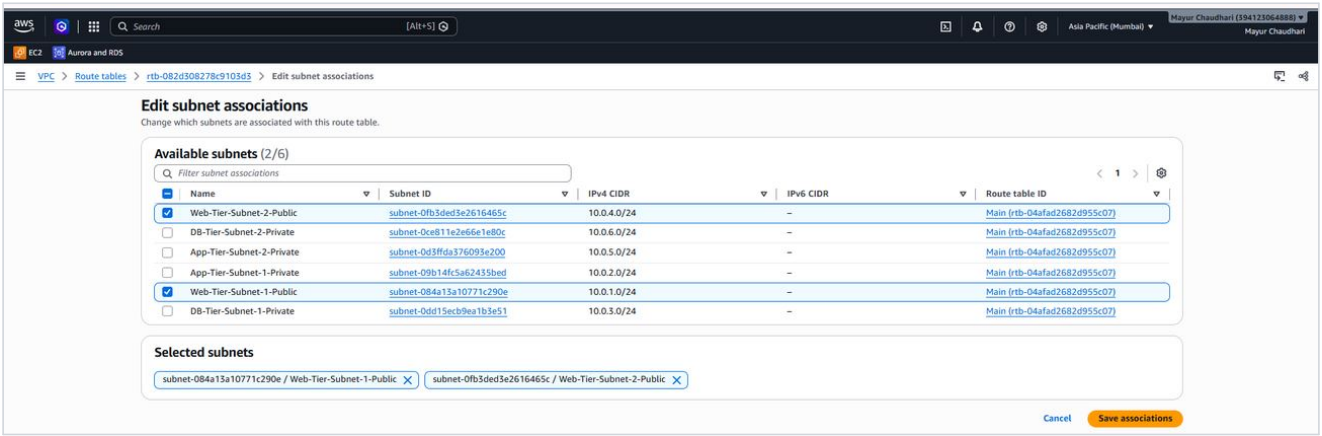
Step 2.15 Web-Tier-RT: edit routes, 0.0.0.0/0 to the Internet Gateway.



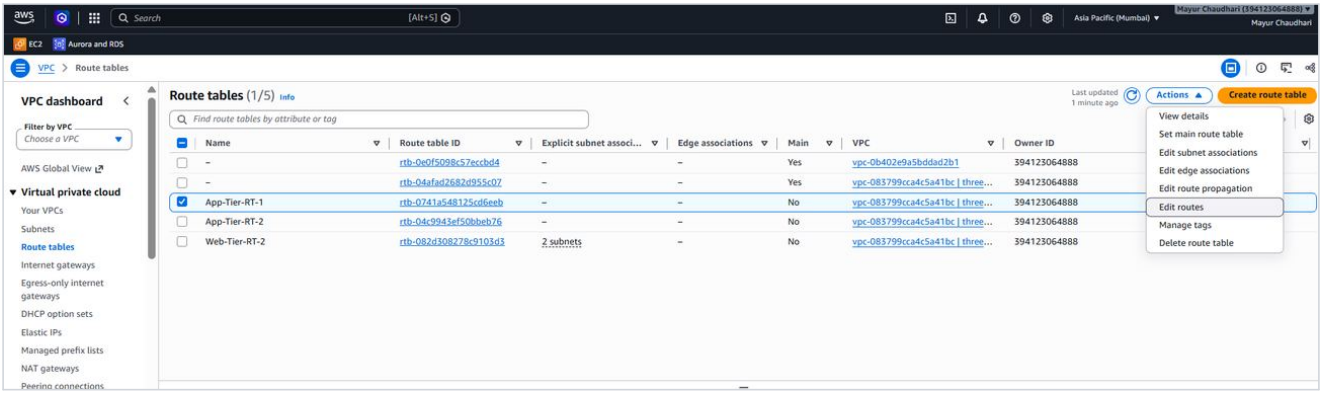
Step 2.16



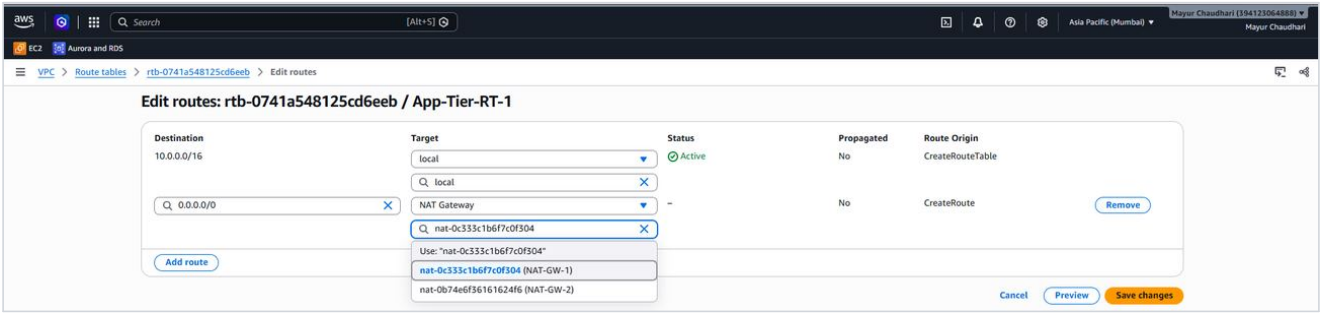
Step 2.17



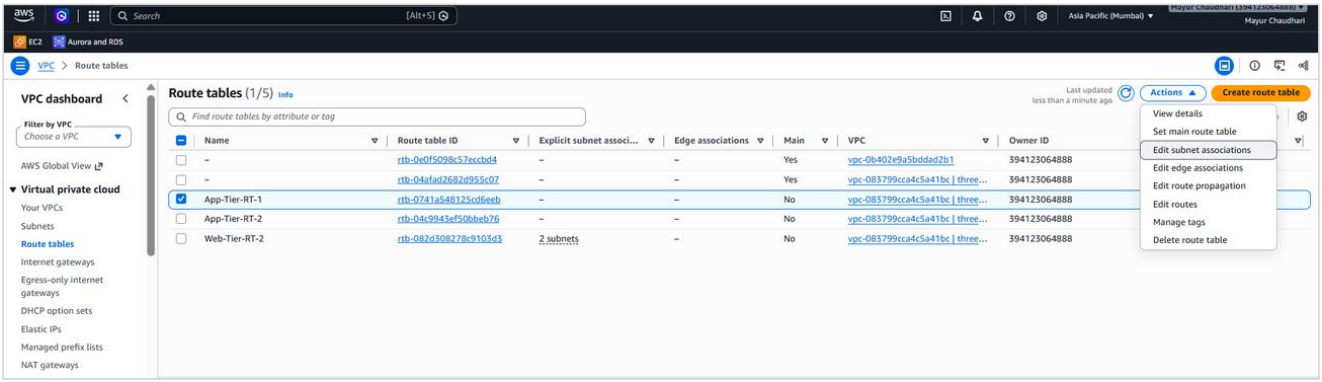
Step 2.18



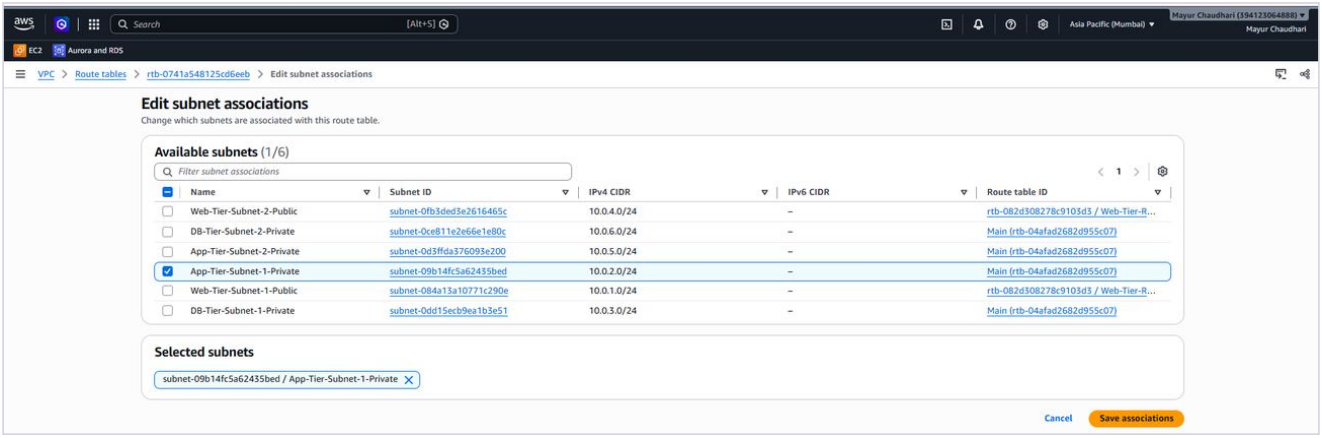
Step 2.19 App-Tier-RT-1: edit routes, 0.0.0.0/0 to NAT Gateway 1.



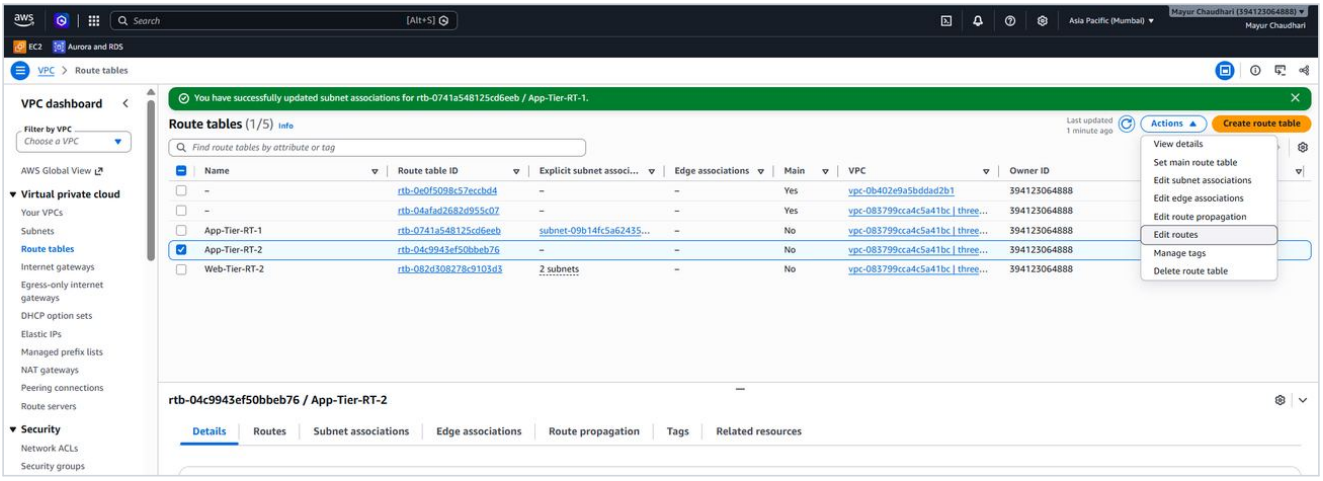
Step 2.20



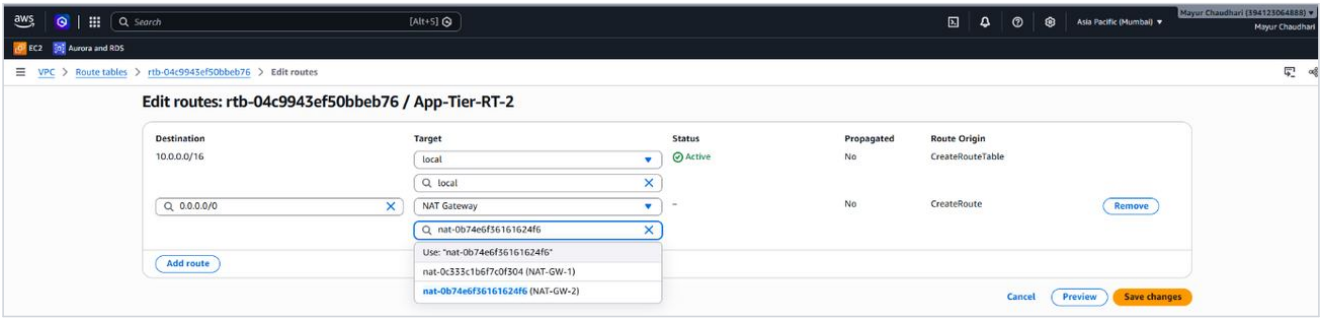
Step 2.21



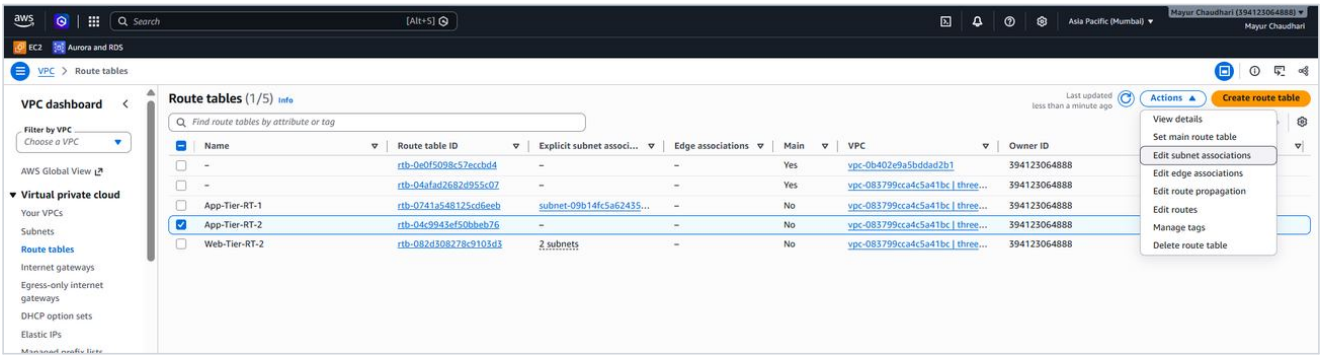
Step 2.22



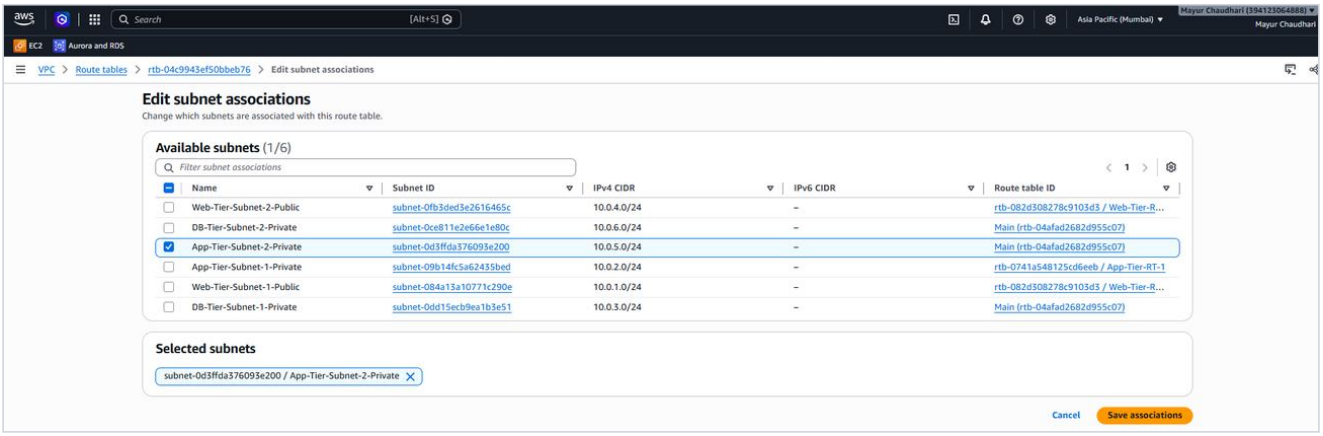
Step 2.23 App-Tier-RT-2: edit routes, 0.0.0.0/0 to NAT Gateway 2.



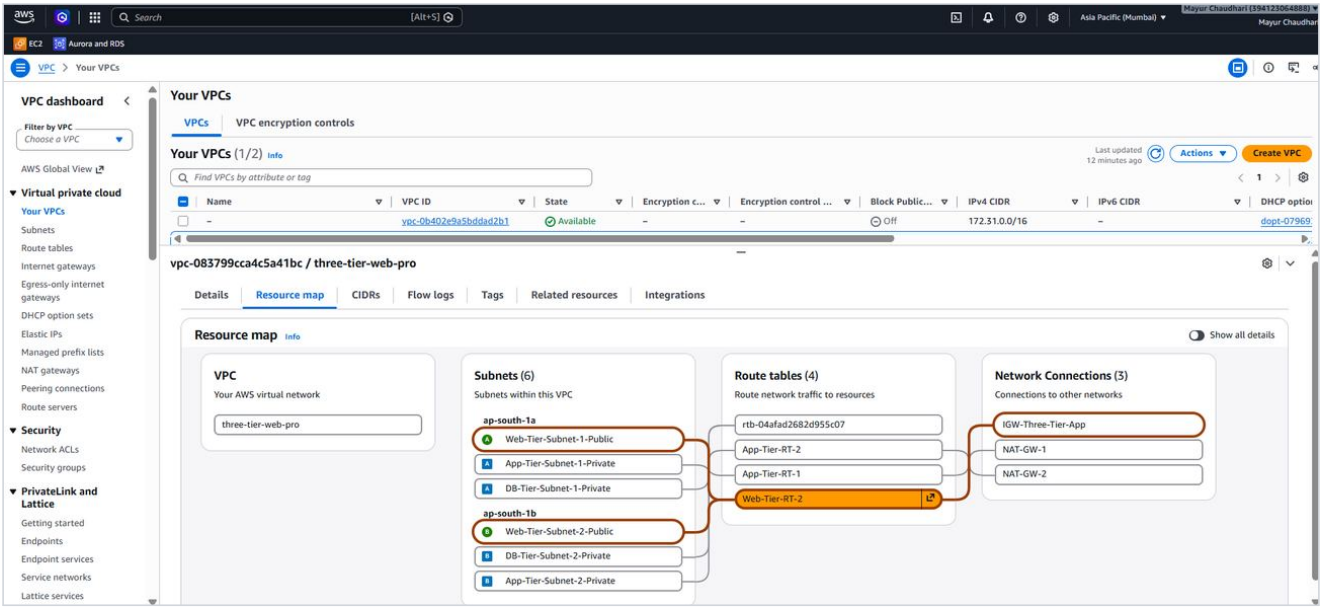
Step 2.24



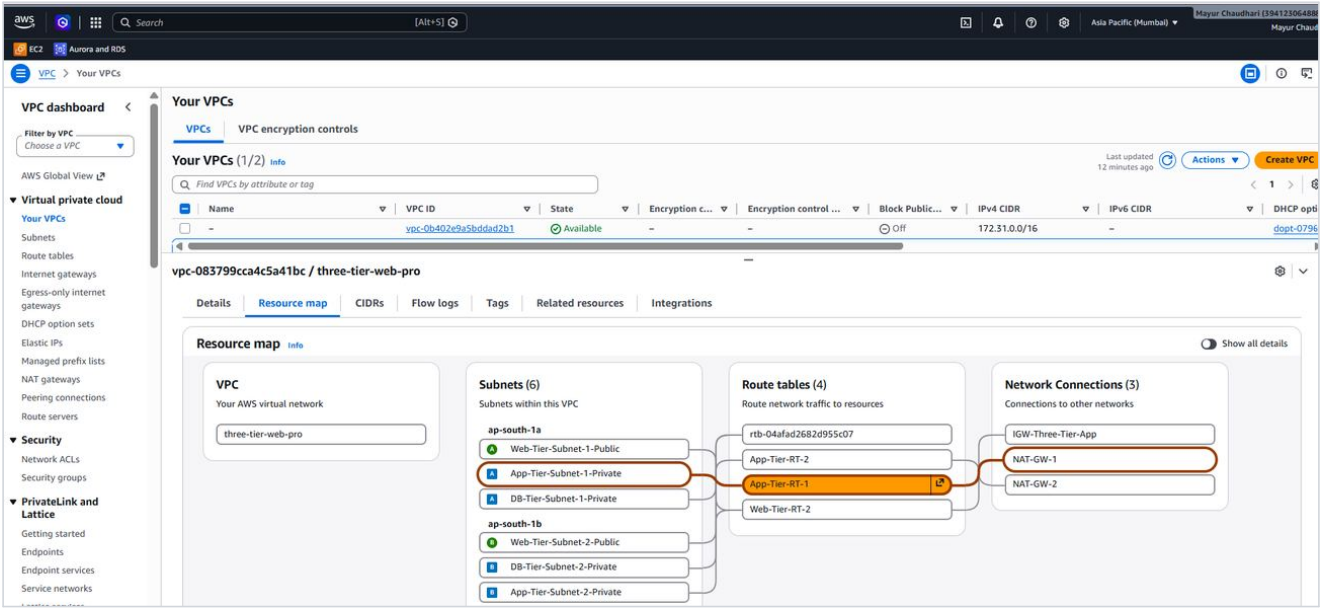
Step 2.25



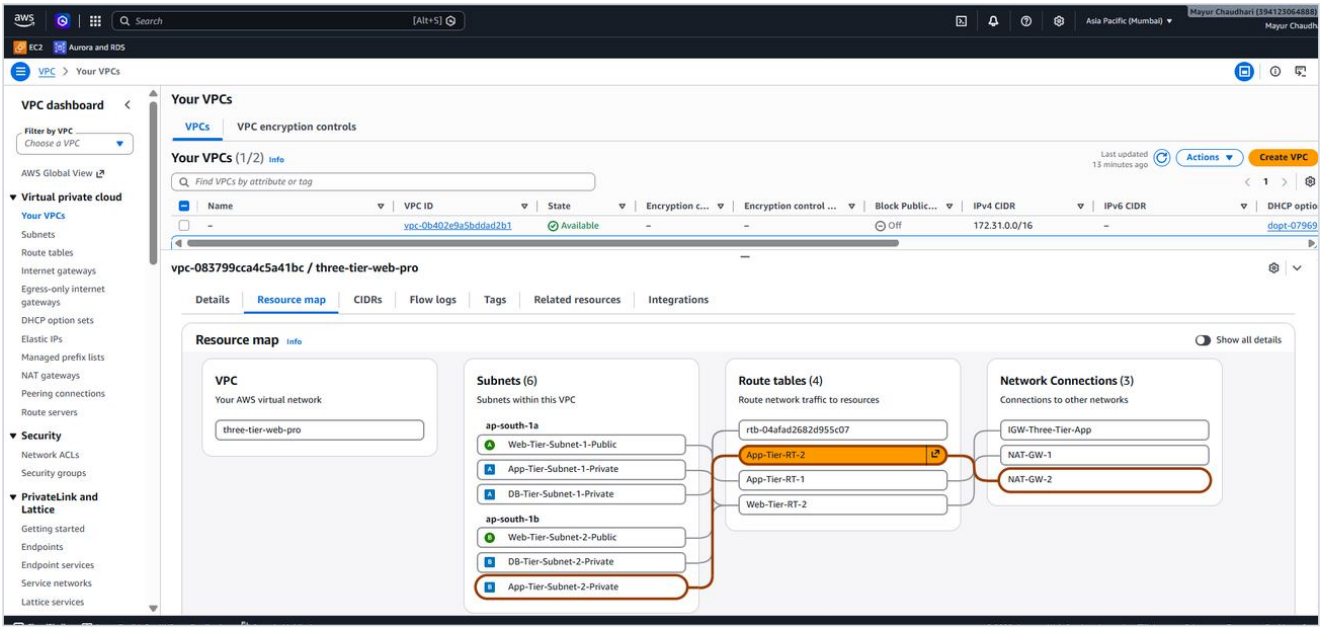
Step 2.26



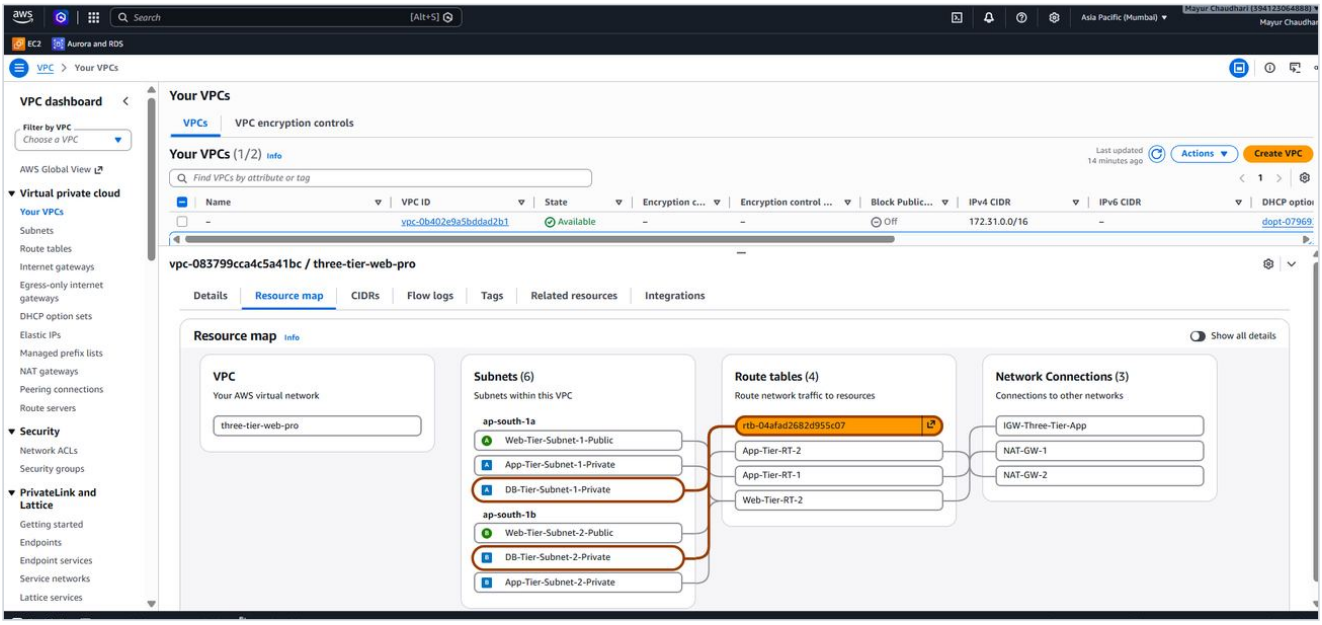
Step 2.27



Step 2.28

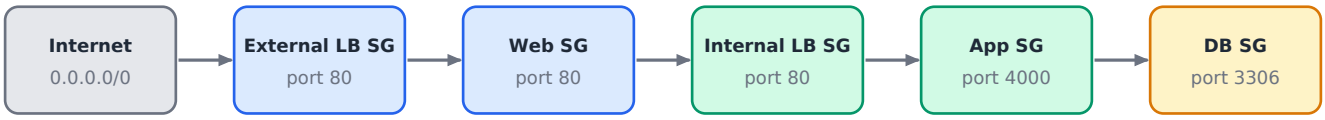


Step 2.29 VPC Resource map: check that the connections match the architecture diagram.



4. Step 3: Security Groups (the guest lists)

Each SG says "only *this* source may knock on *this* port". The clever part: sources are other **security groups**, not IP addresses. So traffic can only walk through the tiers in order:

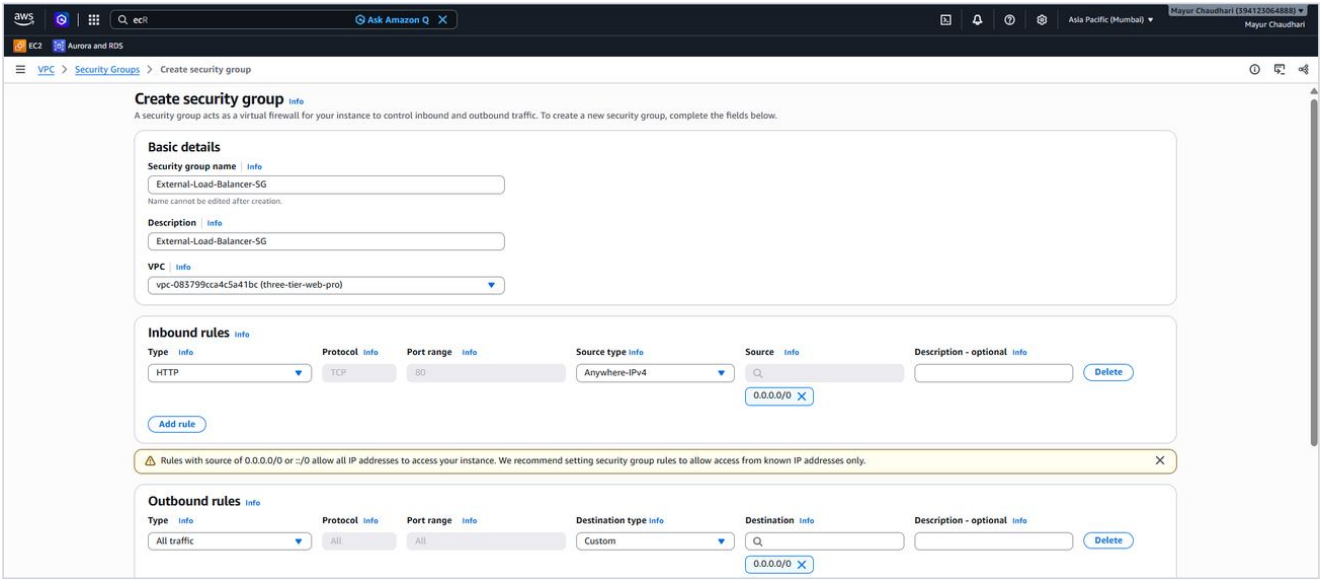


Security Group	Allows inbound	Port	Only from
External-Load-Balancer-SG	HTTP	80	0.0.0.0/0 (whole internet)
Web-Tier-SG	HTTP	80	External-Load-Balancer-SG
Internal-Load-Balancer-SG	HTTP	80	Web-Tier-SG
App-Tier-SG	Custom TCP	4000	Internal-Load-Balancer-SG
DB-Tier-SG	MySQL/Aurora	3306	App-Tier-SG

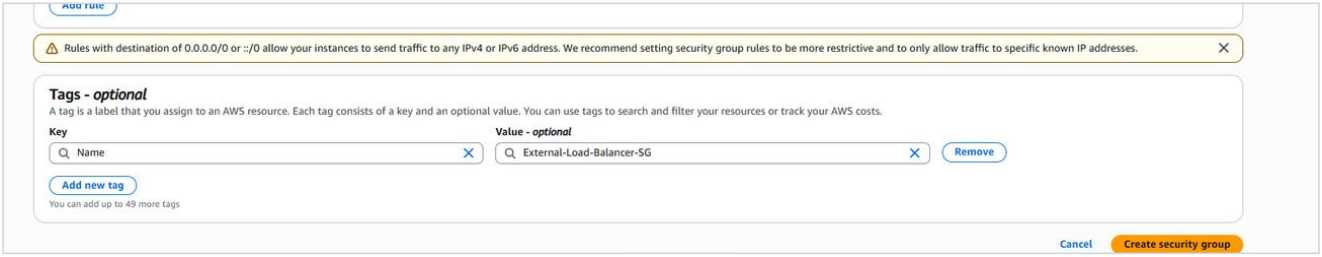
Result: the database cannot be reached by the internet, by the web server, or by anyone except the app servers. That is the core security benefit of 3-tier.

Step 3 - Security groups

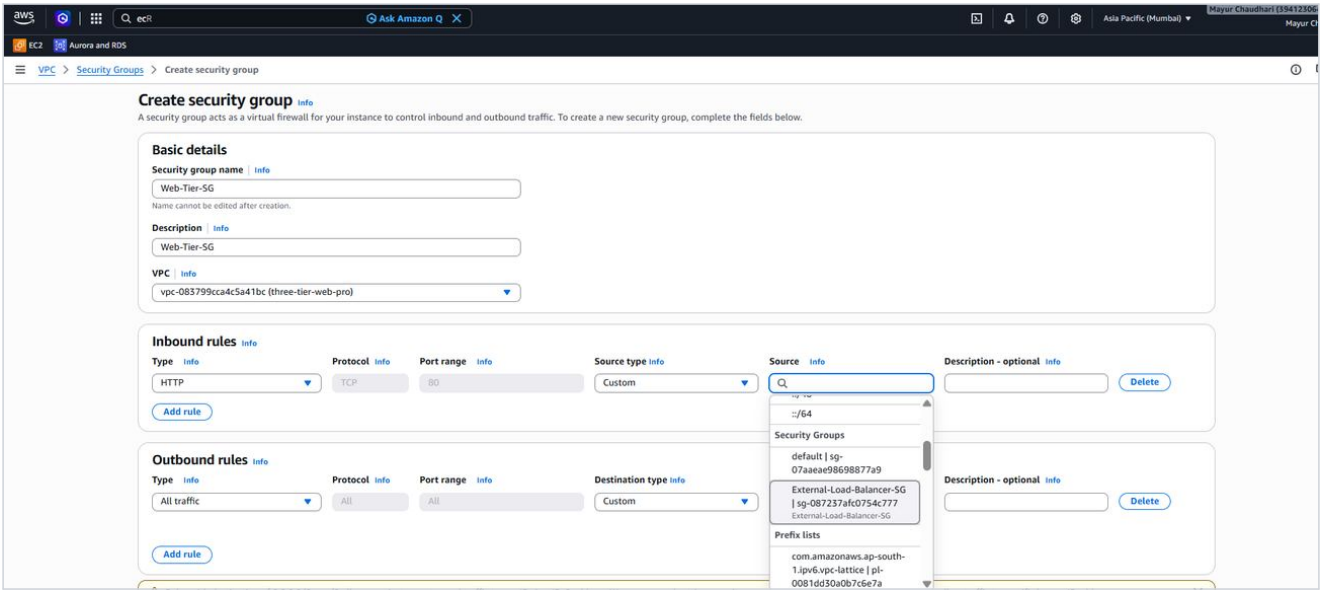
Step 3.1



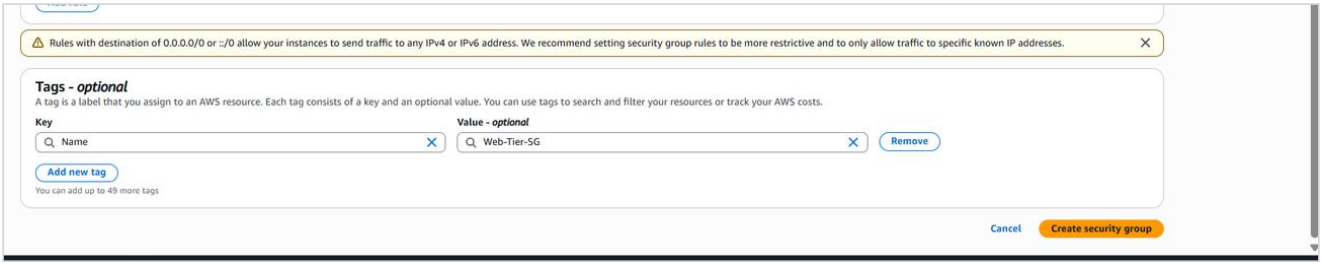
Step 3.2



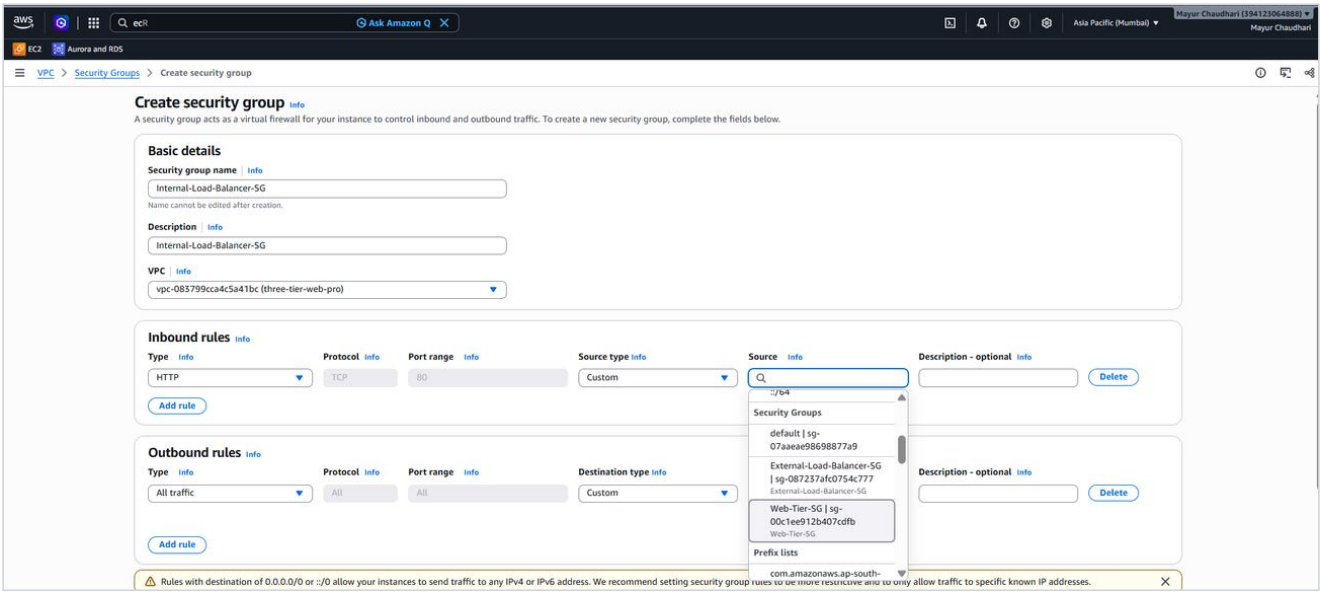
Step 3.3



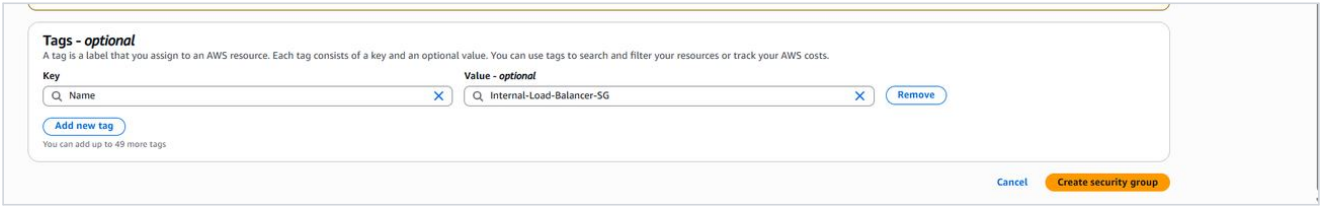
Step 3.4



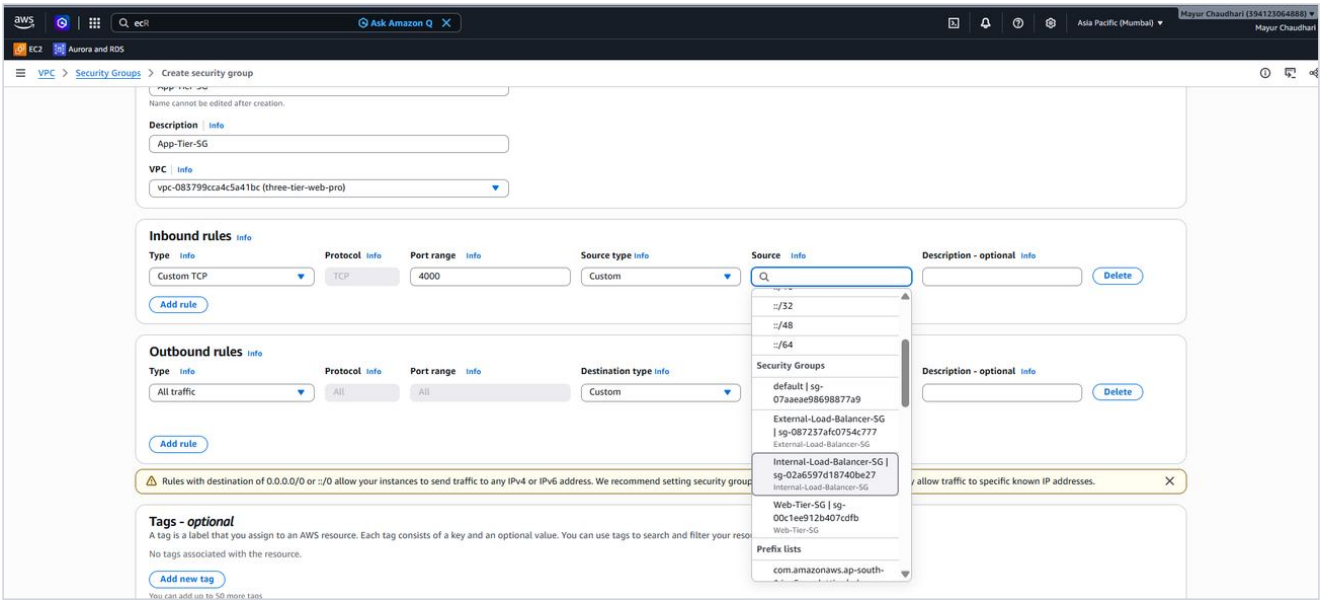
Step 3.5



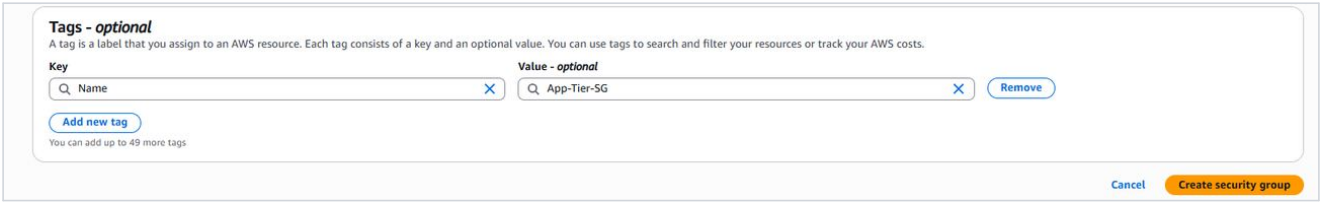
Step 3.6



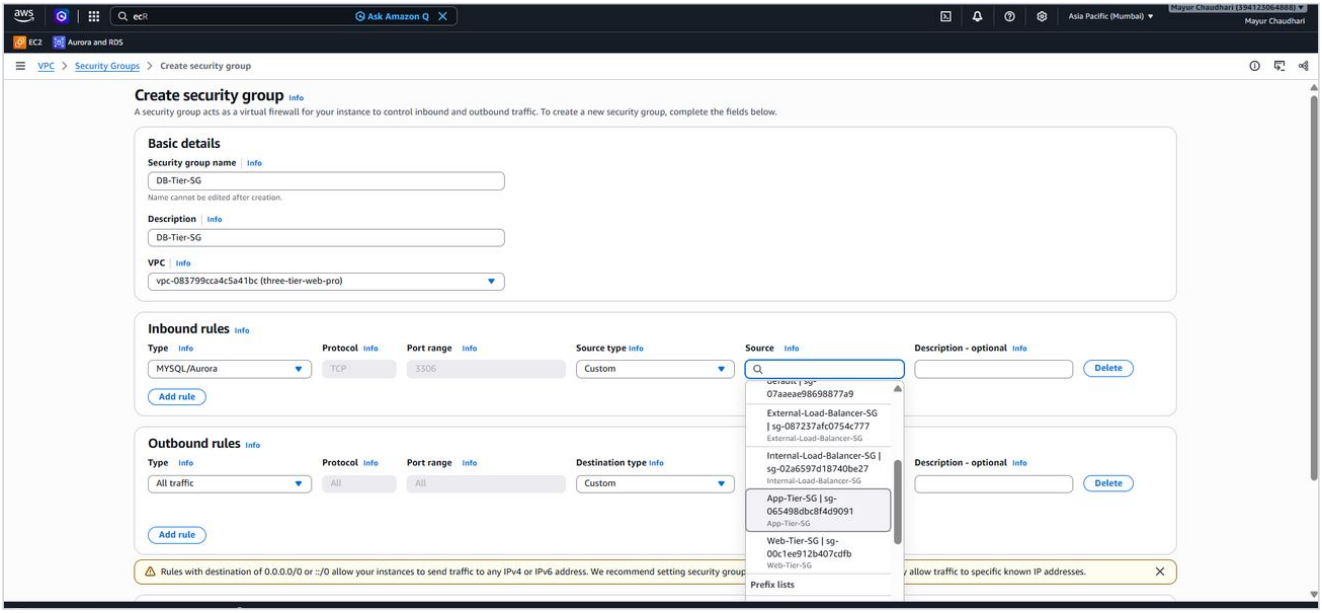
Step 3.7



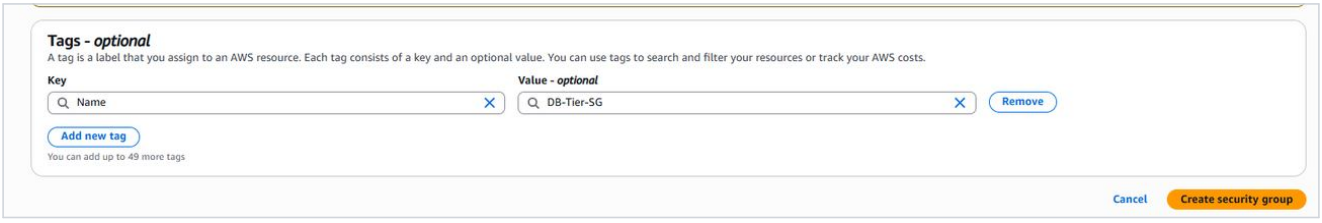
Step 3.8



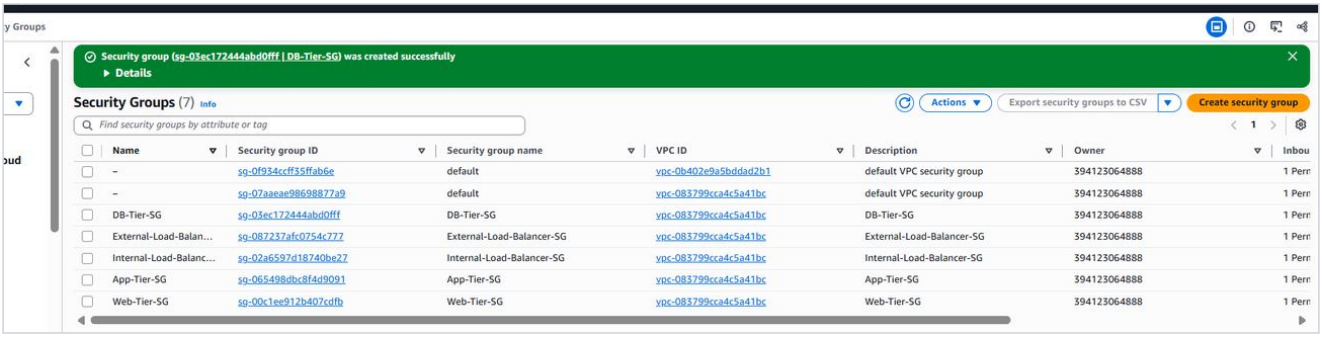
Step 3.9



Step 3.10



Step 3.11 All five security groups created.



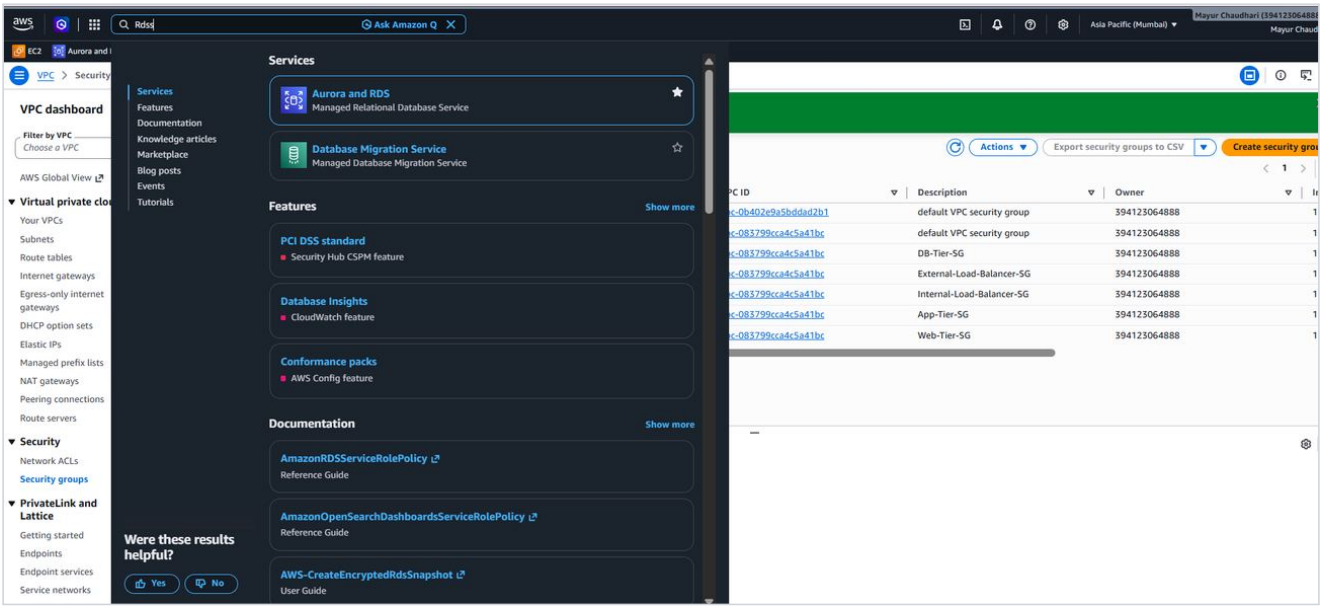
5. Steps 4-6: Database, IAM role, S3

5.1 Database (Amazon RDS / Aurora MySQL)

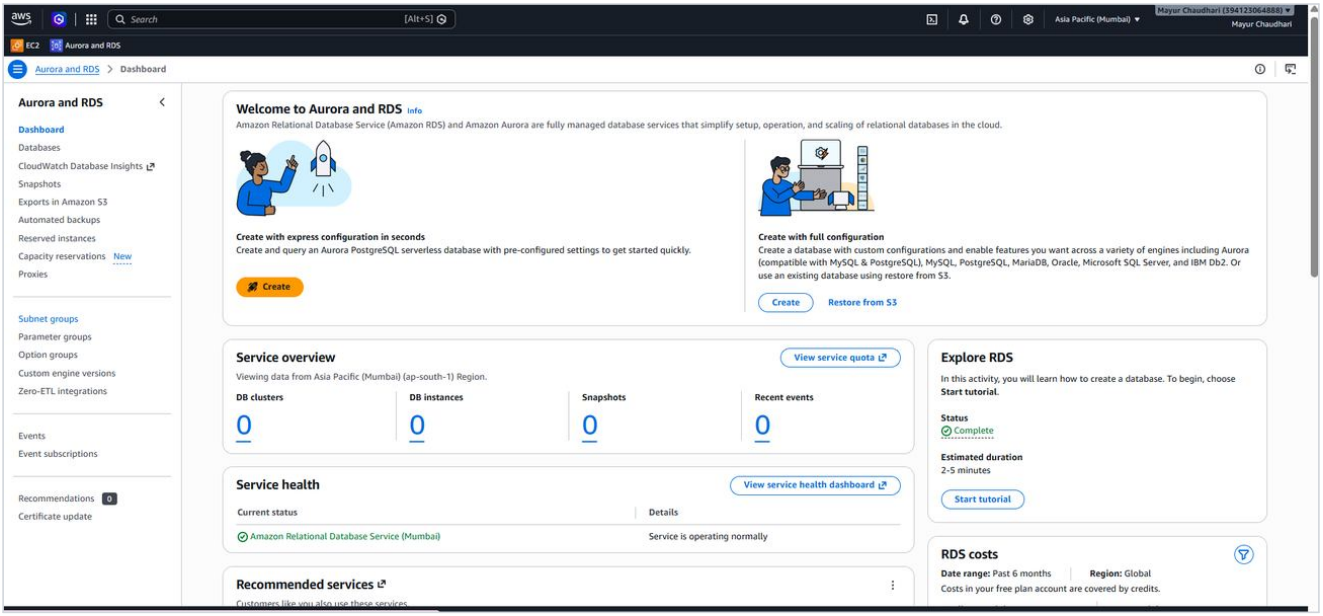
- First create a **DB subnet group** using the two DB subnets (10.0.3.0/24 in 1a and 10.0.6.0/24 in 1b). AWS needs subnets in two AZs so it can place a standby copy for failover.
- Create the database: engine **Aurora (MySQL compatible)**, attach **DB-Tier-SG**, put it in the subnet group, no public access.
- Save the **writer endpoint**, **username** and **password**. The app needs these.
- Cost warning: the DB is the most expensive piece. The doc deletes everything right after the demo.

Step 4 - DB subnet group and database

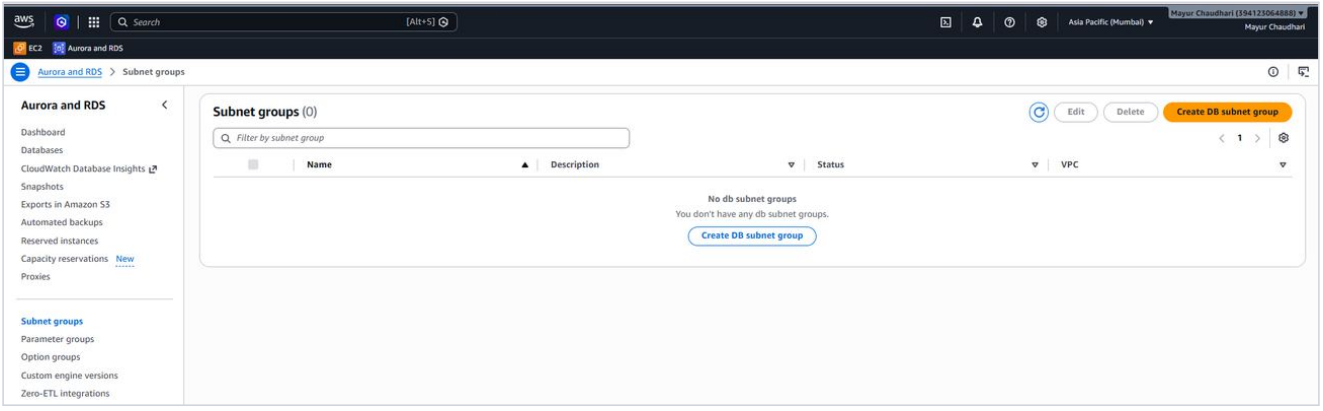
Step 4.1



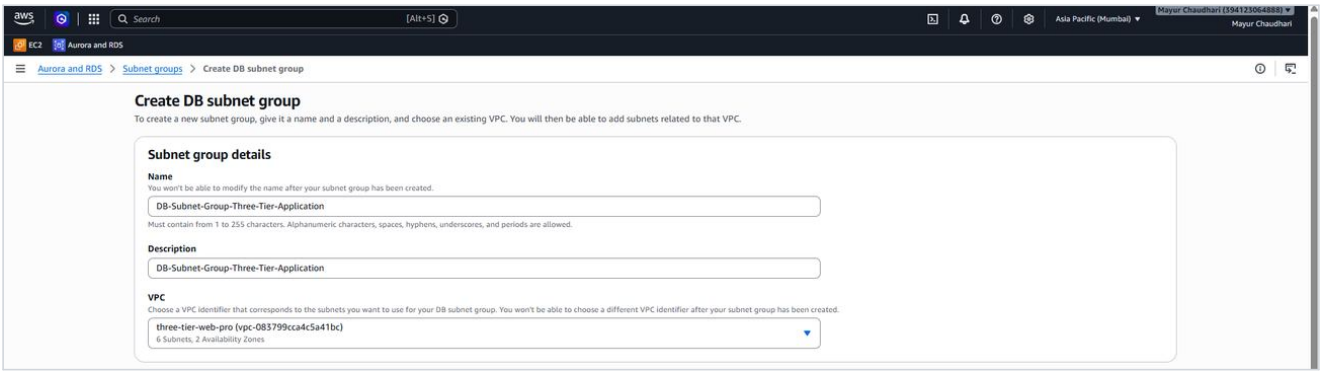
Step 4.2 RDS dashboard.



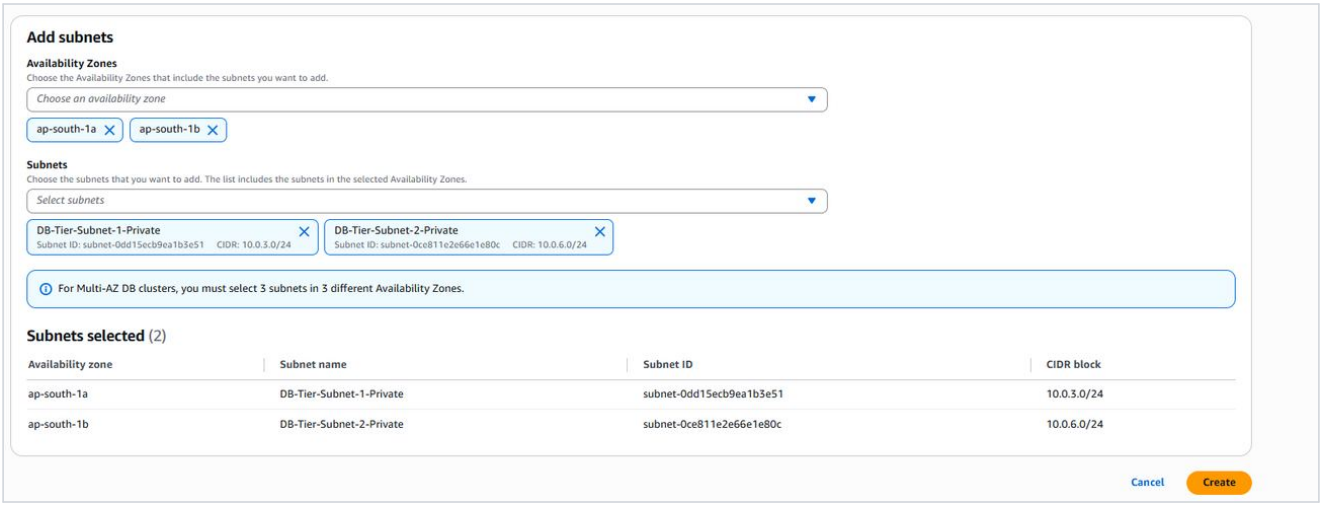
Step 4.3 DB subnet groups.



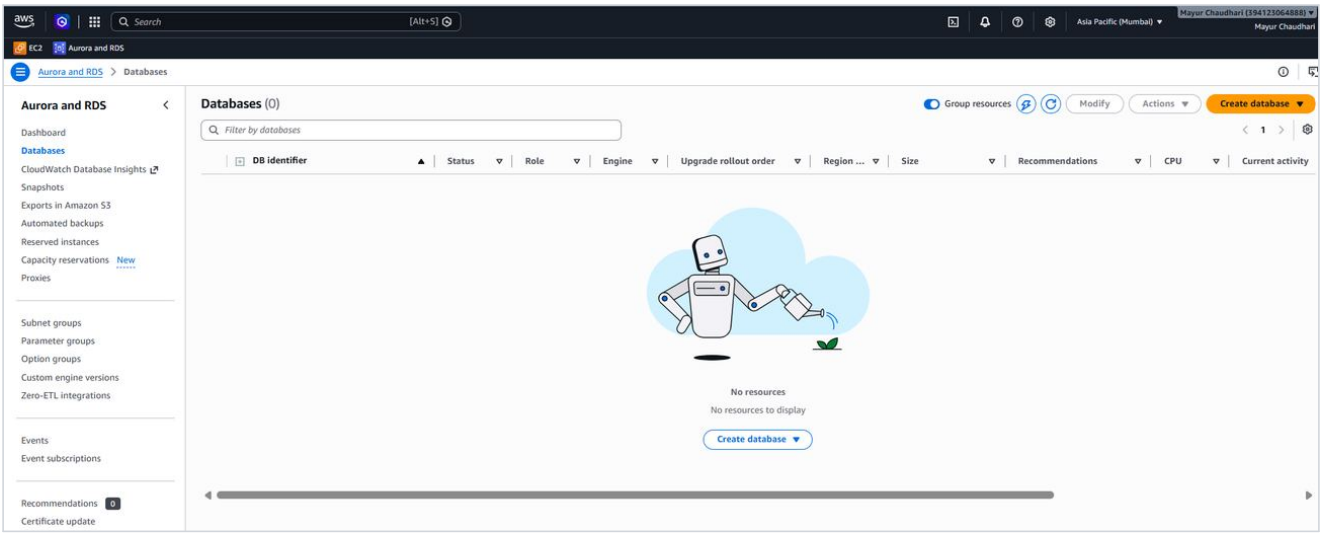
Step 4.4 Create DB subnet group.



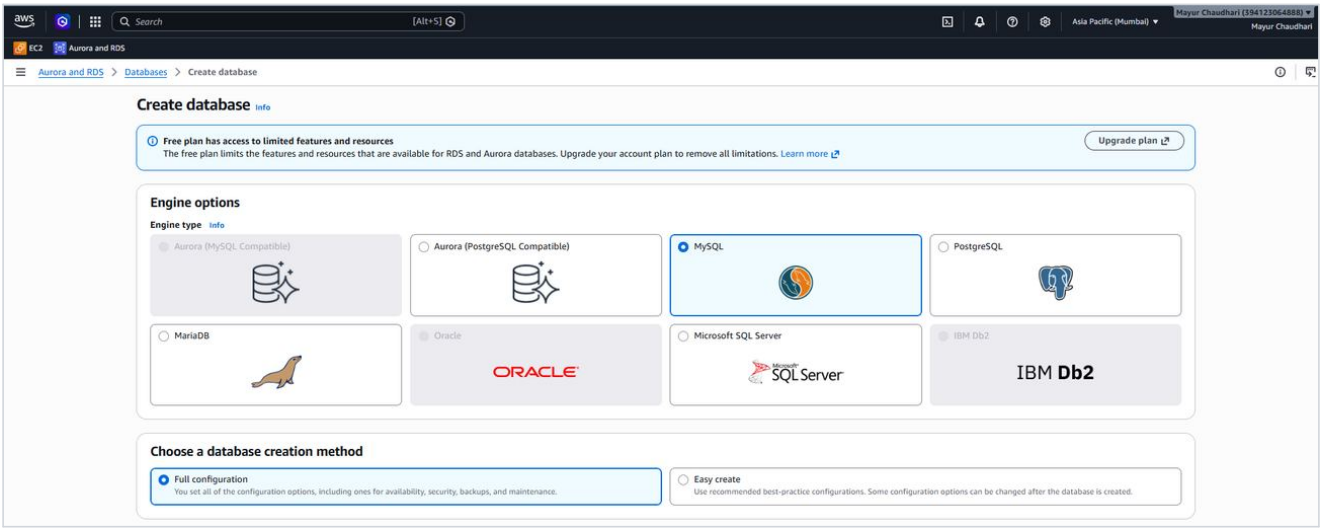
Step 4.5 Choose the two AZs and the two DB subnets.



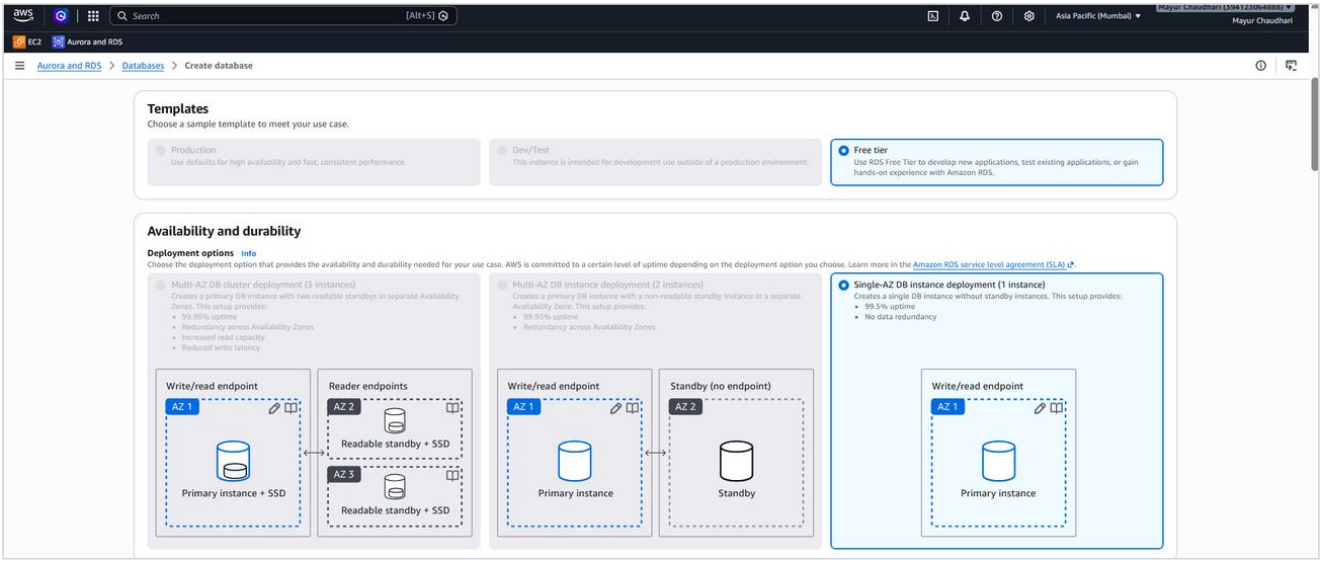
Step 4.6 Databases list.



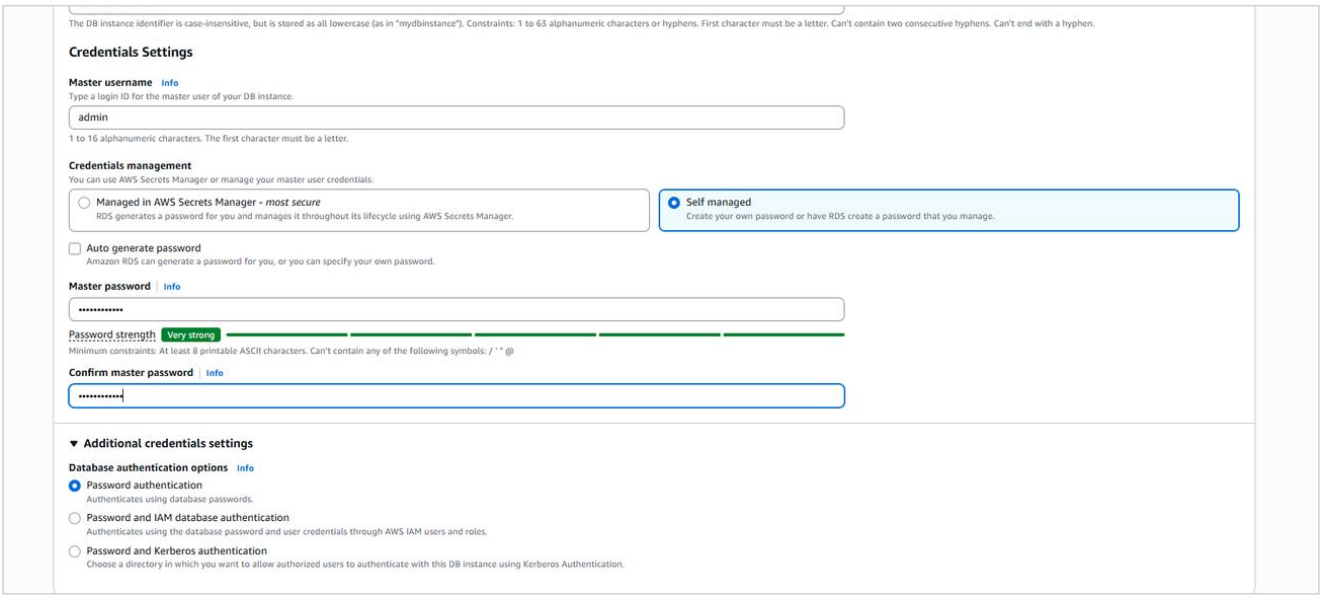
Step 4.7 Create database: choose the engine.



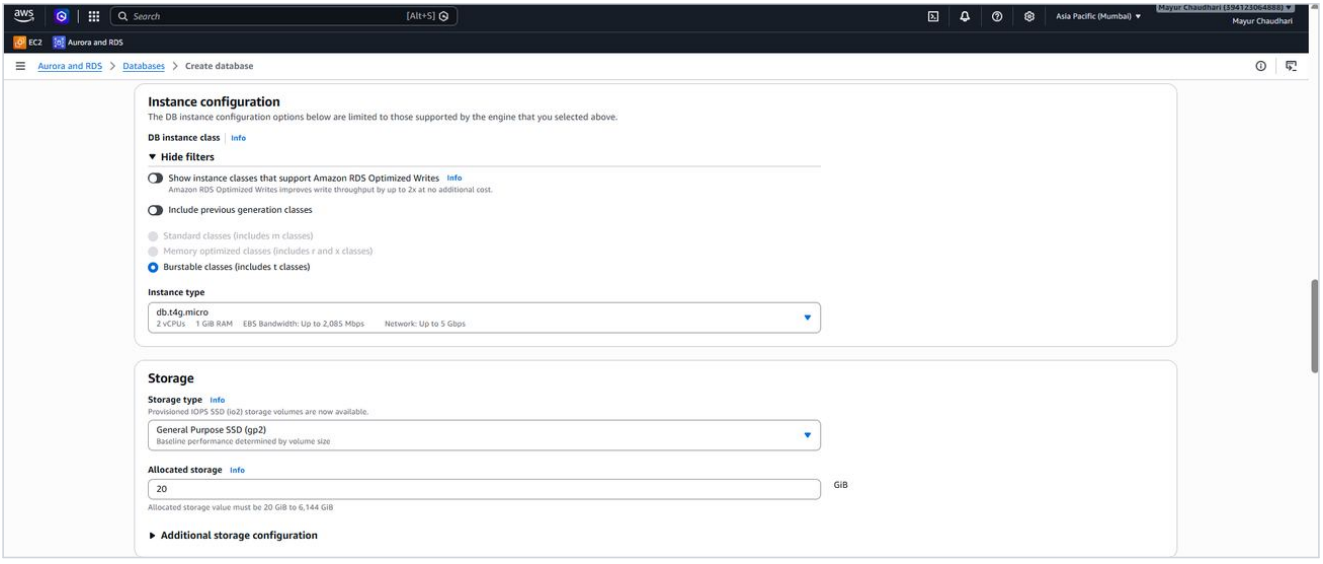
Step 4.8



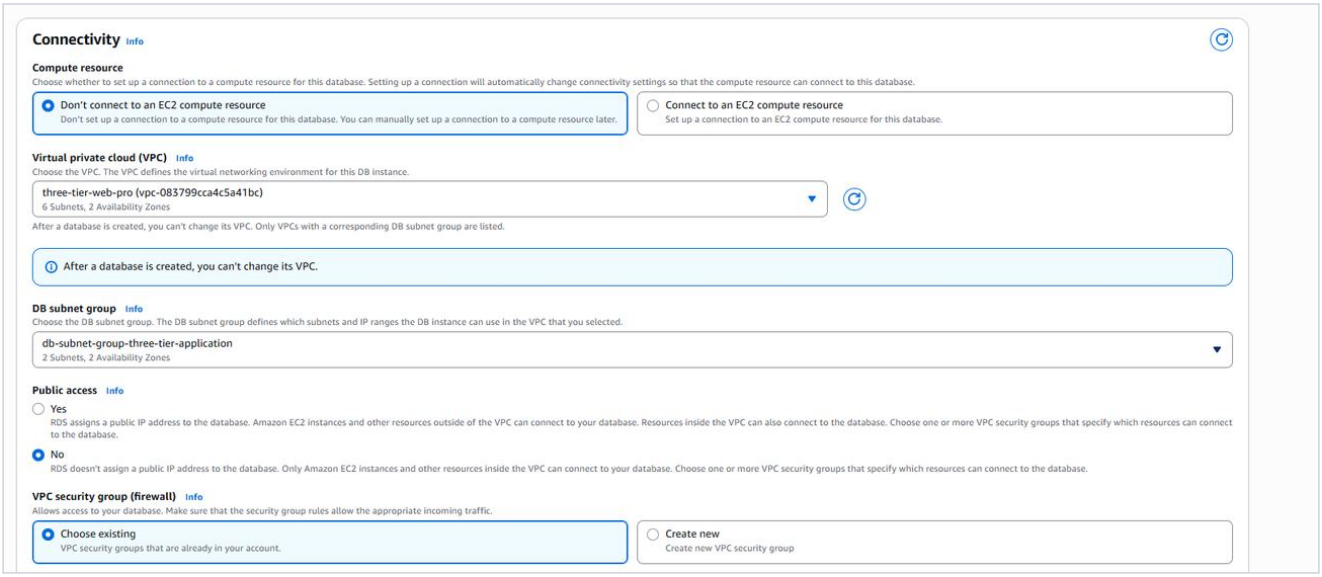
Step 4.9 Database credentials: set master user and password.



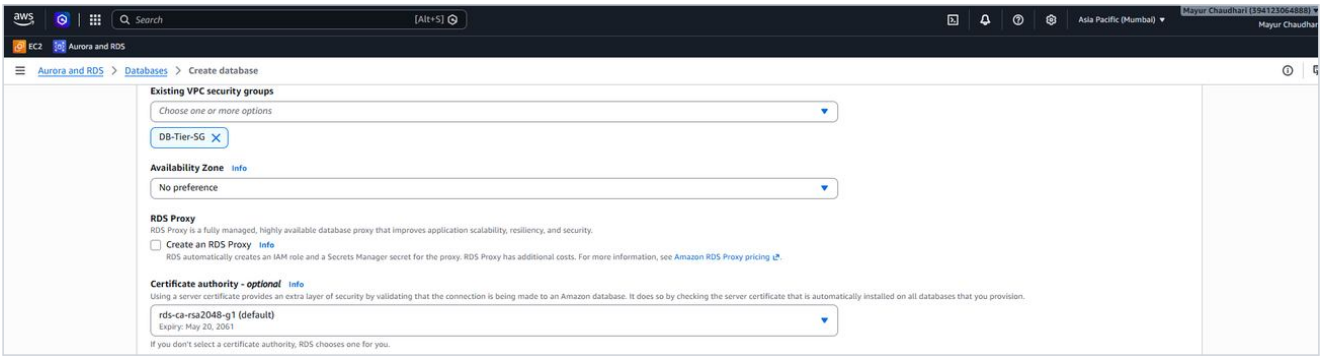
Step 4.10 Connectivity: your VPC, DB subnet group and DB-Tier-SG.



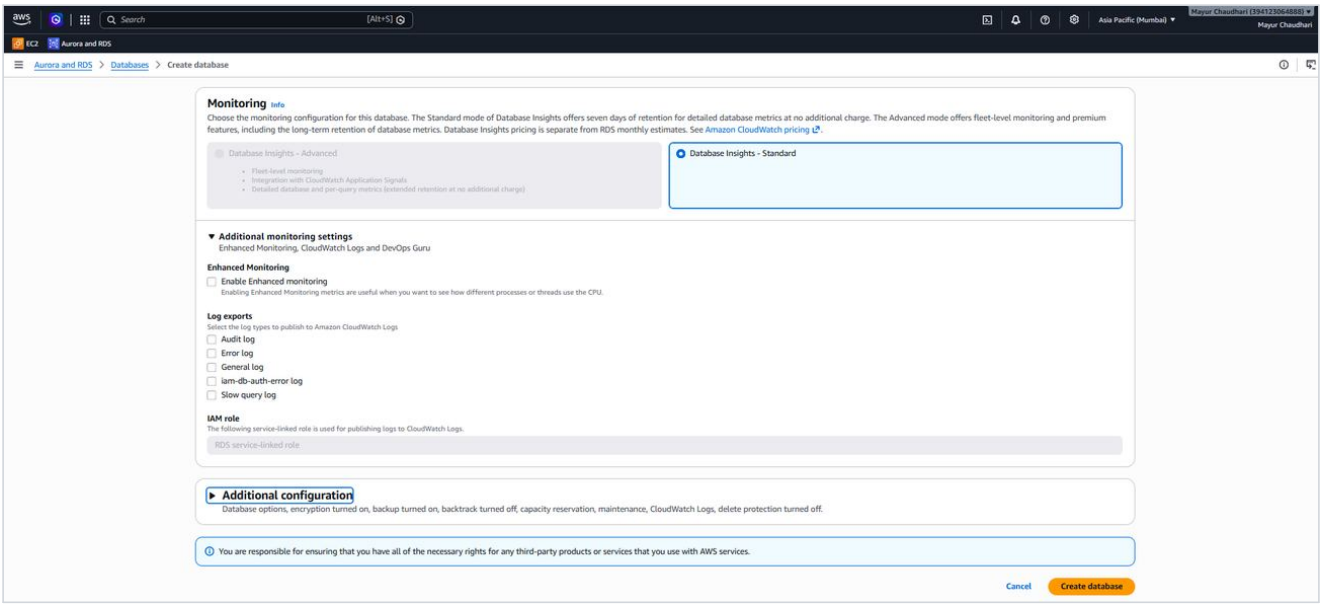
Step 4.11



Step 4.12



Step 4.13



5.2 IAM role (the server's ID card)

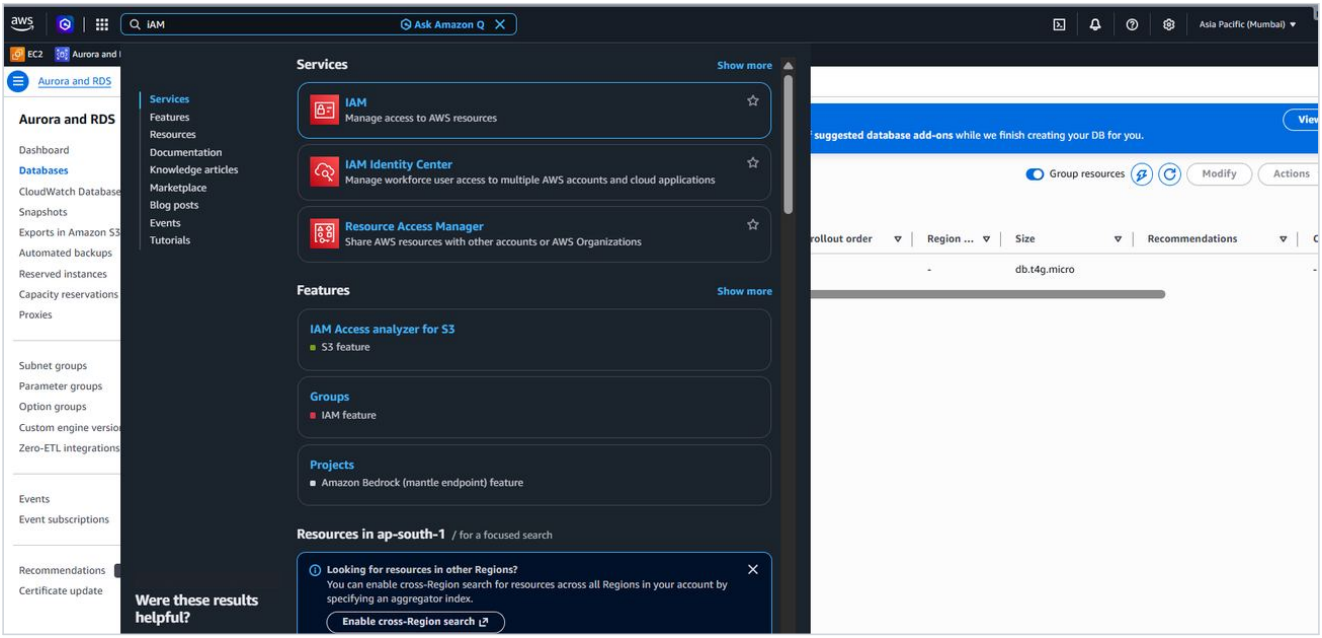
One role, two permissions, attached to every EC2 server:

Policy	Why it is needed
AmazonS3ReadOnlyAccess	So the server can download the app code from the S3 bucket

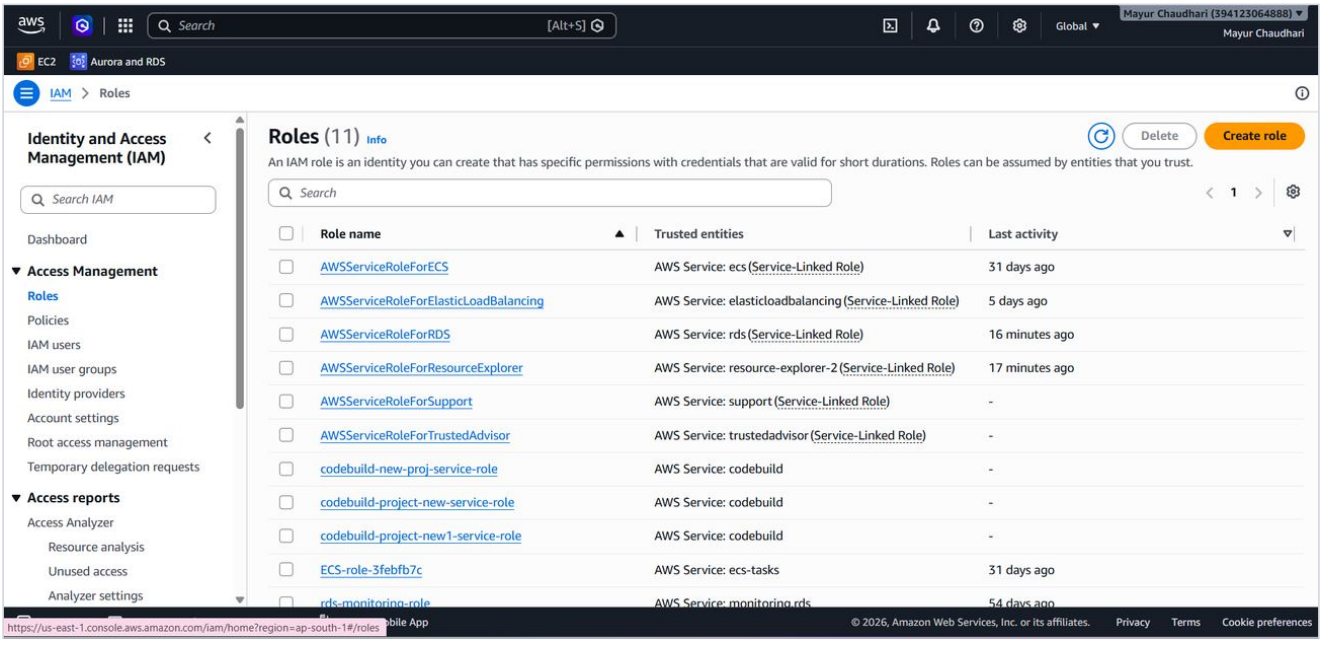
Policy	Why it is needed
AmazonSSMManagedInstanceCore	So you can open a terminal on the private server through Session Manager (no SSH, no key pair, no open port 22)

Step 5 - IAM role

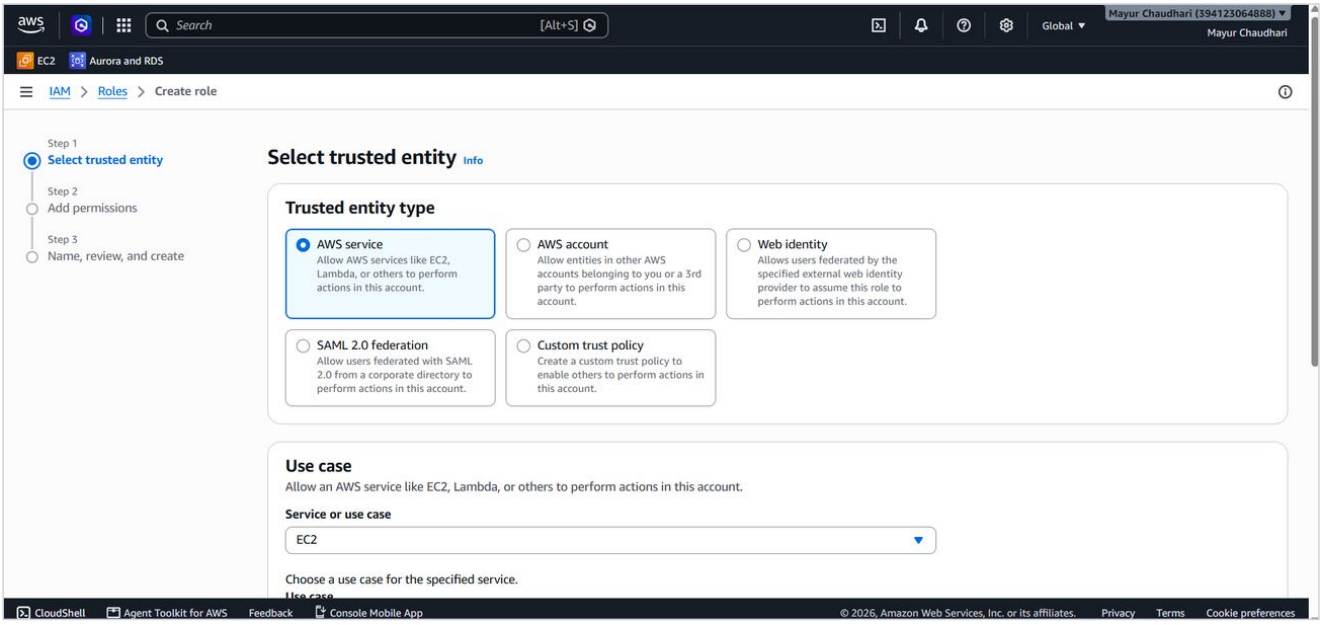
Step 5.1



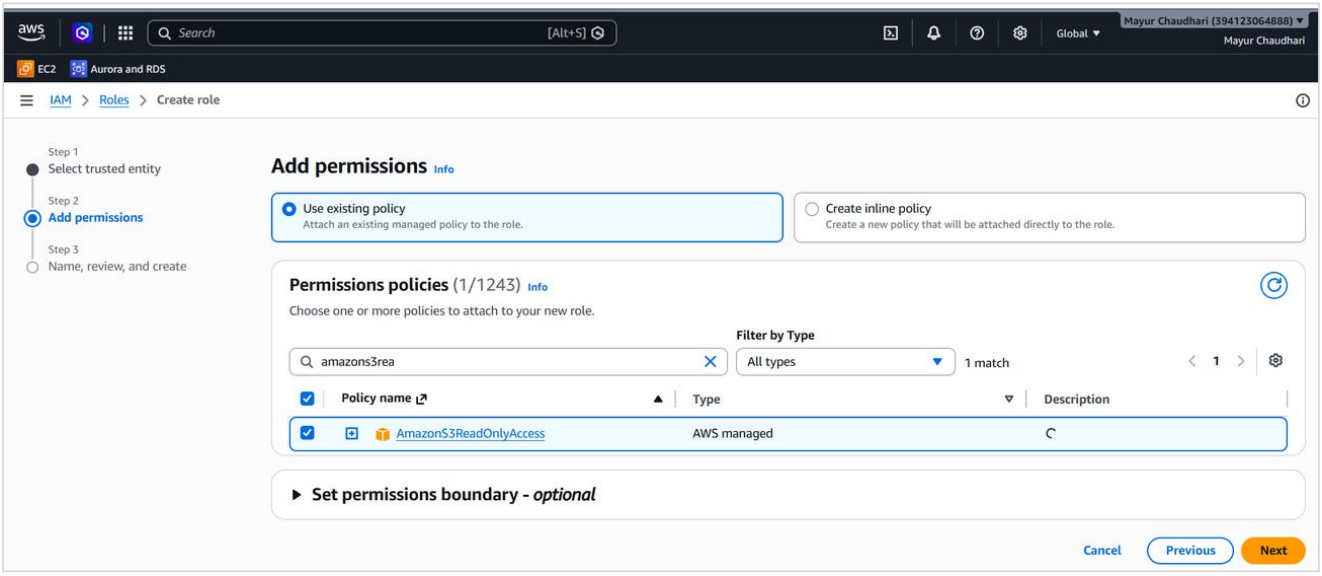
Step 5.2 IAM > Roles.



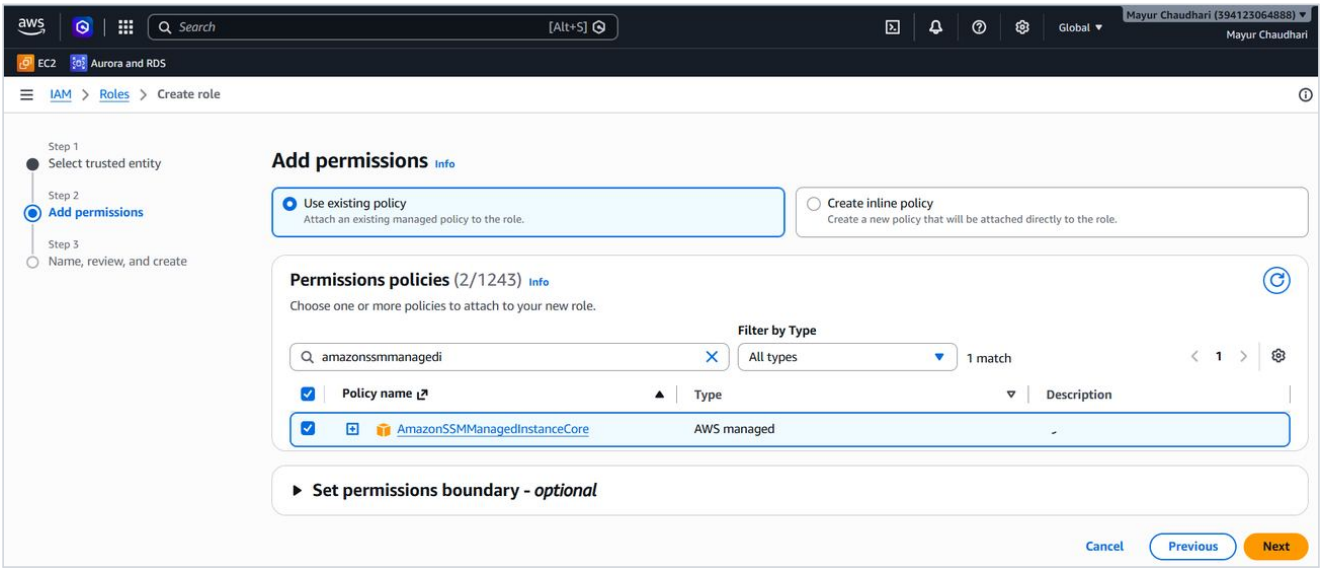
Step 5.3 Trusted entity: AWS service, use case EC2.



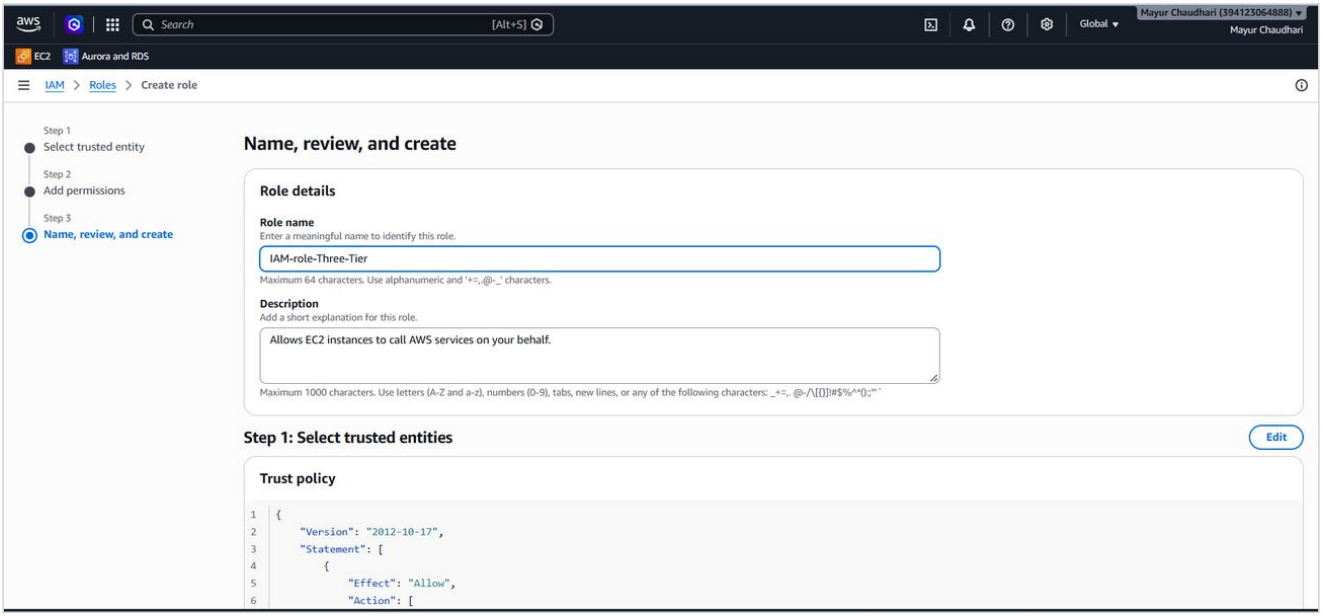
Step 5.4 Add permissions policy.



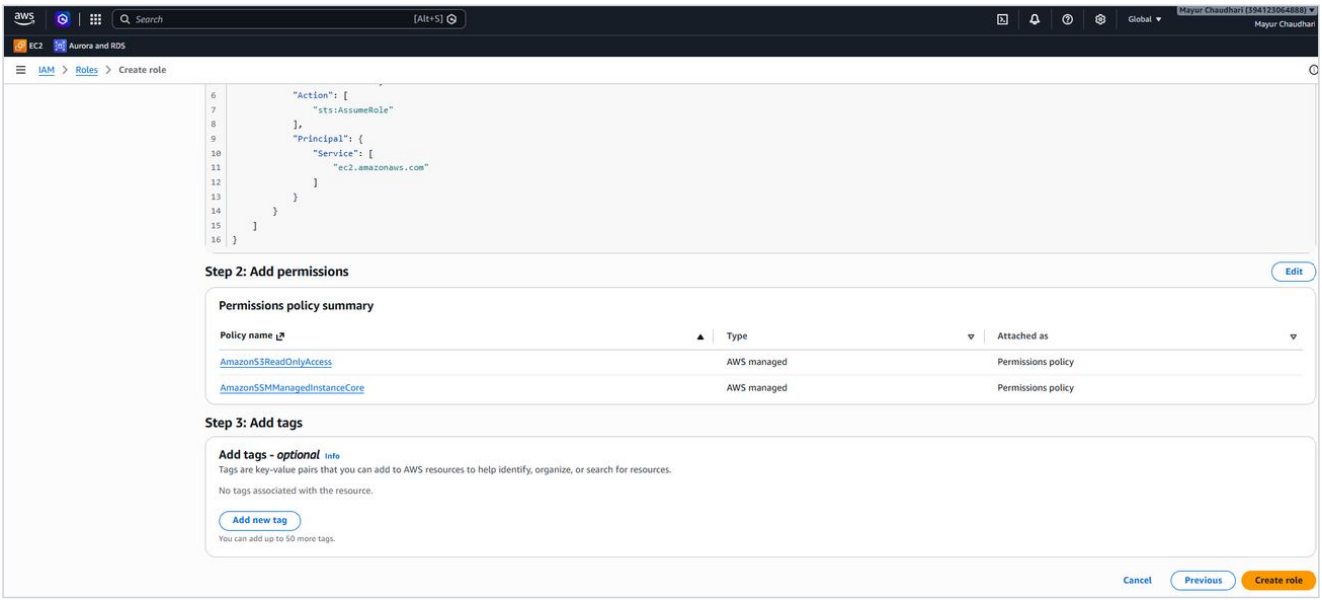
Step 5.5 Add permissions policy.



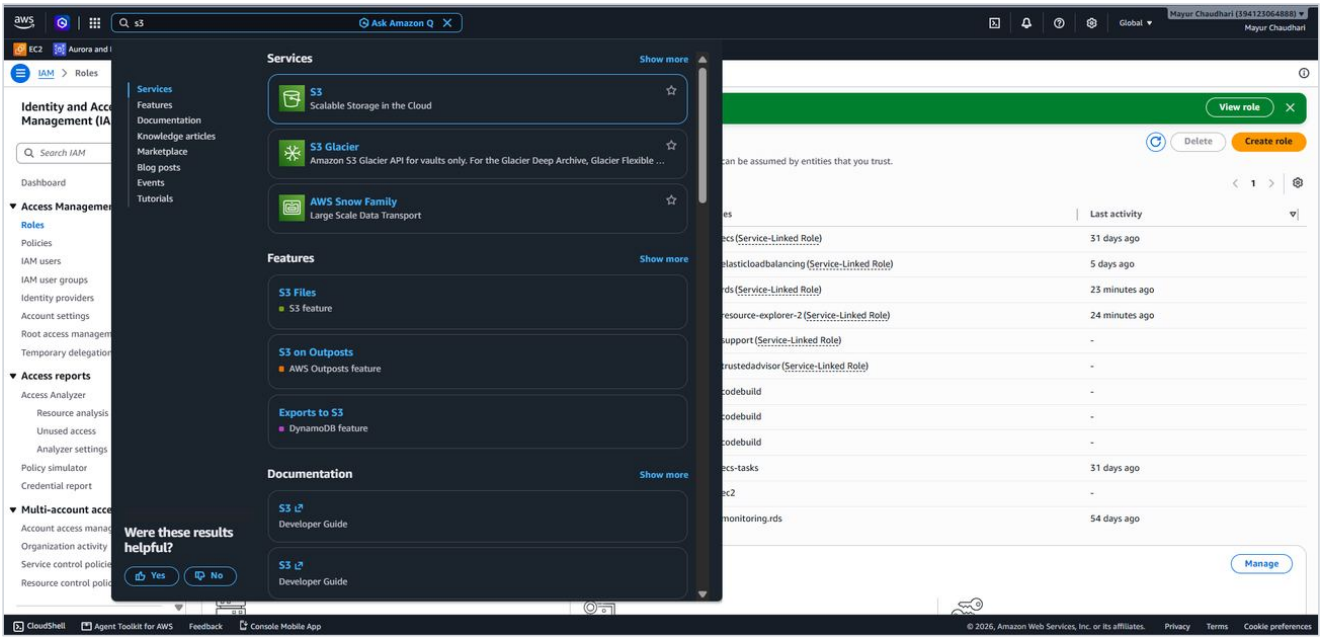
Step 5.6 Name, review and create the role.



Step 5.7 Review the permissions.



Step 5.8 Role created.



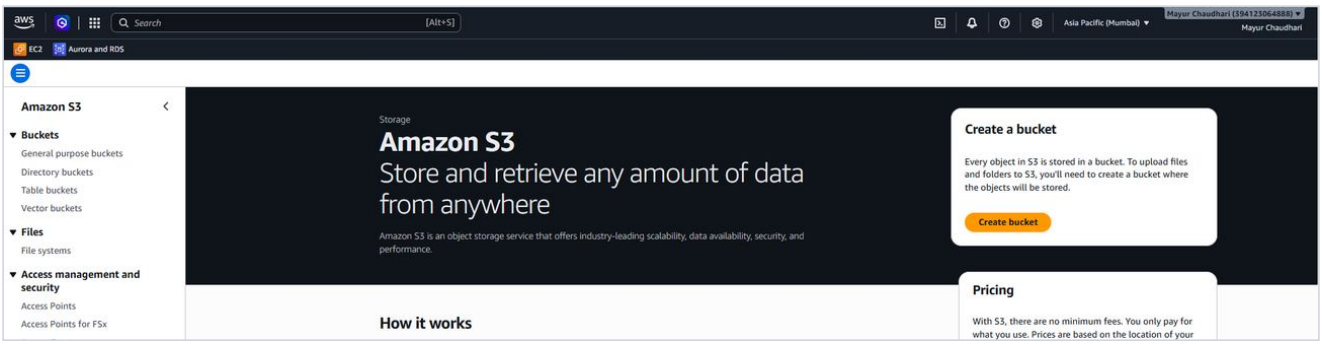
5.3 S3 bucket (the code store)

Create a bucket with a unique name and upload the `application-code` folder (from the GitHub repo *SubodhK143/three-tier-app-on-aws*). New servers simply copy their code from here, which makes setup easy and repeatable. Before uploading, two files are edited:

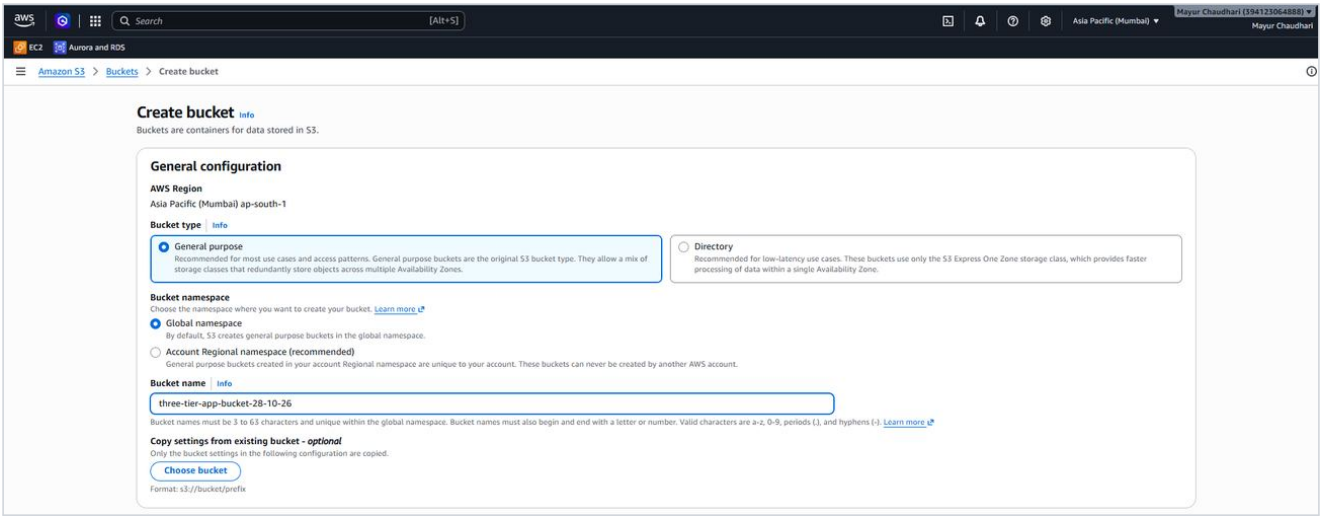
- `app-tier/DbConfig.js` : set DB_HOST (RDS writer endpoint), DB_USER, DB_PWD, DB_DATABASE (= webappdb).
- `nginx.conf` : paste the **Internal Load Balancer DNS name** in the proxy section (done later, in step 10).

Step 6 - S3 bucket and source code

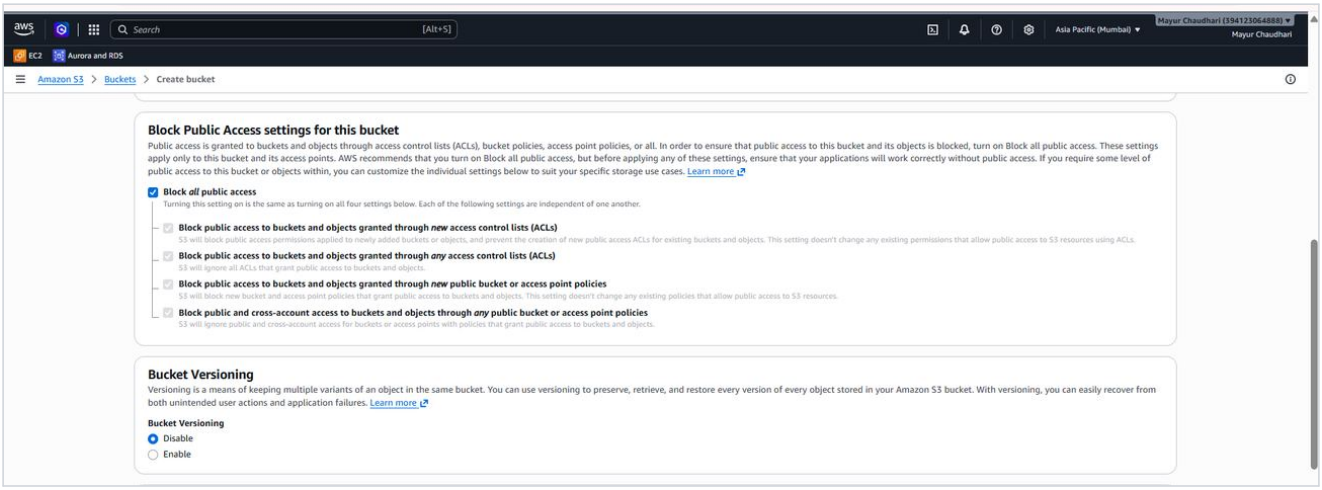
Step 6.1



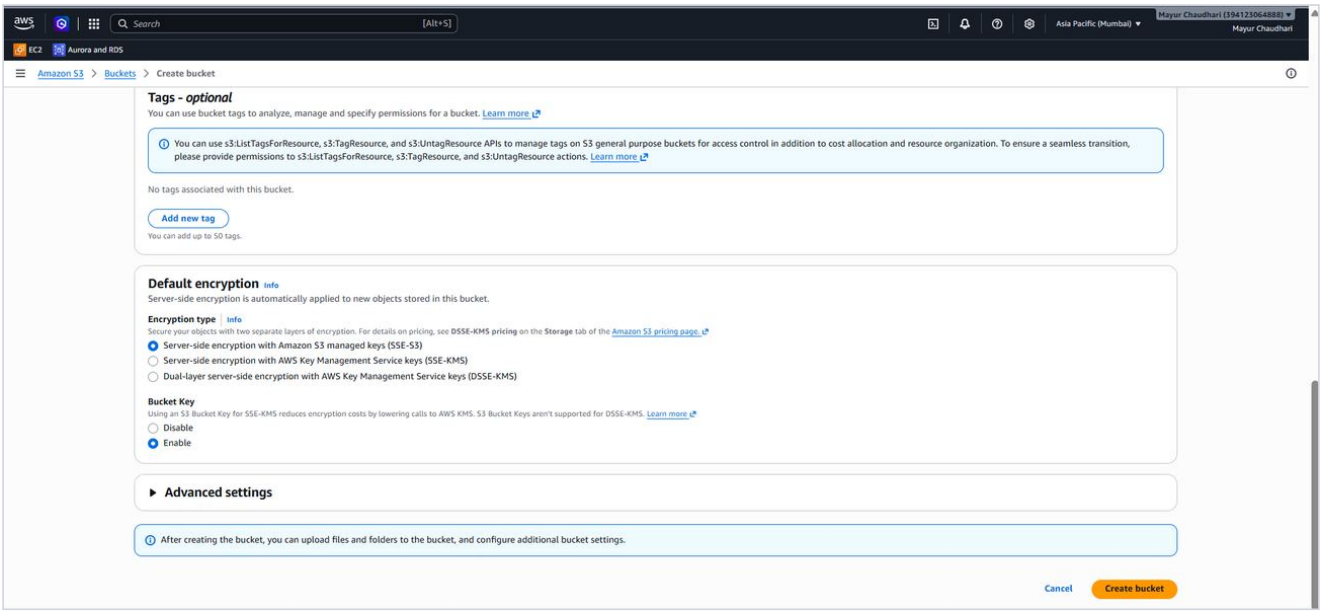
Step 6.2 Create bucket.



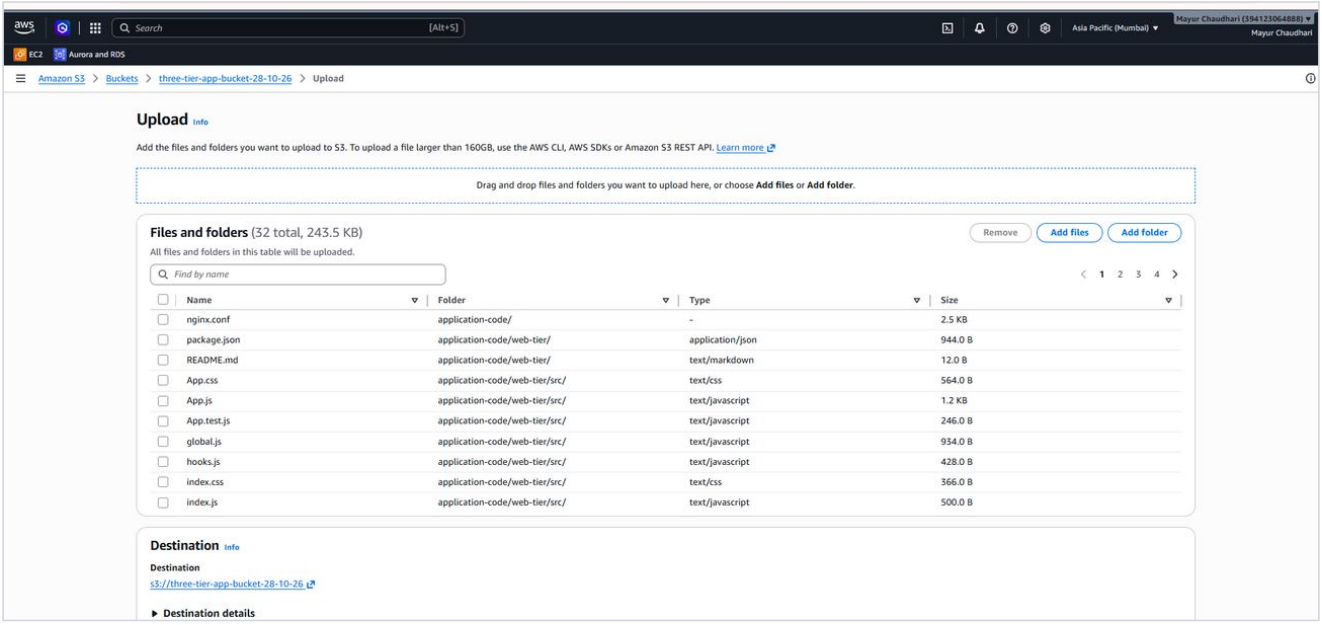
Step 6.3



Step 6.4



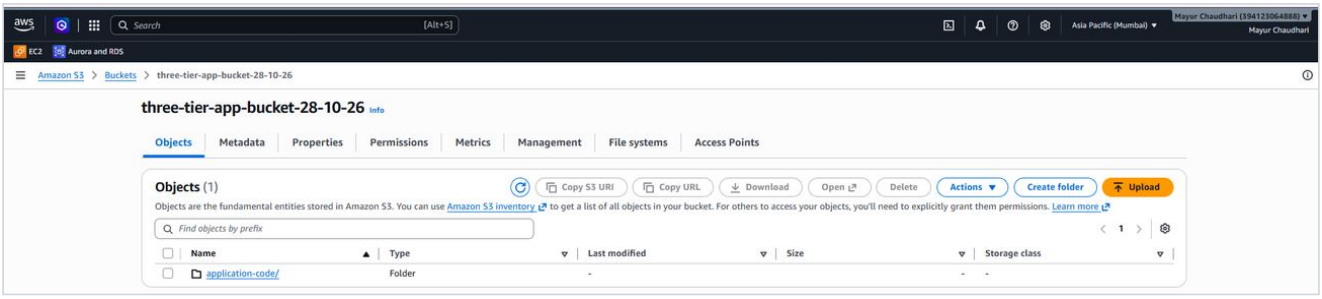
Step 6.5 Upload: add files and folders.



Step 6.6



Step 6.7 Application code uploaded to the bucket.



6. Steps 7-8: Build the App server and save it as an AMI

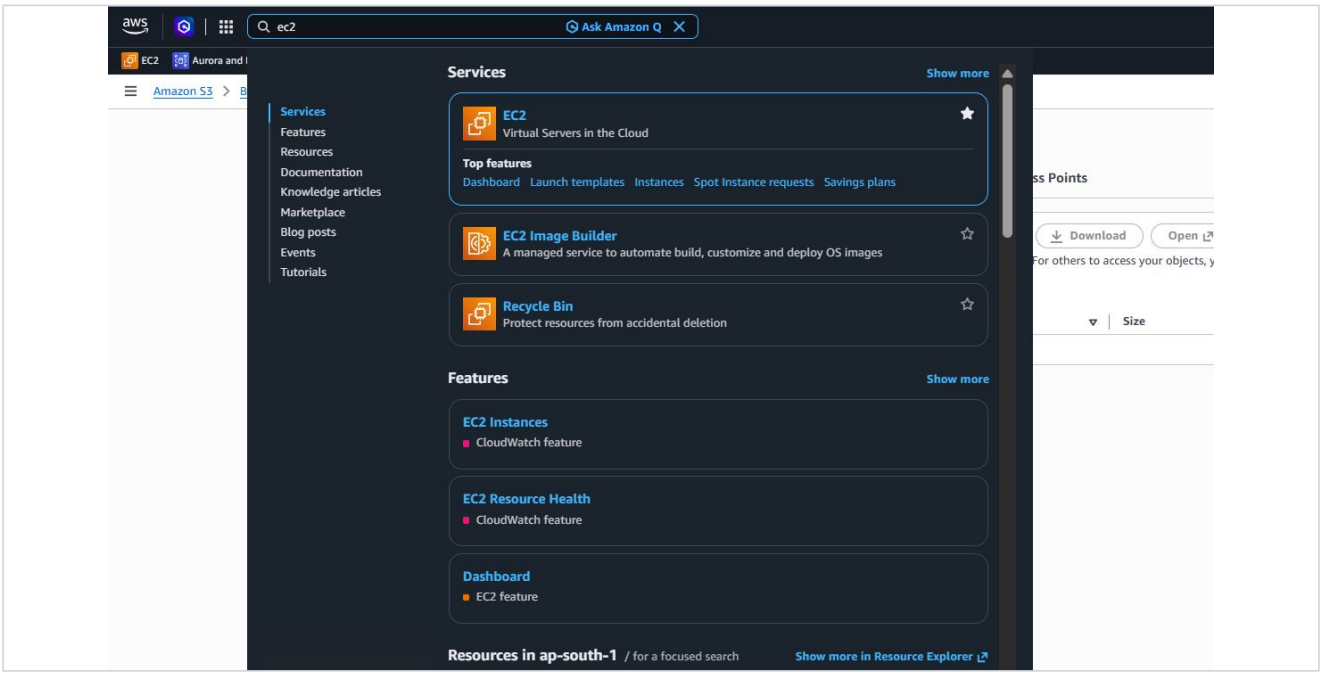
Idea: set up **one** server by hand, take a "photo" (AMI) of it, then let AWS clone that photo as many times as needed.

- Launch **Test-App-Server** (EC2) in App-Tier-Subnet-1-Private, with App-Tier-SG and the IAM role.
- Connect with **Session Manager** (works even though the server is private).
- Install the MySQL client, connect to RDS, create database `webappdb` and table `transactions` (id, amount, description) and insert one sample row.
- Copy the app code from S3, install **Node.js 16** with nvm, run `npm install`.
- Start the app with **PM2** (a process manager that restarts the app if it crashes and starts it on reboot): `pm2 start index.js`, then `pm2 startup` and `pm2 save`.

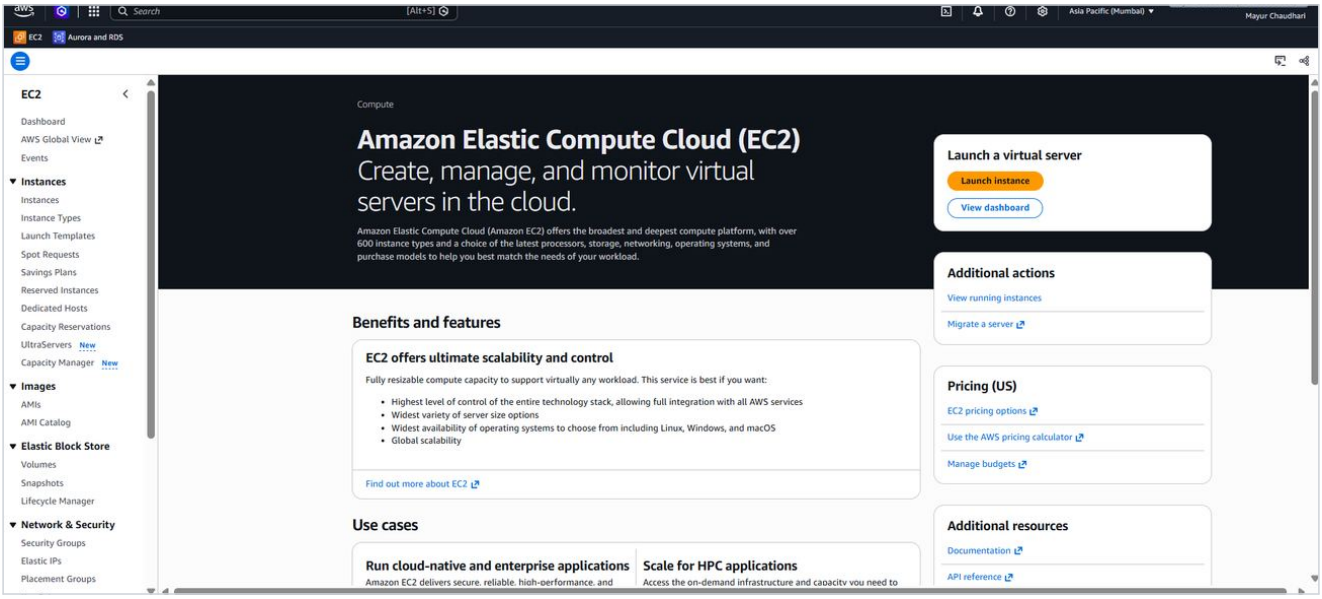
- Test: `curl http://localhost:4000/health` should answer OK.
- Create the AMI: **Actions > Image and templates > Create image**. Name it App-Server-Image.

Step 7 - Test App Server

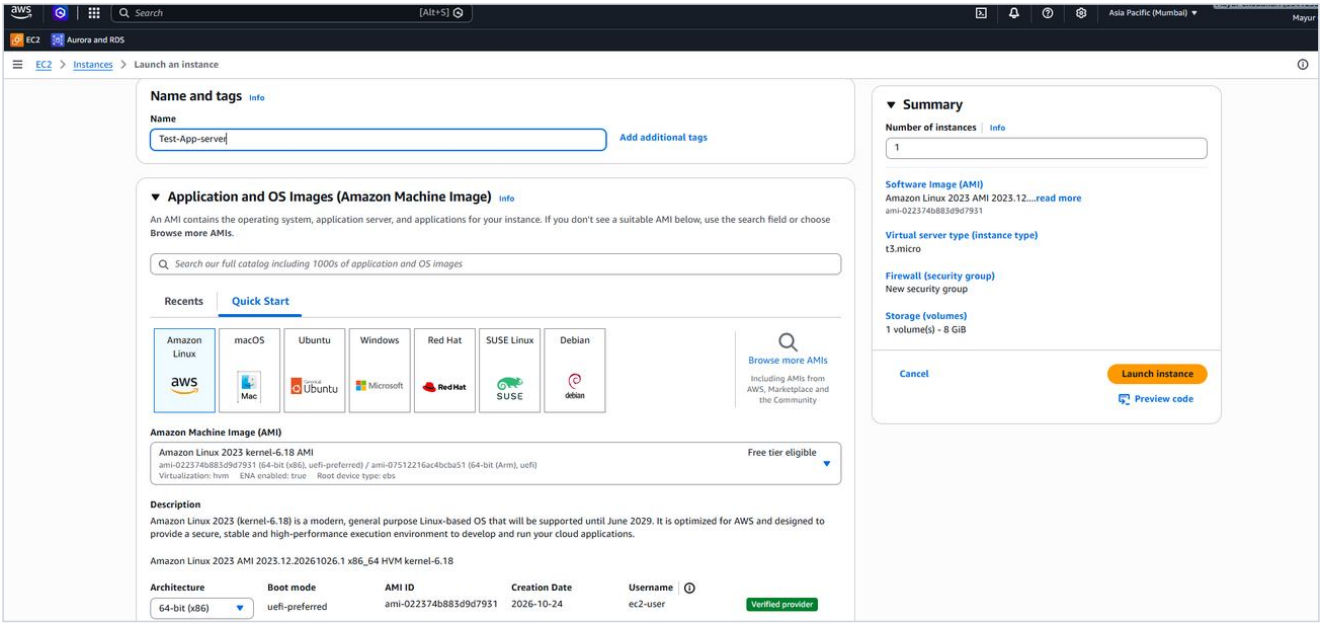
Step 7.1 Search for EC2.



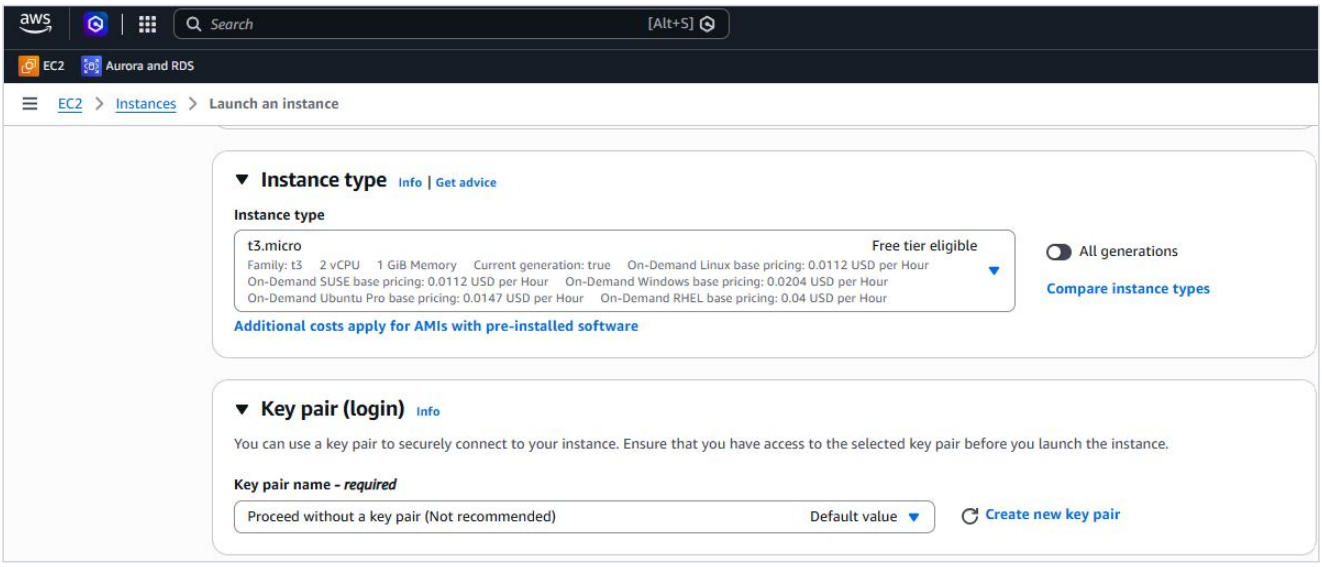
Step 7.2



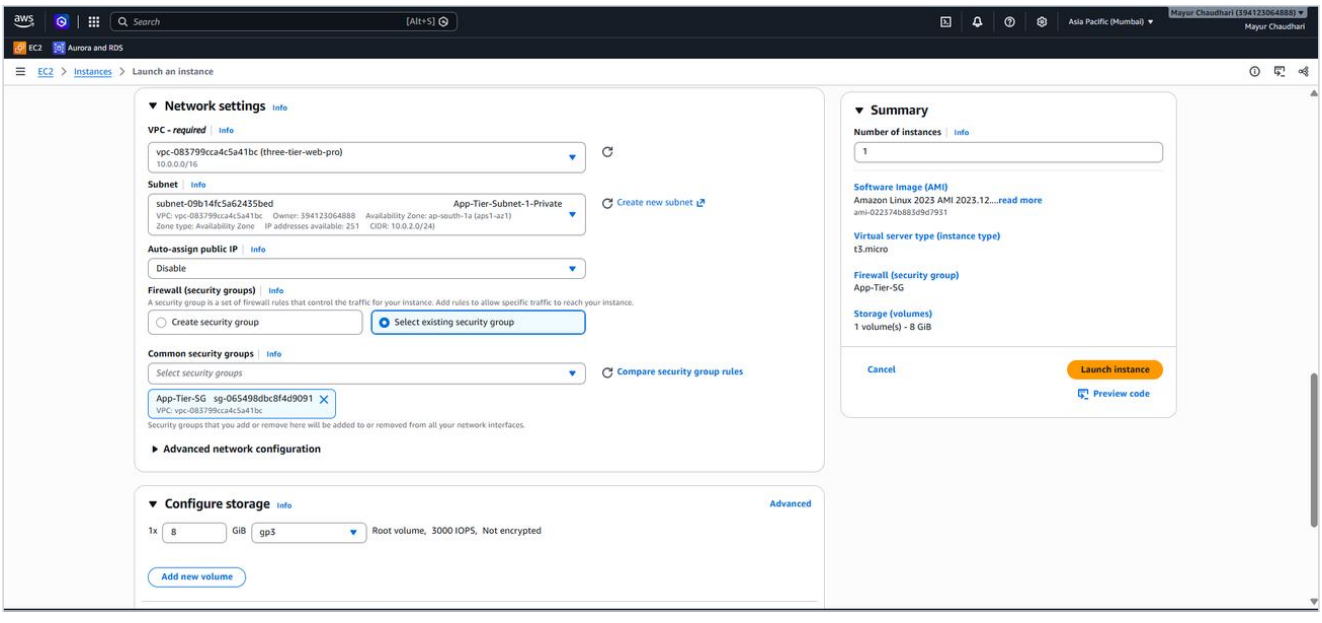
Step 7.3 Launch instance: name and Amazon Linux AMI.



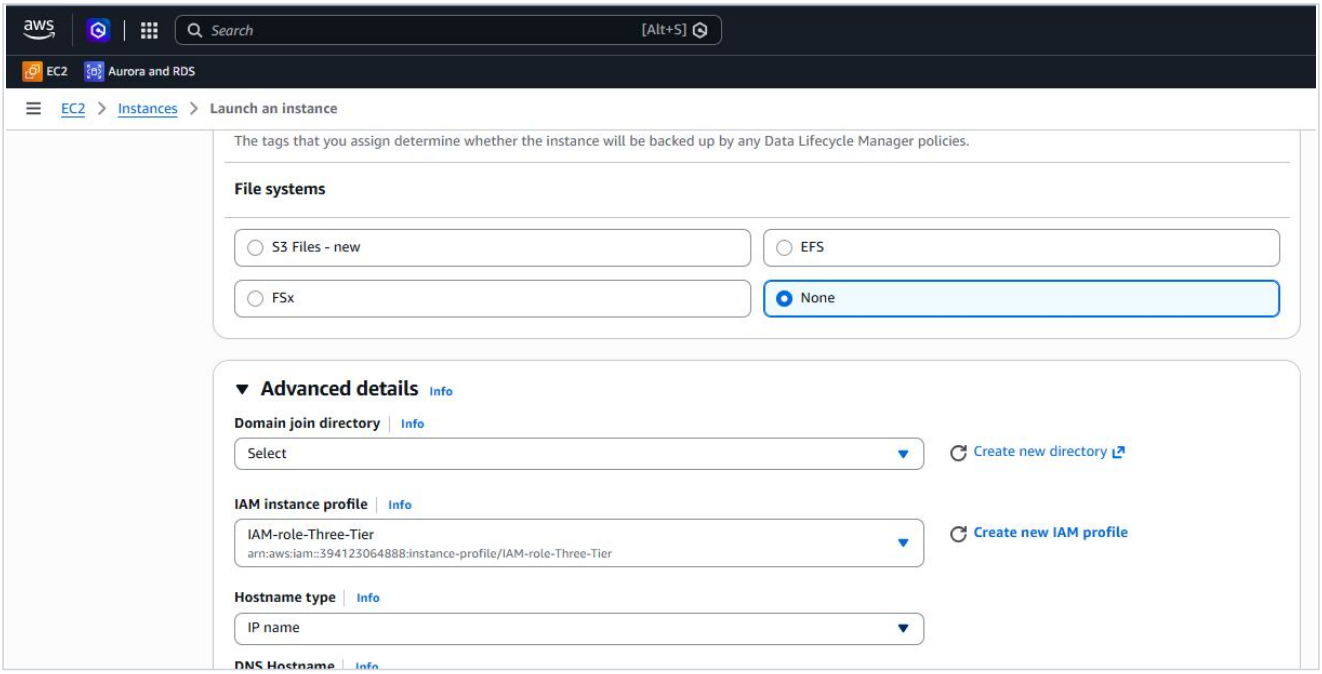
Step 7.4



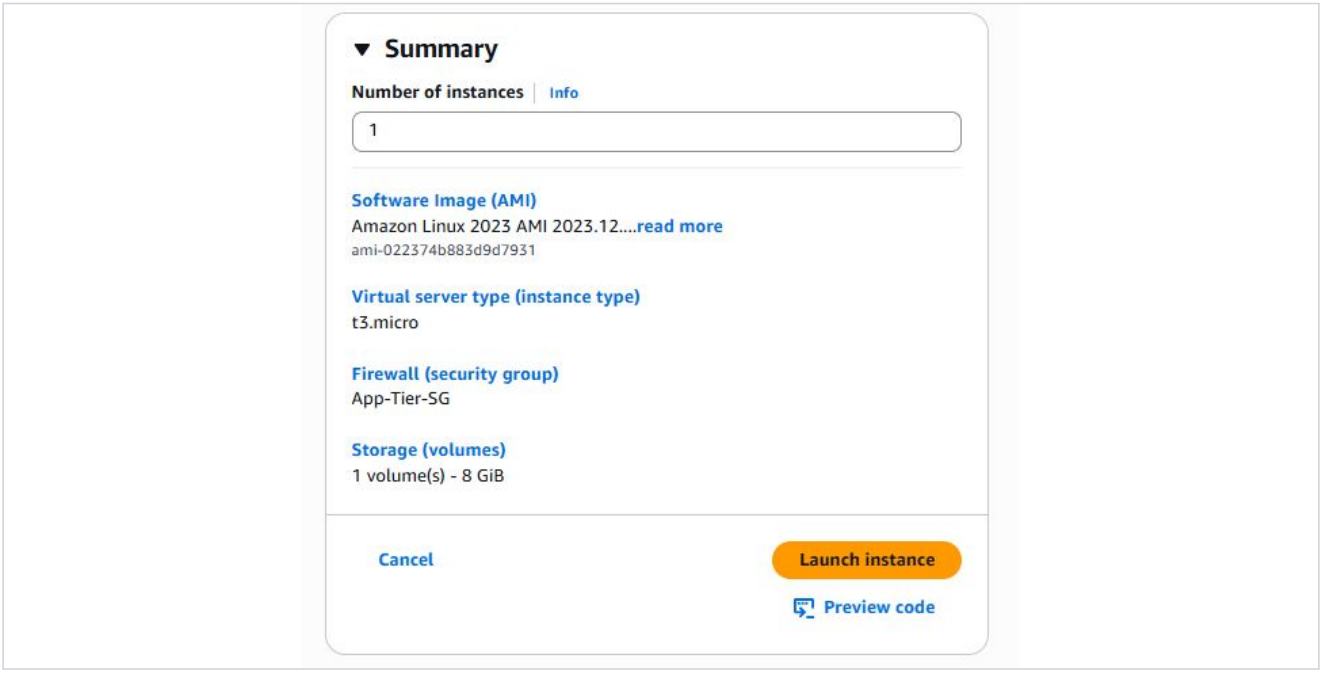
Step 7.5 Network settings: your VPC, private app subnet, App-Tier-SG.



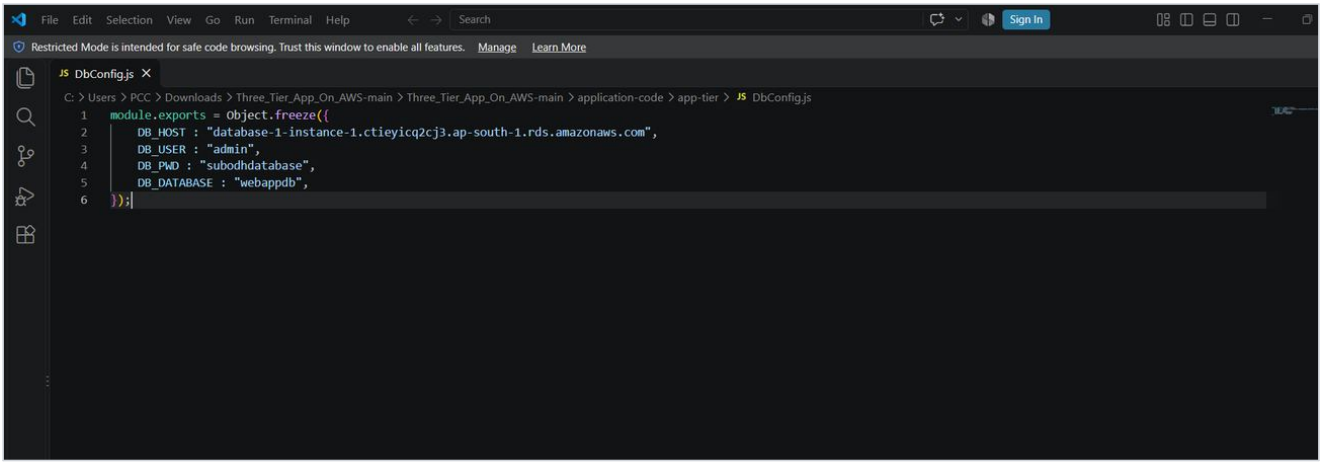
Step 7.6 Advanced details: select the IAM instance profile.



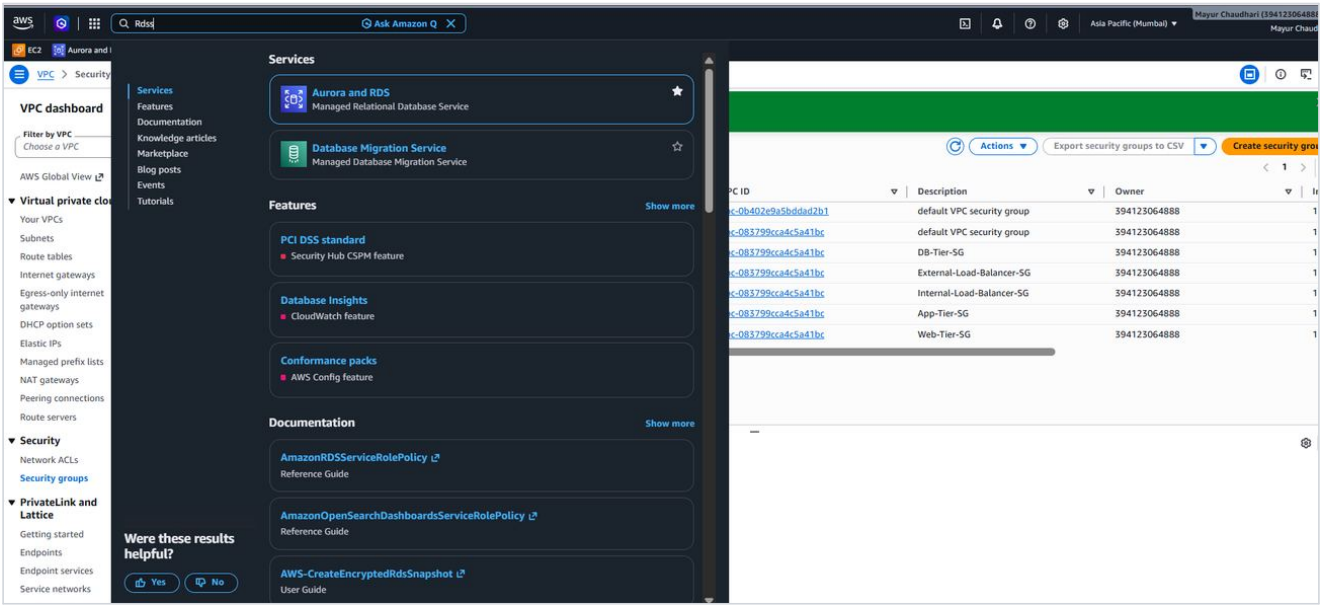
Step 7.7 Launch instance.



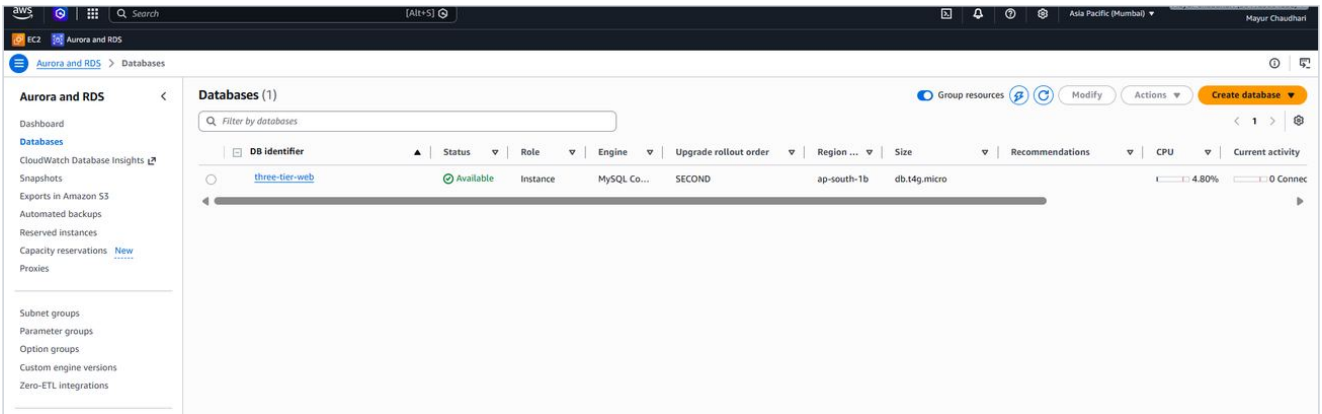
Step 7.8 DbConfig.js edited with your database details.



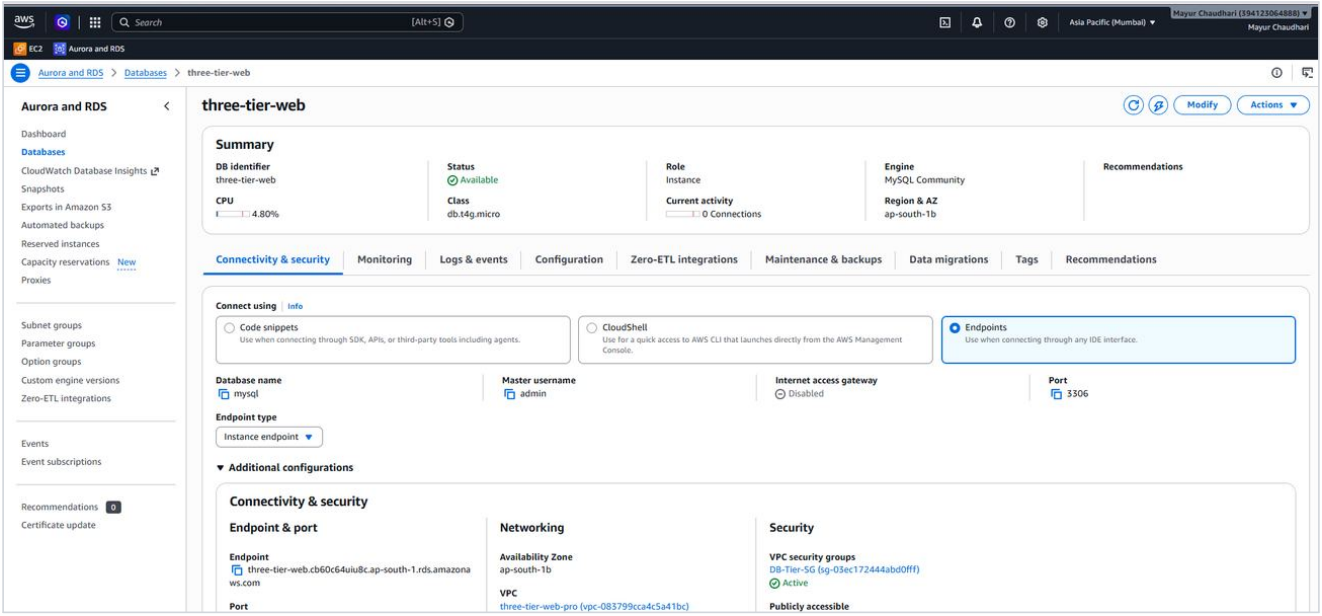
Step 7.9



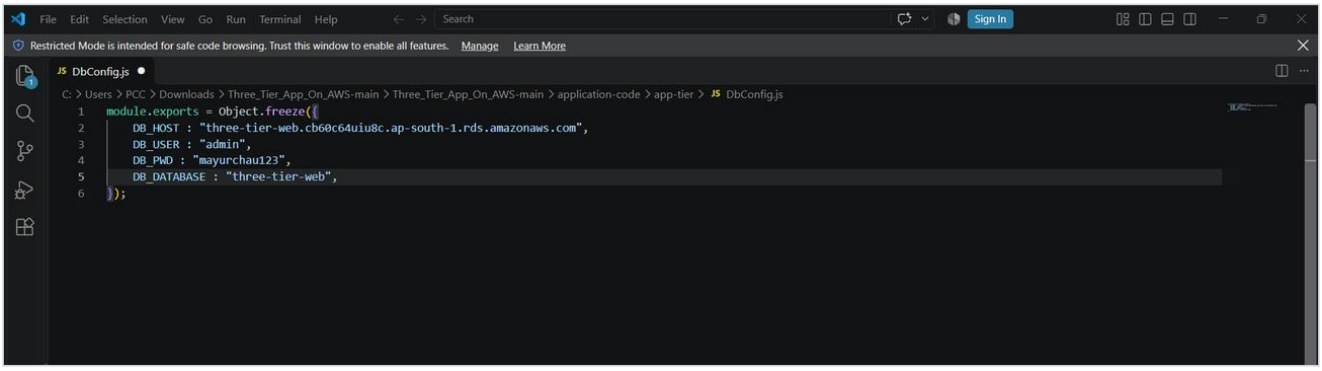
Step 7.10 RDS databases list.



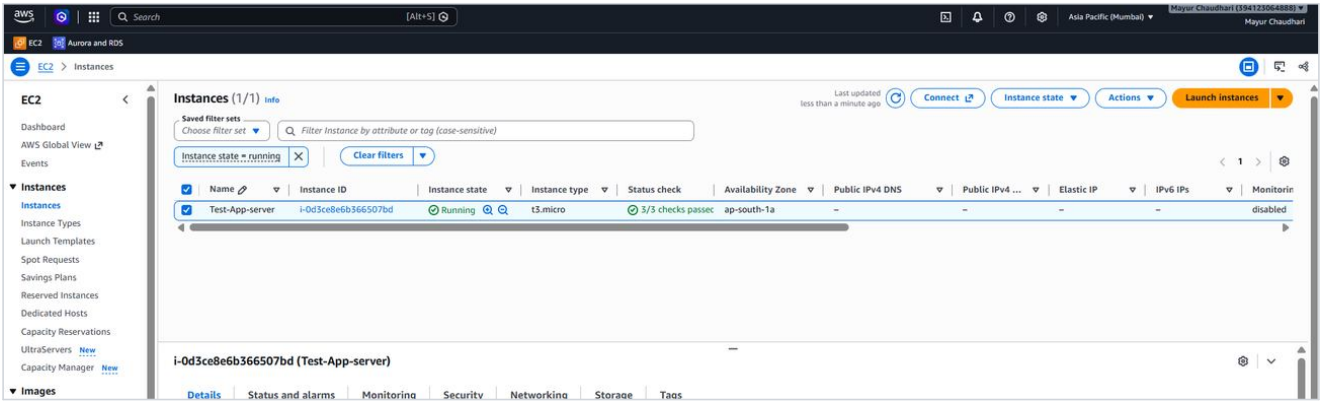
Step 7.11 Copy the writer endpoint (DB_HOST).



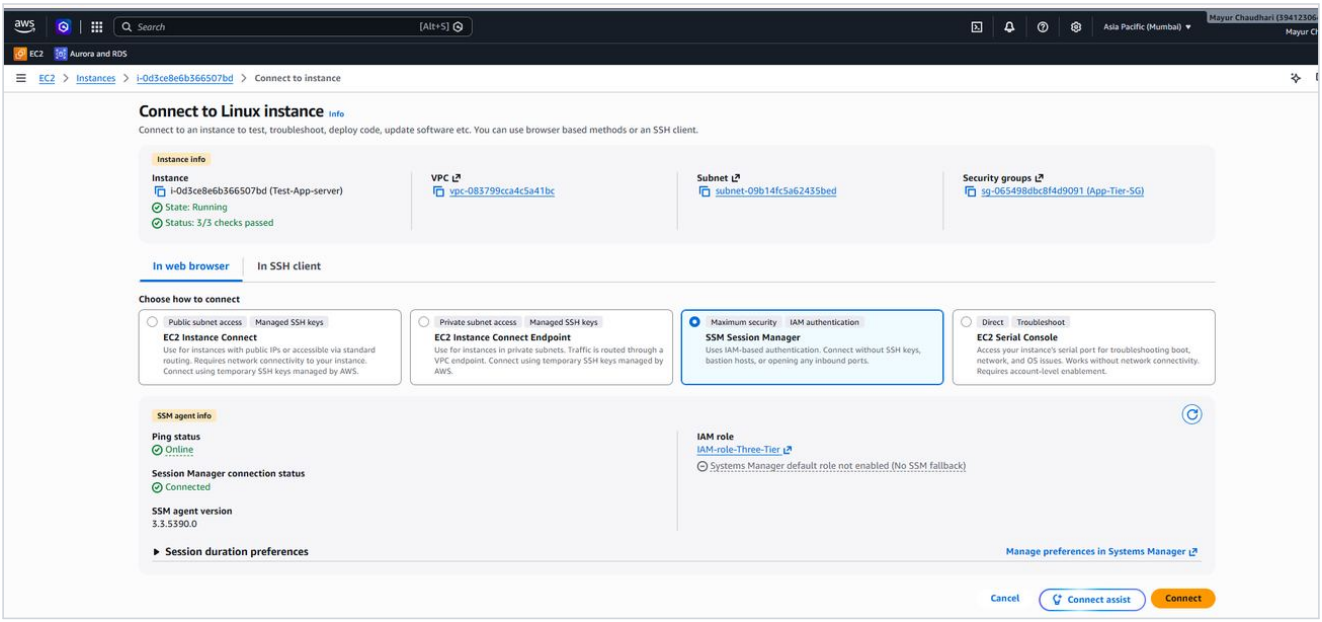
Step 7.12 DbConfig.js ready to upload to S3.



Step 7.13 Test app server running.



Step 7.14 Connect > Session Manager.



Step 7.15 MySQL client installed (mysql --version).

```
[ec2-user@ip-10-0-2-89 bin]$ mysql --version
mysql  Ver 8.0.46 for Linux on x86_64 (MySQL Community Server - GPL)
[ec2-user@ip-10-0-2-89 bin]$
```

Step 7.16 Connect to the database with mysql -h -u admin -p.

```
[ec2-user@ip-10-0-2-89 bin]$ mysql -h three-tier-web.cb60c64uiu8c.ap-south-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 48
Server version: 8.4.11 Source distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

Step 7.17 Create the database and select it (your database is named three-tier-web).

```
mysql> CREATE DATABASE `three-tier-web`;
Query OK, 1 row affected (0.02 sec)

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| three-tier-web     |
+--------------------+
5 rows in set (0.01 sec)

mysql> use three-tier-web;
Database changed
mysql>
```

Step 7.18 Create the transactions table.

```
mysql> CREATE TABLE IF NOT EXISTS transactions (
    ->     id INT NOT NULL AUTO_INCREMENT,
    ->     amount DECIMAL(10,2),
    ->     description VARCHAR(100),
    ->     PRIMARY KEY (id)
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> show tables;
+--------------------------+
| Tables_in_three-tier-web |
+--------------------------+
| transactions             |
+--------------------------+
1 row in set (0.05 sec)

mysql>
```

Step 7.19 Insert a sample row and select it.

```
mysql> INSERT INTO transactions (amount, description)
    -> VALUES (90000, 'Laptop');
Query OK, 1 row affected (0.18 sec)

mysql> SELECT * FROM transactions;
+----+----------+-------------+
| id | amount   | description |
+----+----------+-------------+
|  1 | 90000.00 | Laptop      |
+----+----------+-------------+
1 row in set (0.00 sec)
```

Step 7.20 Copy the app-tier code from S3 and set permissions.

```
[ec2-user@ip-10-0-2-89 bin]$ cd /home/ec2-user
[ec2-user@ip-10-0-2-89 ~]$ sudo aws s3 cp s3://three-tier-app-bucket-28-10-26/application-code/app-tier/ ./app-tier/ --recursive
download: s3://three-tier-app-bucket-28-10-26/application-code/app-tier/package.json to app-tier/package.json
download: s3://three-tier-app-bucket-28-10-26/application-code/app-tier/DbConfig.js to app-tier/DbConfig.js
download: s3://three-tier-app-bucket-28-10-26/application-code/app-tier/README.md to app-tier/README.md
download: s3://three-tier-app-bucket-28-10-26/application-code/app-tier/TransactionService.js to app-tier/TransactionService.js
download: s3://three-tier-app-bucket-28-10-26/application-code/app-tier/index.js to app-tier/index.js
download: s3://three-tier-app-bucket-28-10-26/application-code/app-tier/package-lock.json to app-tier/package-lock.json
[ec2-user@ip-10-0-2-89 ~]$ cd app-tier
[ec2-user@ip-10-0-2-89 app-tier]$ sudo chown -R ec2-user:ec2-user /home/ec2-user/app-tier
[ec2-user@ip-10-0-2-89 app-tier]$ sudo chmod-R 755 /home/ec2-user/app-tier
sudo: chmod-R: command not found
[ec2-user@ip-10-0-2-89 app-tier]$ sudo chmod -R 755 /home/ec2-user/app-tier
[ec2-user@ip-10-0-2-89 app-tier]$
```

Step 7.21 Install PM2 and start index.js.

```
[ec2-user@ip-10-0-2-89 app-tier]$ pm2 start index.js

                        -------------

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```

Step 7.22 PM2 shows the app online.

```
[PM2] Spawning PM2 daemon with pm2_home=/home/ec2-user/.pm2
[PM2] PM2 Successfully daemonized
[PM2] Starting /home/ec2-user/app-tier/index.js in fork_mode (1 instance)
[PM2] Done.
```

id	name	namespace	version	mode	pid	uptime	↺	status	cpu	mem	user	watching
0	index	default	1.0.0	fork	28501	0s	0	online	0%	28.9mb	ec2-user	disabled

```
[ec2-user@ip-10-0-2-89 app-tier]$
```

Step 7.23 PM2 logs: backend is listening on port 4000.

```
[ec2-user@ip-10-0-2-89 app-tier]$ pm2 logs
[TAILING] Tailing last 15 lines for [all] processes (change the value with --lines option)
/home/ec2-user/.pm2/pm2.log last 15 lines:
PM2        | 2026-10-28T09:38:18: PM2 log: Node.js version      : 16.20.2
PM2        | 2026-10-28T09:38:18: PM2 log: Current arch         : x64
PM2        | 2026-10-28T09:38:18: PM2 log: PM2 home             : /home/ec2-user/.pm2
PM2        | 2026-10-28T09:38:18: PM2 log: PM2 PID file         : /home/ec2-user/.pm2/pm2.pid
PM2        | 2026-10-28T09:38:18: PM2 log: RPC socket file      : /home/ec2-user/.pm2/rpc.sock
PM2        | 2026-10-28T09:38:18: PM2 log: BUS socket file      : /home/ec2-user/.pm2/pub.sock
PM2        | 2026-10-28T09:38:18: PM2 log: Application log path : /home/ec2-user/.pm2/logs
PM2        | 2026-10-28T09:38:18: PM2 log: Worker Interval      : 30000
PM2        | 2026-10-28T09:38:18: PM2 log: Process dump file    : /home/ec2-user/.pm2/dump.pm2
PM2        | 2026-10-28T09:38:18: PM2 log: Concurrent actions   : 2
PM2        | 2026-10-28T09:38:18: PM2 log: SIGTERM timeout      : 1600
PM2        | 2026-10-28T09:38:18: PM2 log: Runtime Binary       : /home/ec2-user/.nvm/versions/node/v16.20.2/bin/node
PM2        | 2026-10-28T09:38:18: PM2 log: ================================================================================
PM2        | 2026-10-28T09:38:18: PM2 log: App [index:0] starting in -fork mode-
PM2        | 2026-10-28T09:38:18: PM2 log: App [index:0] online

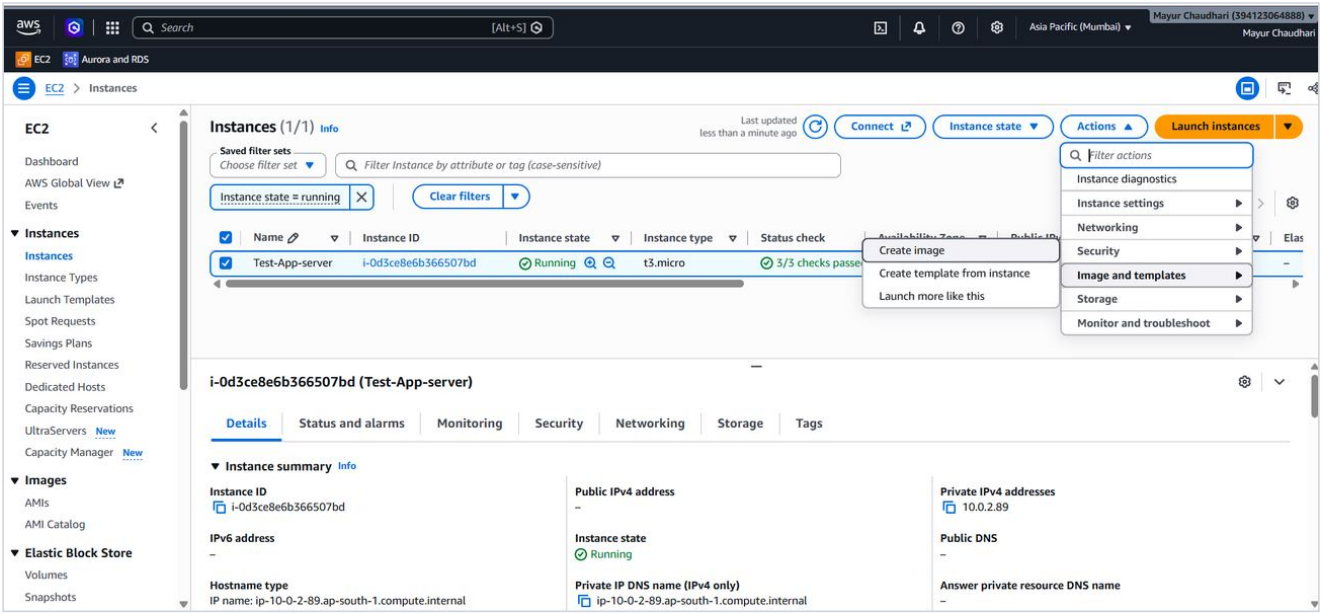
/home/ec2-user/.pm2/logs/index-error.log last 15 lines:
/home/ec2-user/.pm2/logs/index-out.log last 15 lines:
0|index    | AB3 backend app listening at http://localhost:4000
```

Step 7.24 pm2 save, then curl localhost:4000/health answers OK.

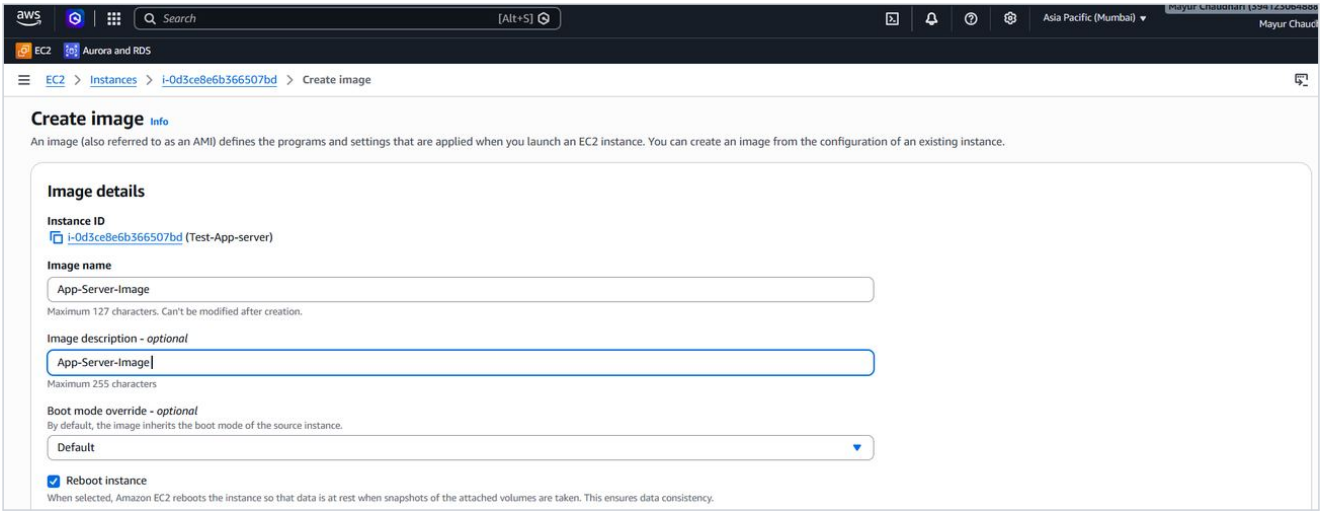
```
[PM2] Remove init script via:
$ pm2 unstartup systemd
[ec2-user@ip-10-0-2-89 app-tier]$ pm2 save
[PM2] Saving current process list...
[PM2] Successfully saved in /home/ec2-user/.pm2/dump.pm2
[ec2-user@ip-10-0-2-89 app-tier]$ curl http://localhost:4000/health
"This is the health check"[ec2-user@ip-10-0-2-89 app-tier]$
```

Step 8 - Create the App-Server AMI

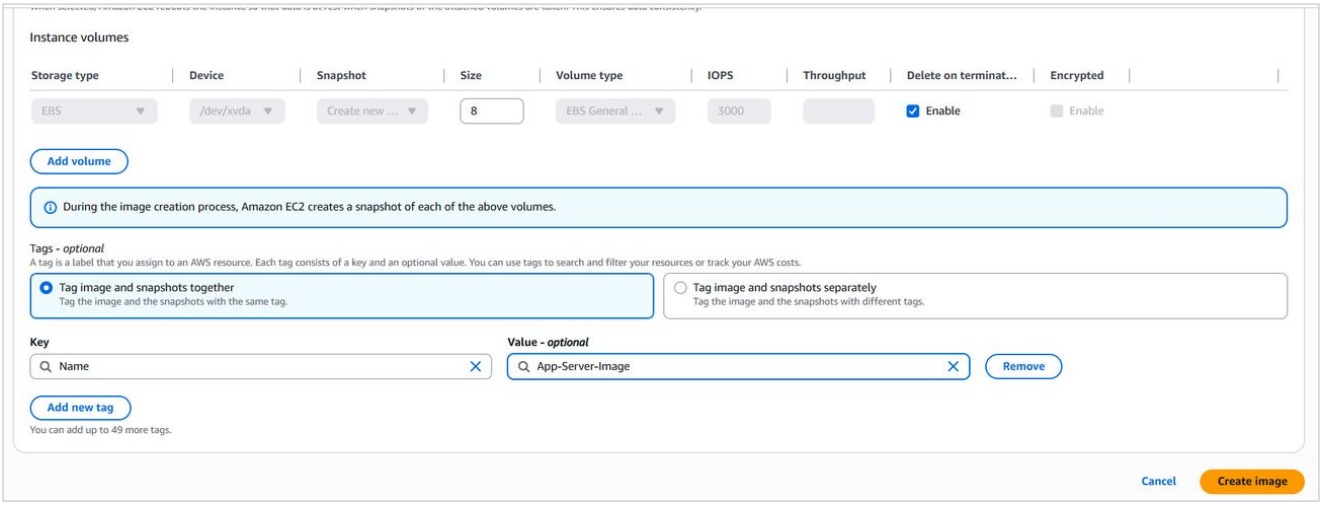
Step 8.1 Instance > Actions > Image and templates > Create image.



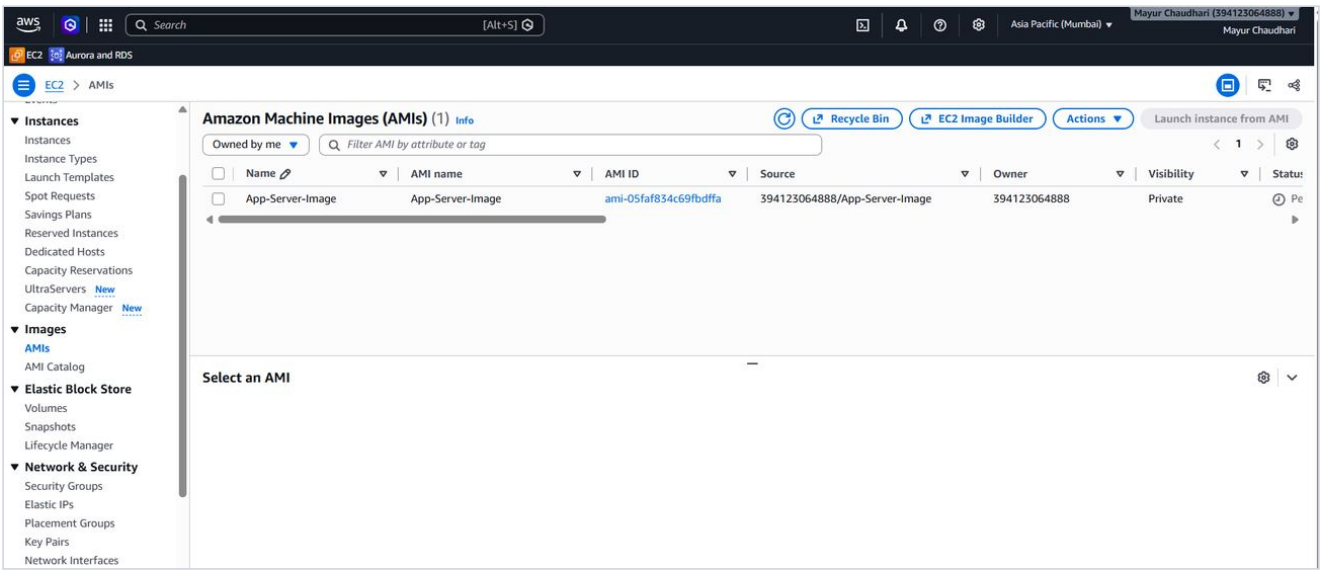
Step 8.2 Create image form.



Step 8.3 Image tags.

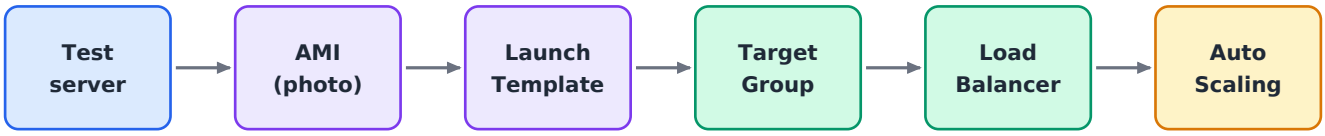


Step 8.4 App server AMI created.



7. Steps 9-13: Launch template, load balancer, auto scaling

The same five-piece chain is repeated for the app tier and then the web tier:



Piece	App tier	Web tier
Target Group	TCP/HTTP port **4000**, health check path `/health`	HTTP port **80**
Launch Template	App-Server-Image AMI + App-Tier-SG + IAM role	Web-Server-Image AMI + Web-Tier-SG + IAM role
Load Balancer	**Internal** (private) ALB in app subnets, Internal-LB-SG, listener 80 to app target group	**Internet-facing** ALB in public subnets, External-LB-SG, listener 80 to web target group
Auto Scaling Group	Subnets: App 1 + App 2, attach to app target group, desired = 2	Subnets: Web 1 + Web 2, attach to web target group, desired = 2

The web server in one paragraph

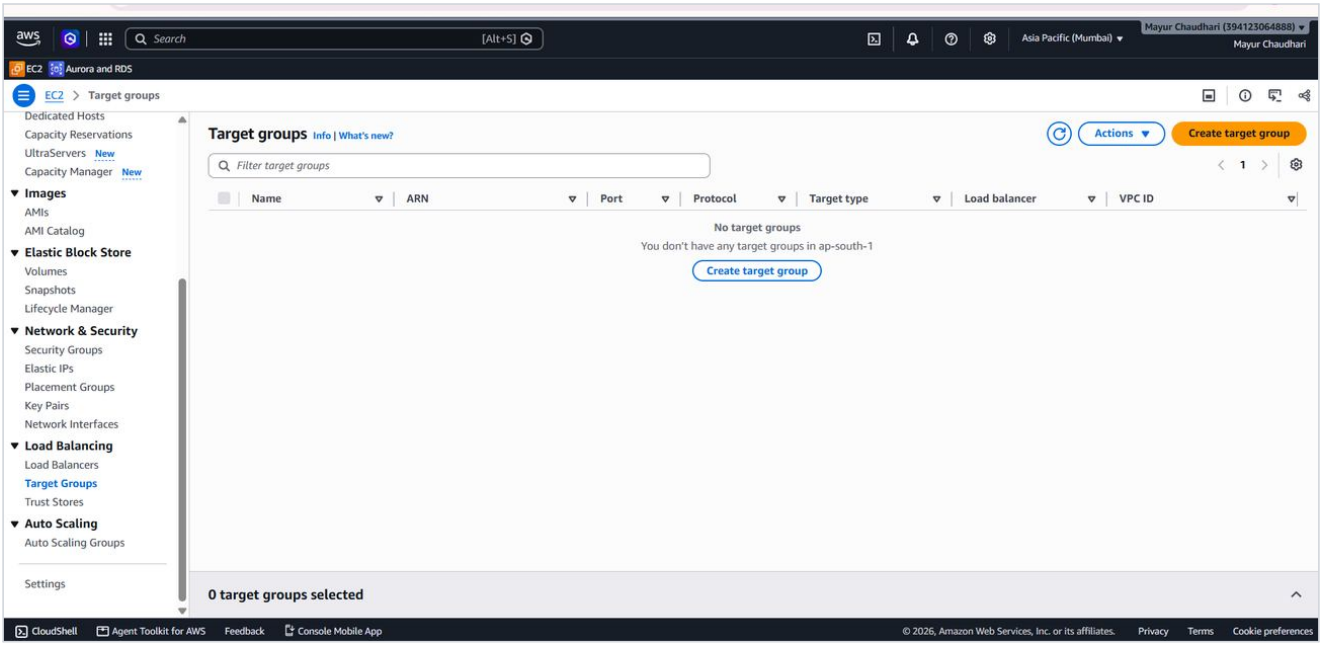
The web server runs **Nginx**. It serves the React frontend (built with `npm run build`) and, for any request starting with `/api`, it **forwards (reverse-proxies)** to the Internal Load Balancer. That is why the internal LB's DNS name is pasted into `nginx.conf`. The web tier never touches the database directly.

Build order matters

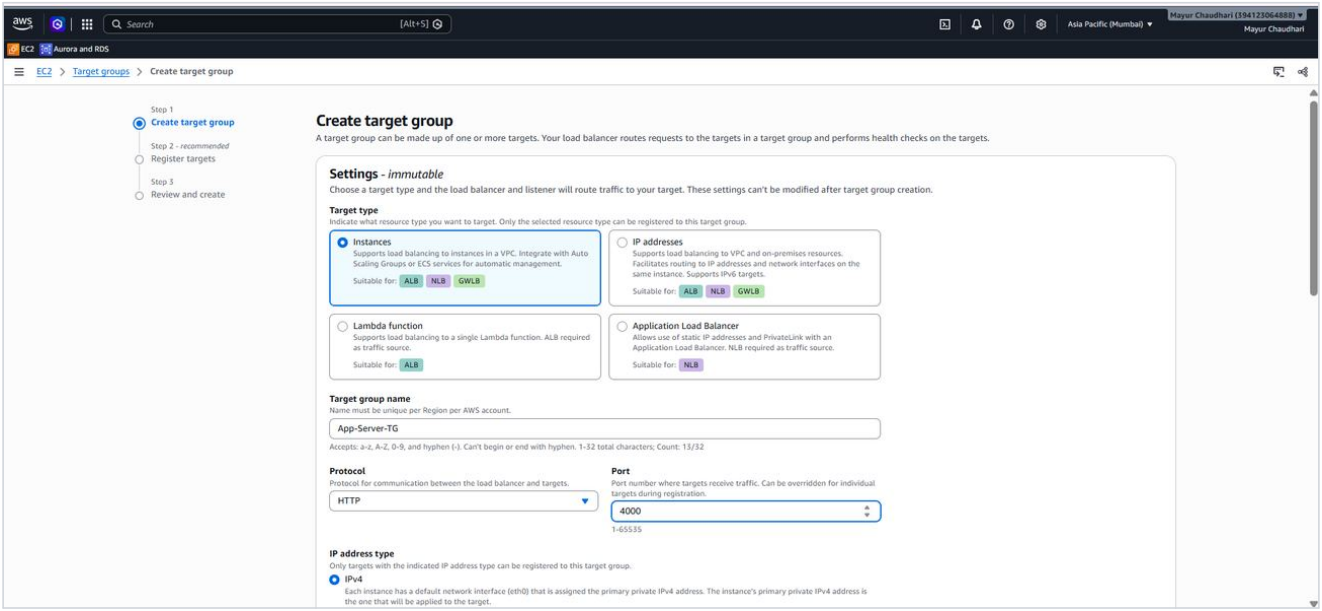
- Do the **app tier first**: you need the internal LB's DNS before you can finish the web server's nginx.conf.
- After both ASGs create their servers you will see **4 servers** (2 web + 2 app). Delete the two *Test* servers; they were only for making the AMIs.

Step 9 - App tier: target group, launch template, internal load balancer

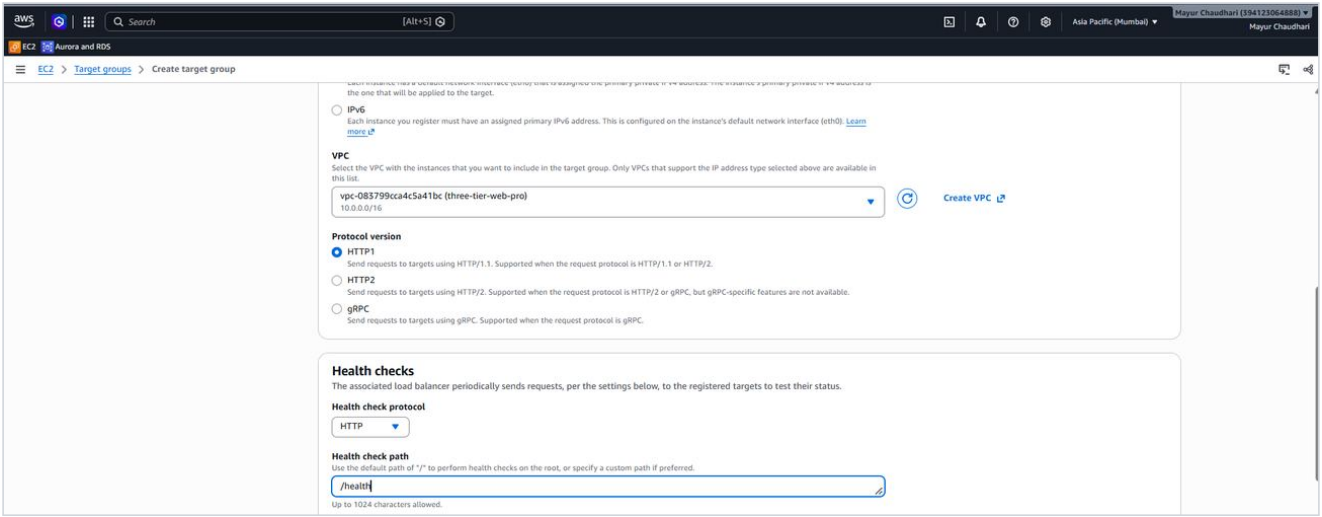
Step 9.1 Target groups page.



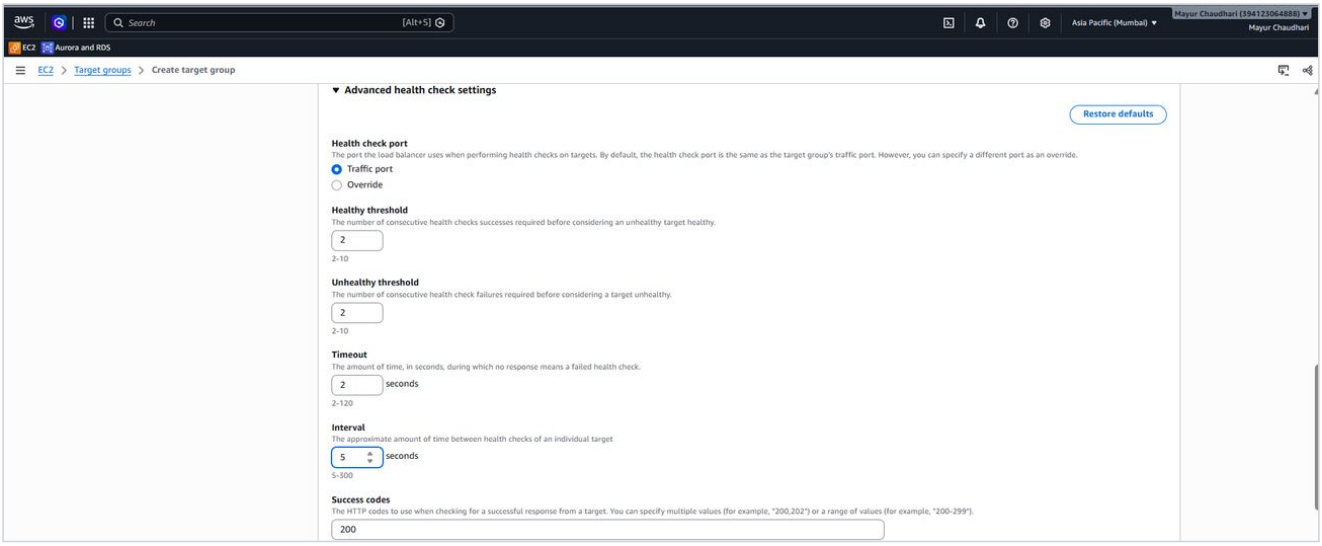
Step 9.2 Create target group: instances, port 4000.



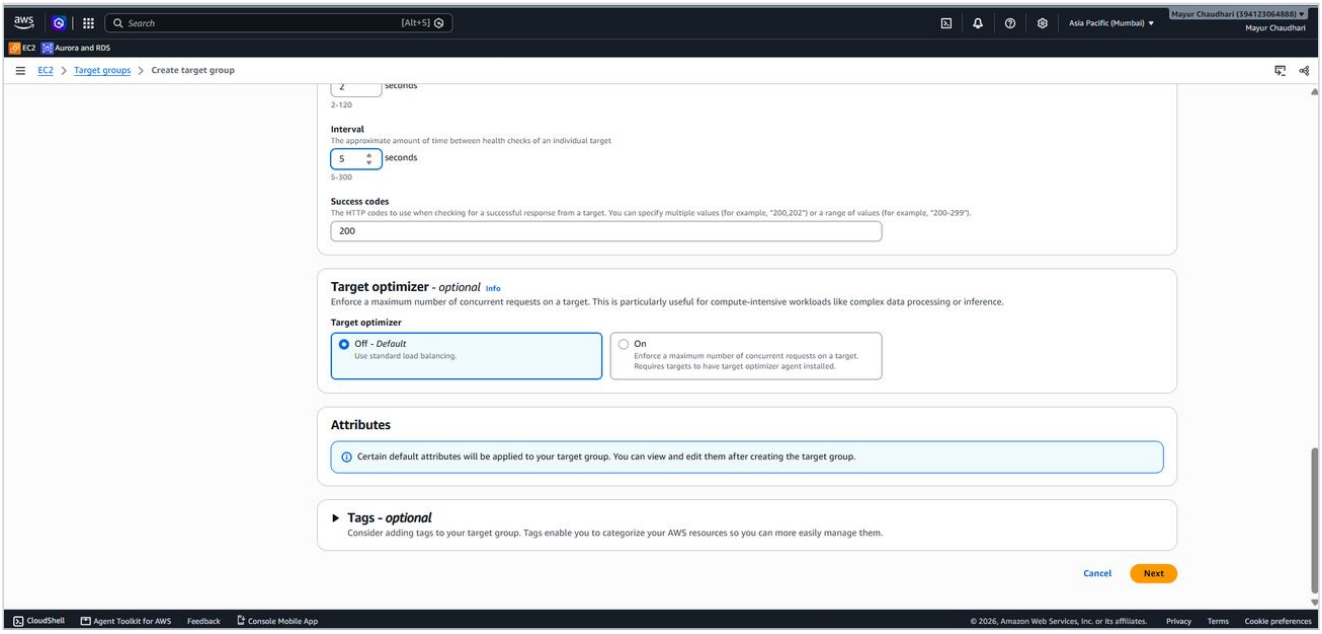
Step 9.3 Target group health check settings.



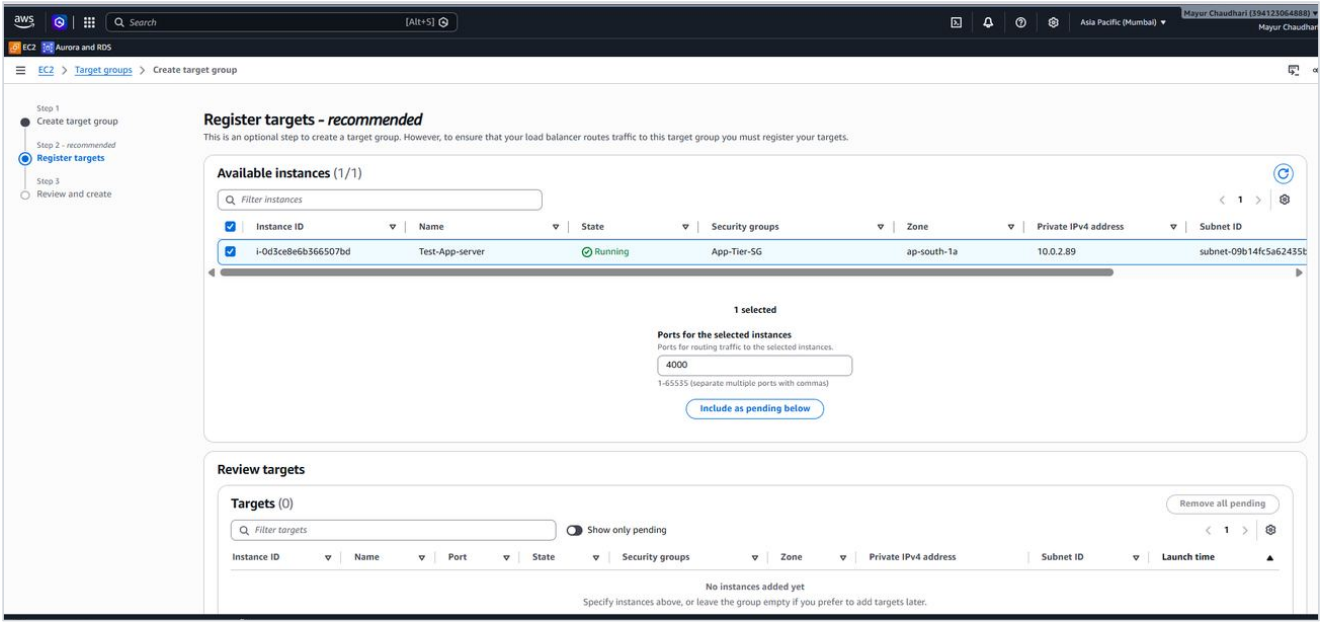
Step 9.4



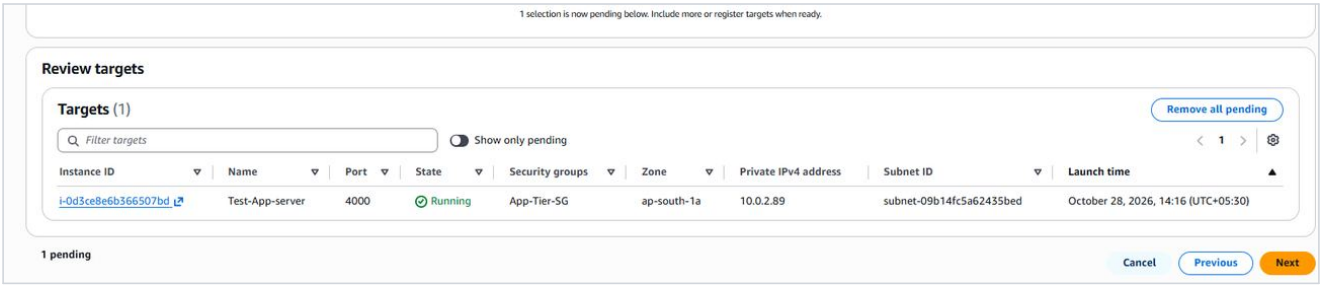
Step 9.5



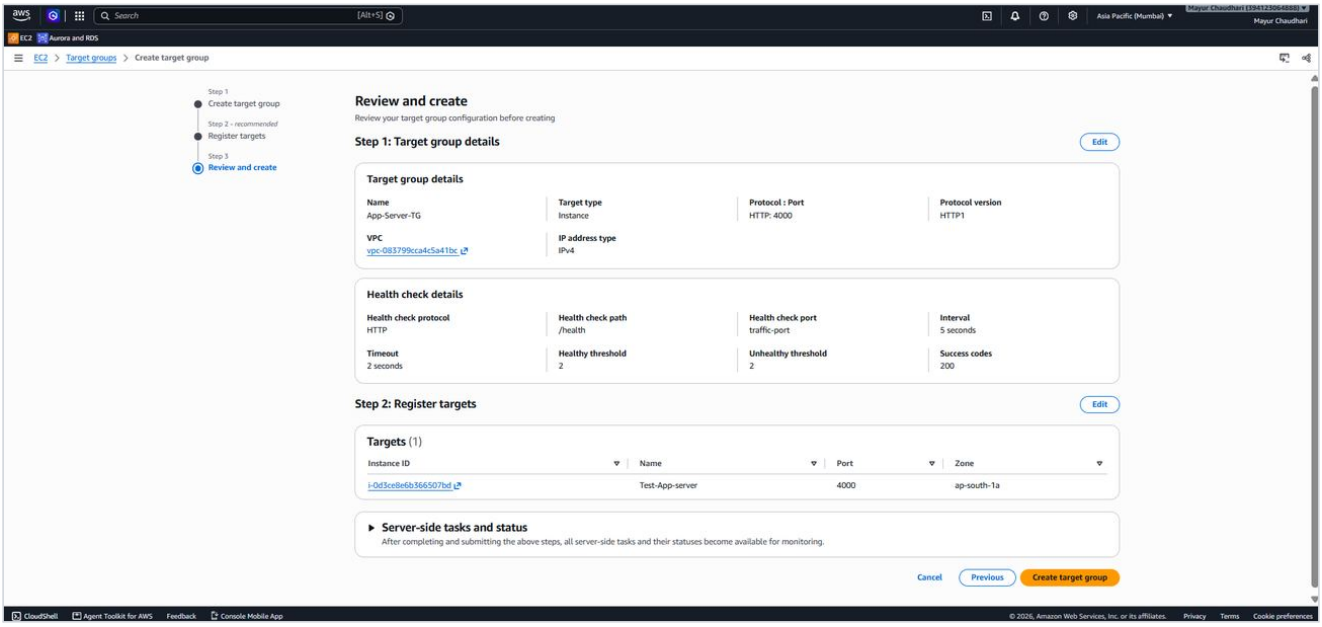
Step 9.6



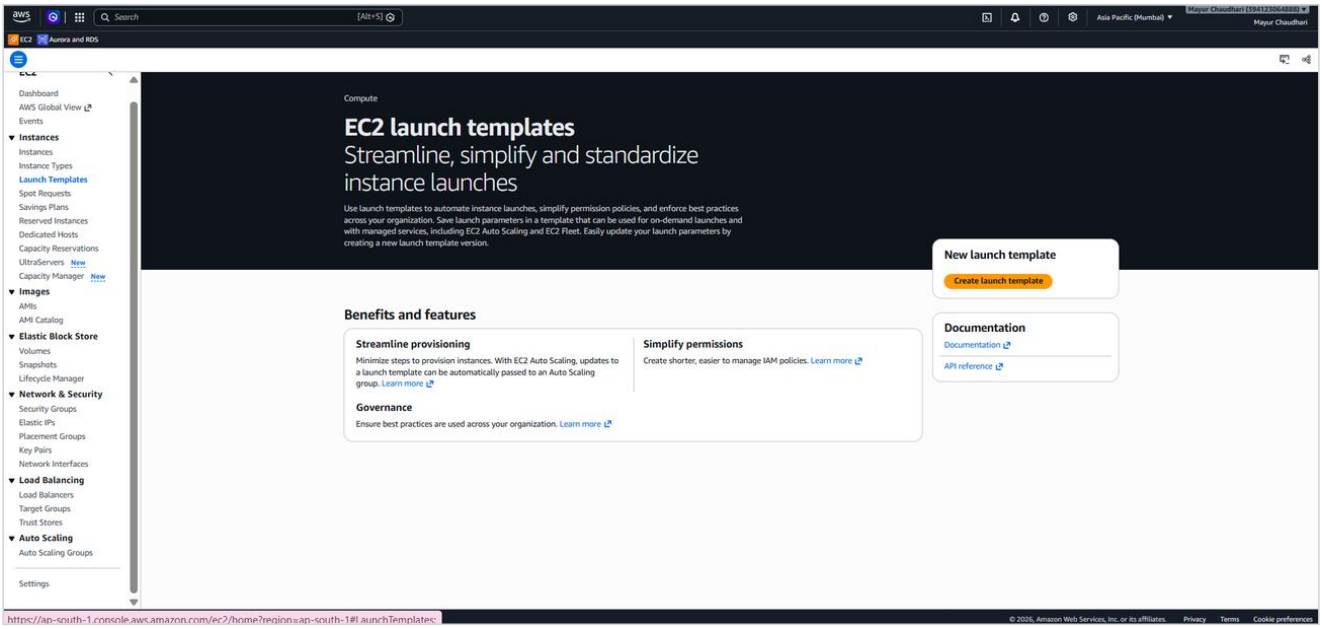
Step 9.7



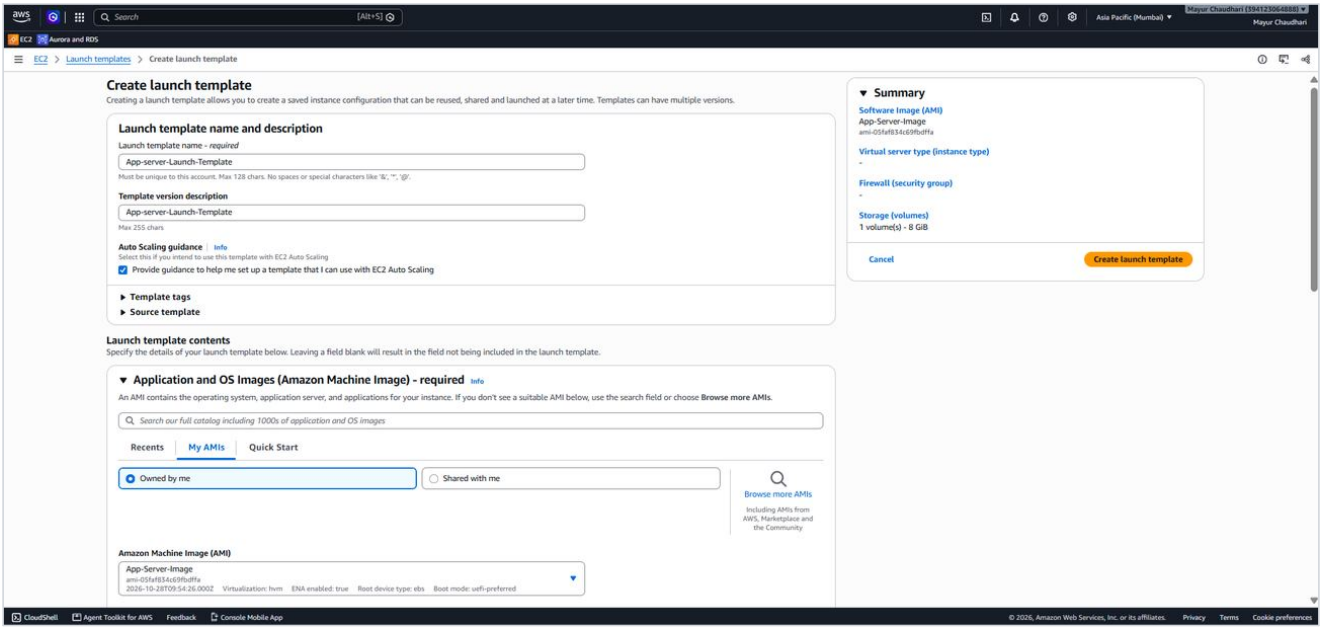
Step 9.8



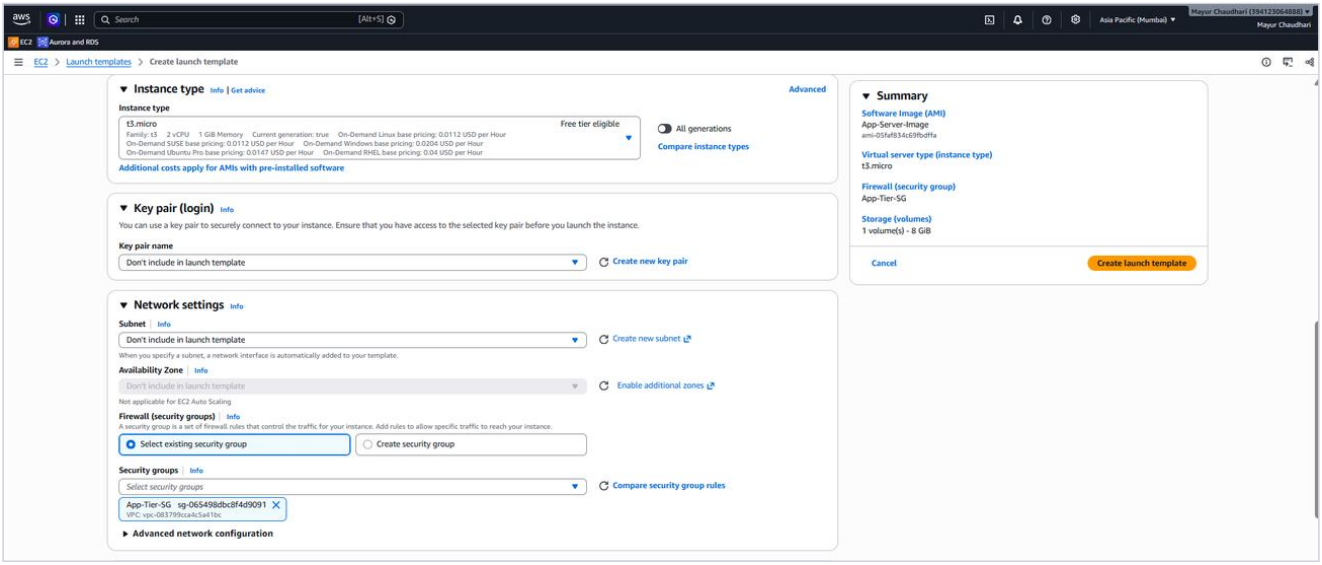
Step 9.9 Launch templates page.



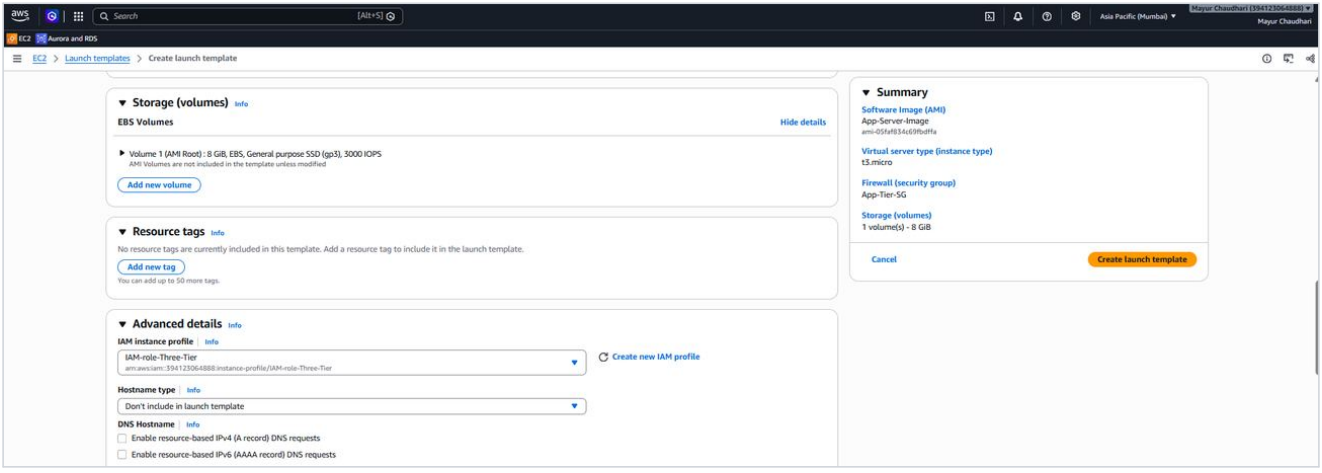
Step 9.10 Create launch template: choose the app AMI.



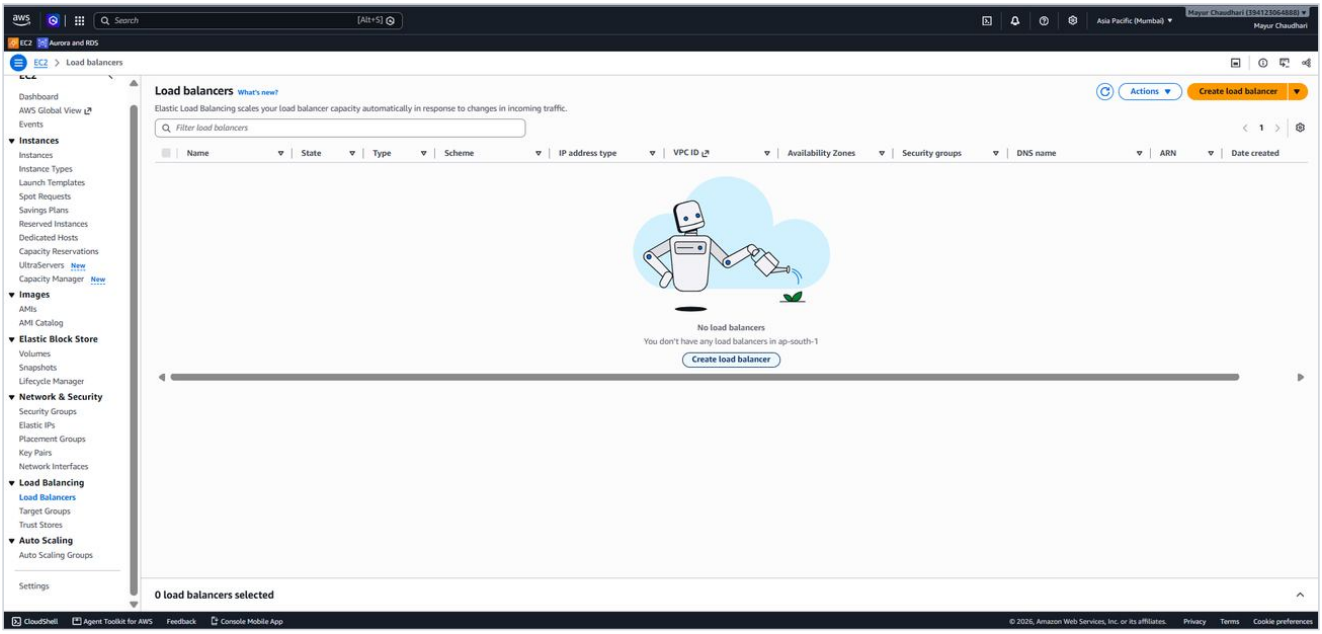
Step 9.11



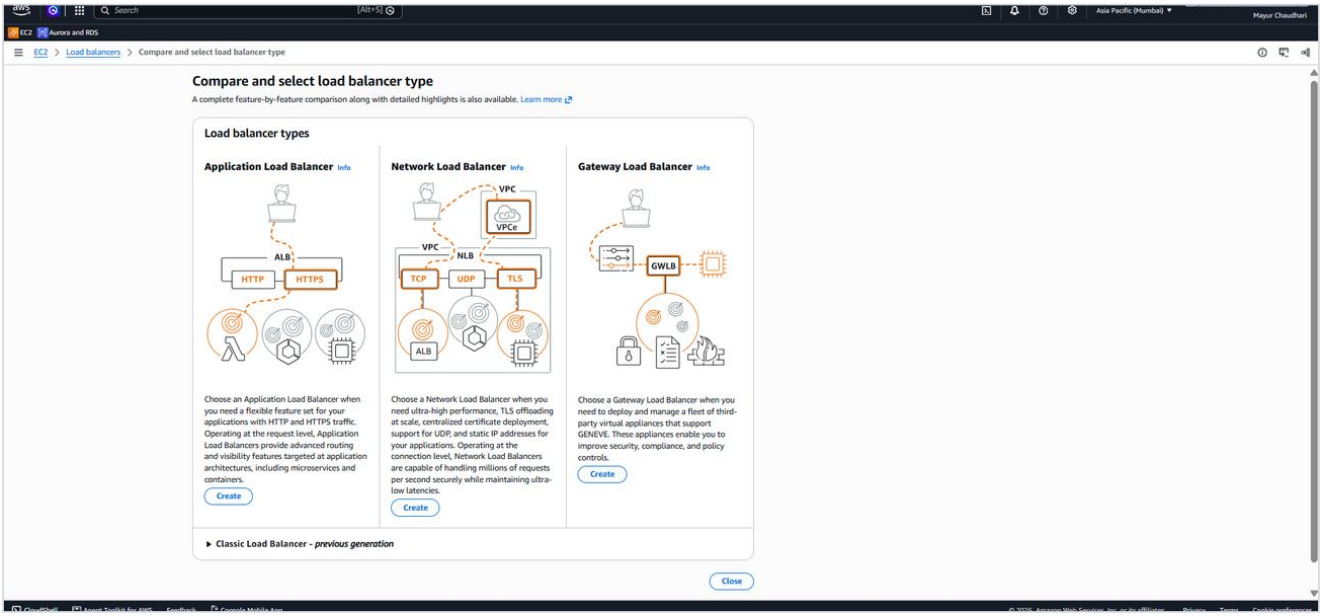
Step 9.12



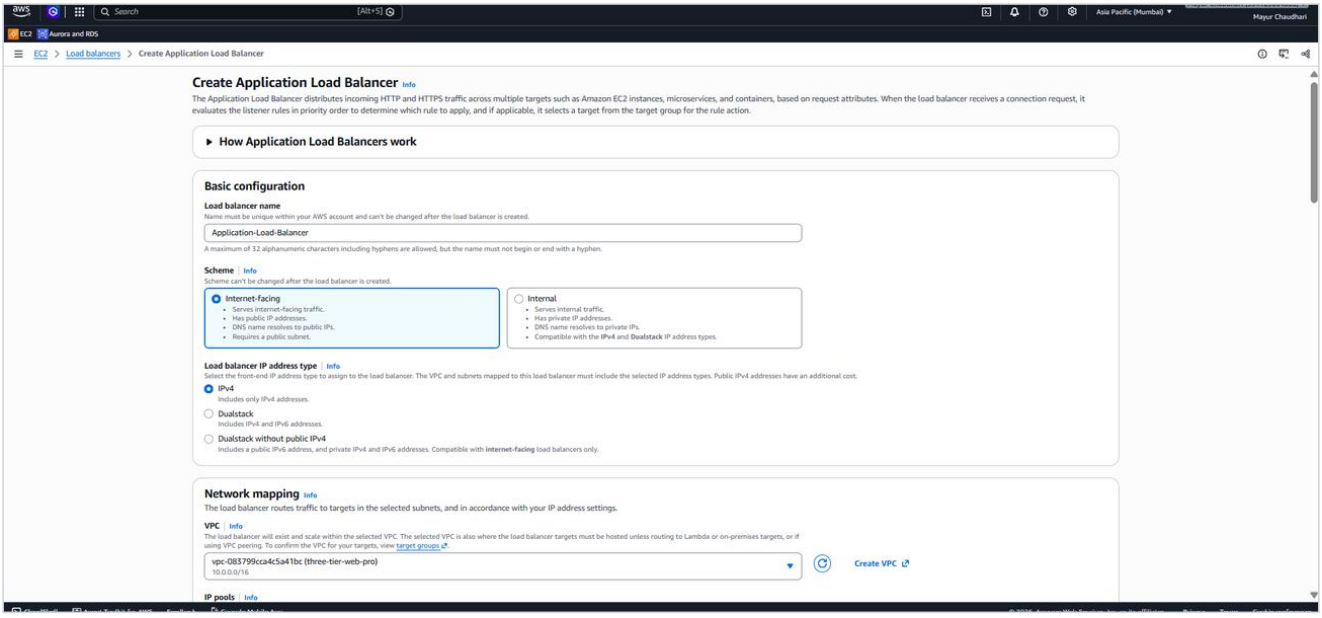
Step 9.13 Load balancers page.



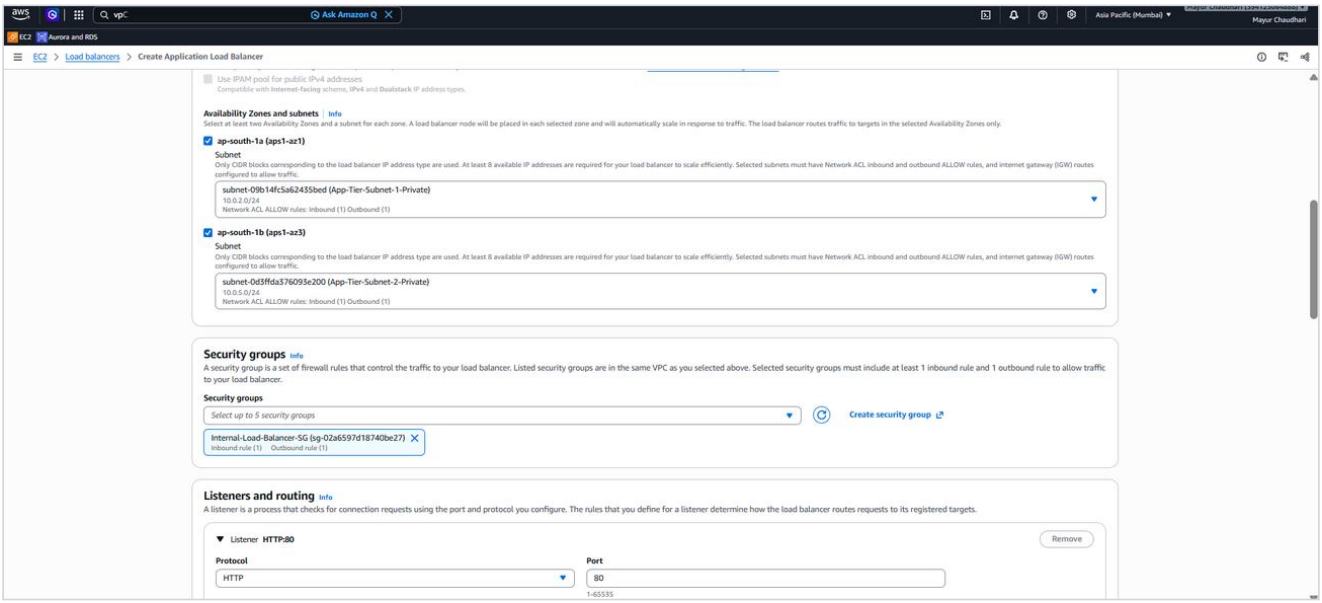
Step 9.14 Choose Application Load Balancer.



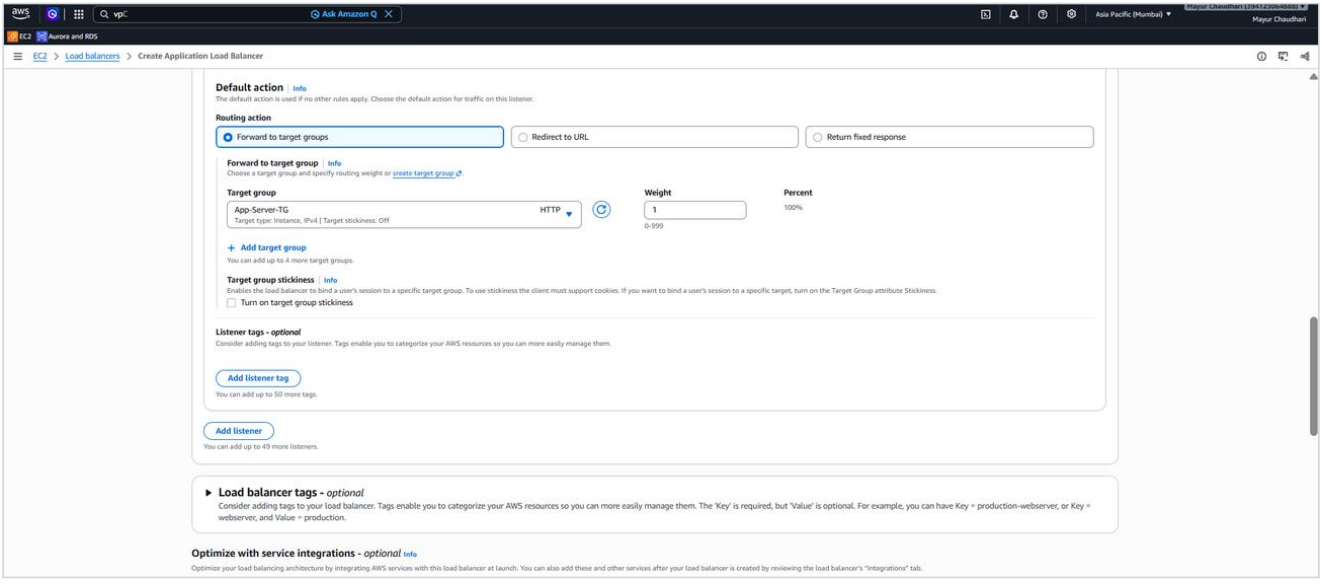
Step 9.15 Create internal Application Load Balancer.



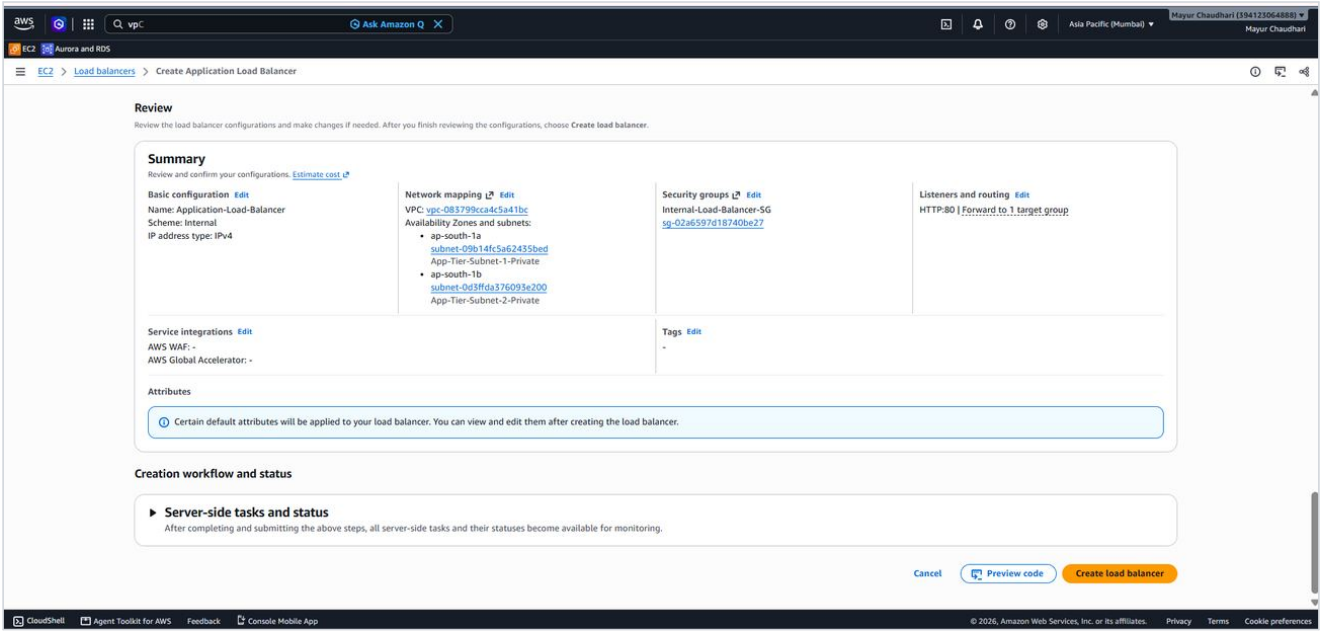
Step 9.16



Step 9.17

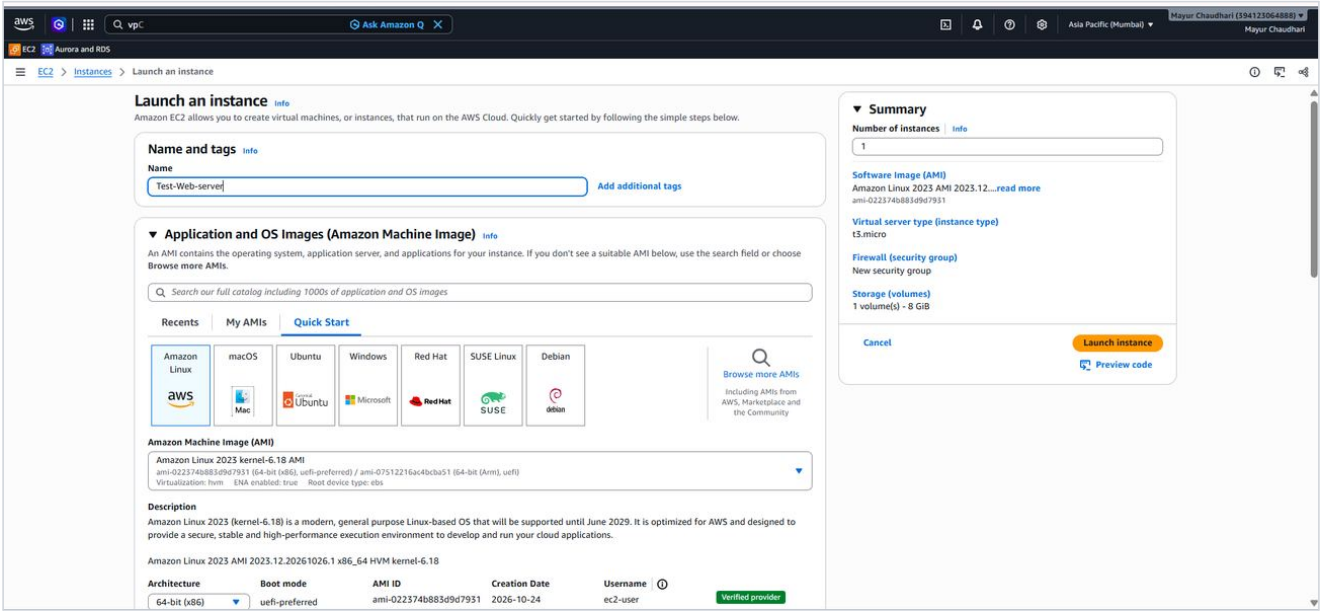


Step 9.18 Review and create the internal load balancer.

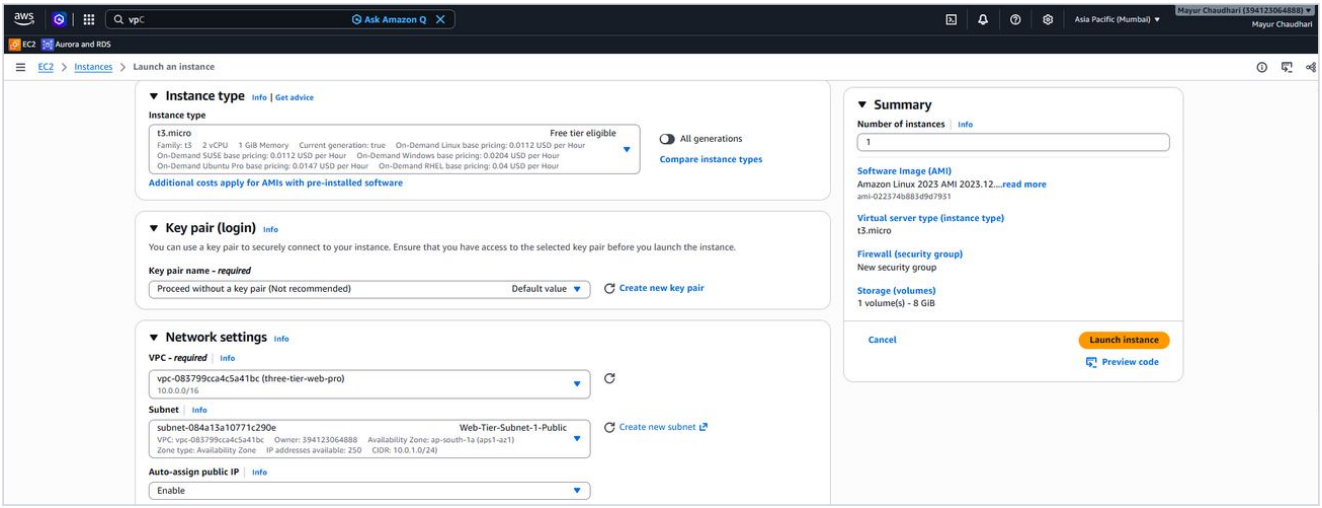


Step 10 - Test Web Server

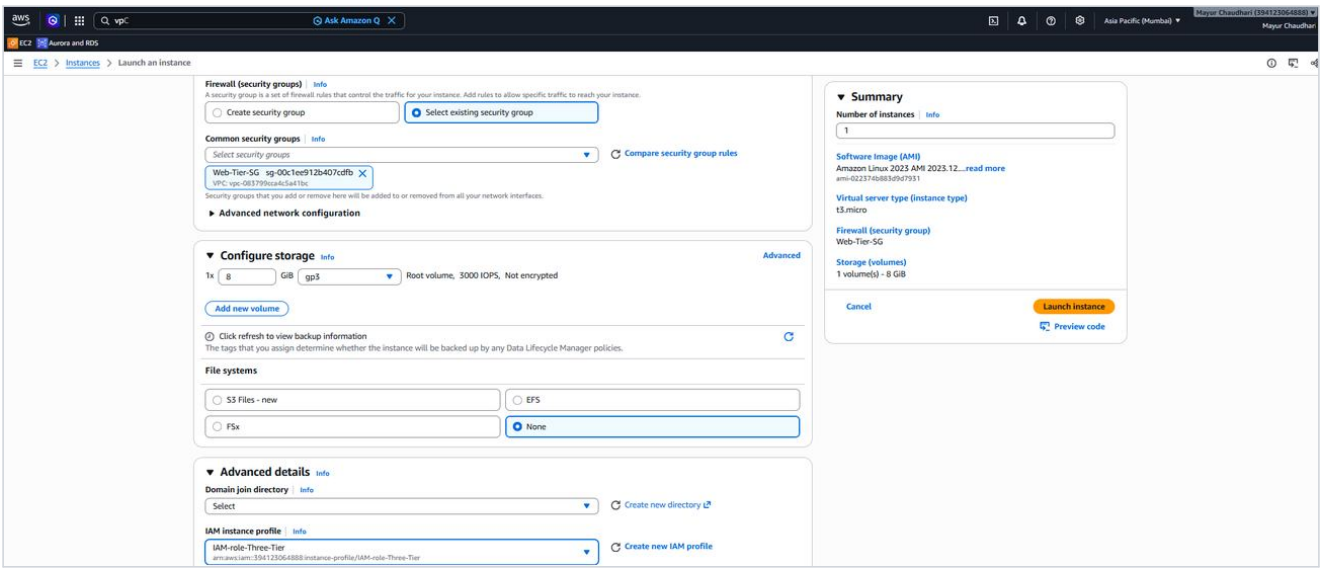
Step 10.1 Launch Test-Web-Server (public web subnet, Web-Tier-SG, IAM role).



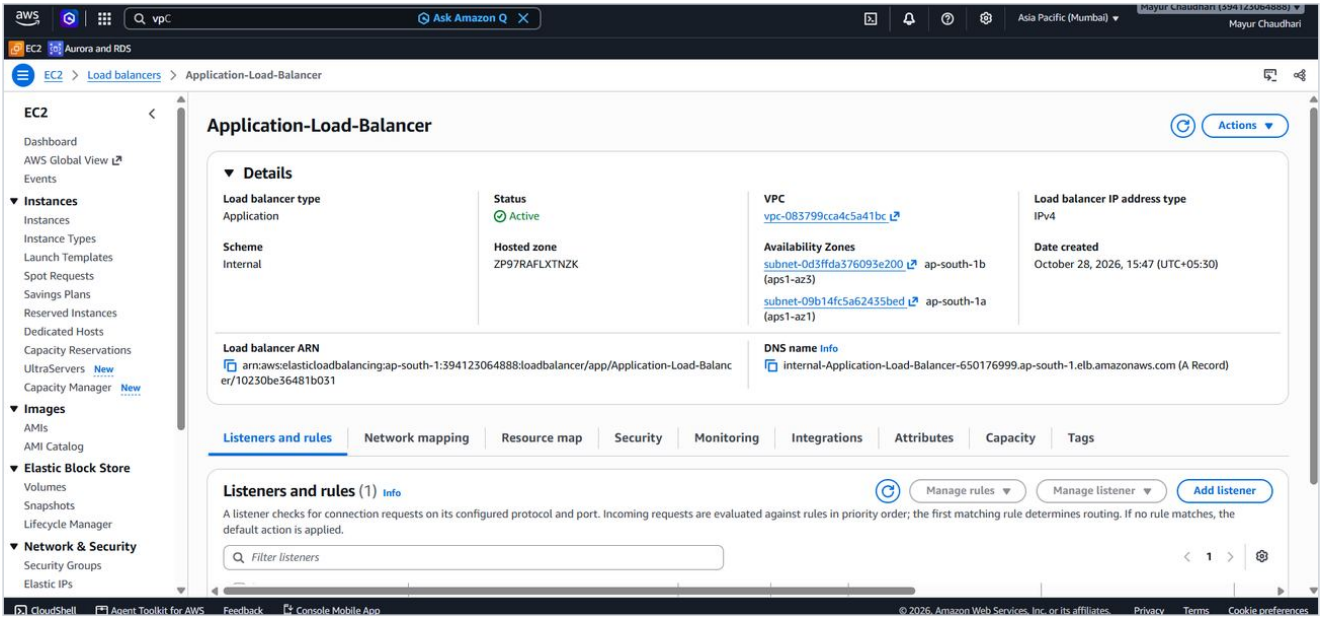
Step 10.2



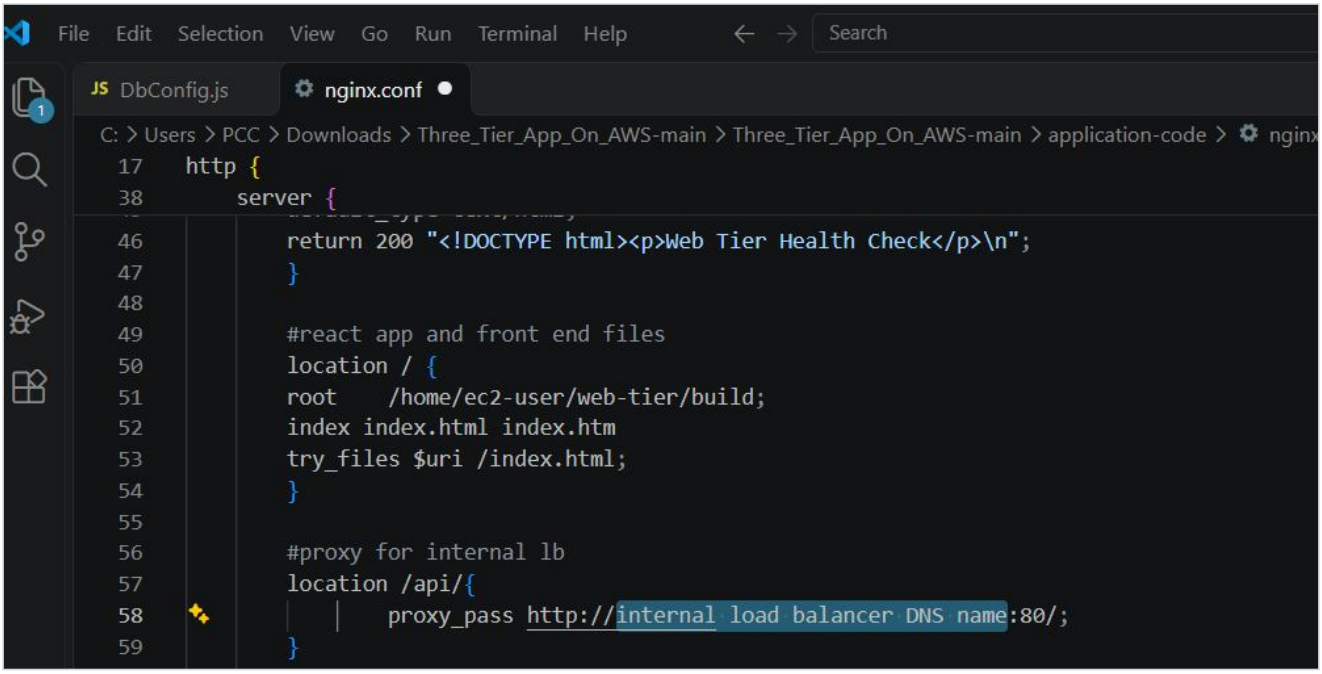
Step 10.3



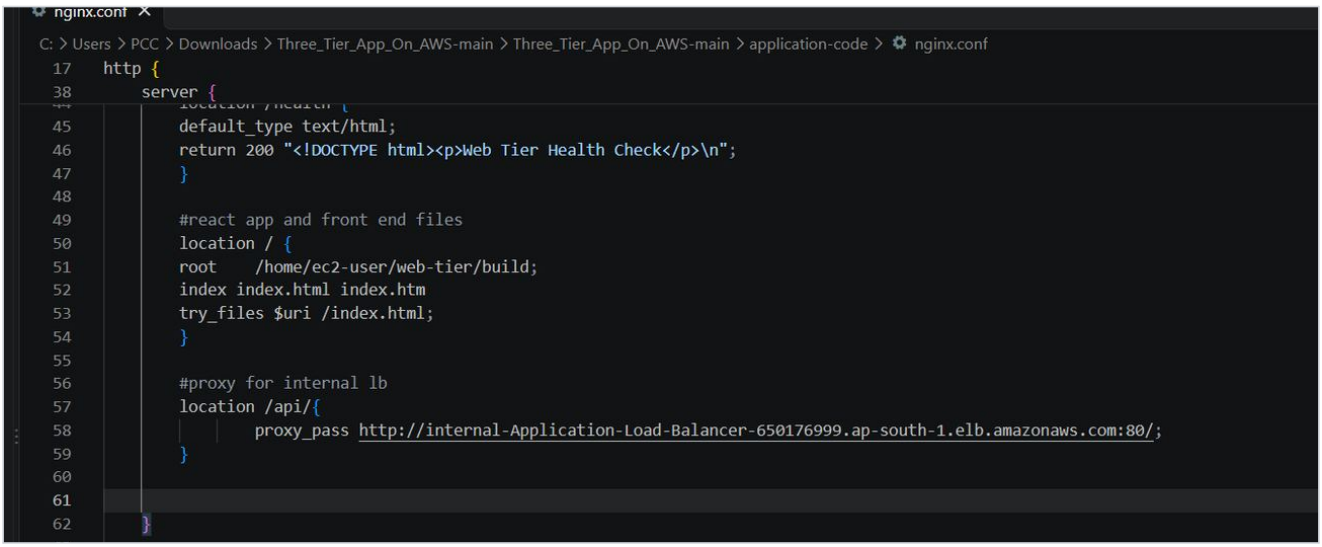
Step 10.4 Copy the Internal Load Balancer DNS name.



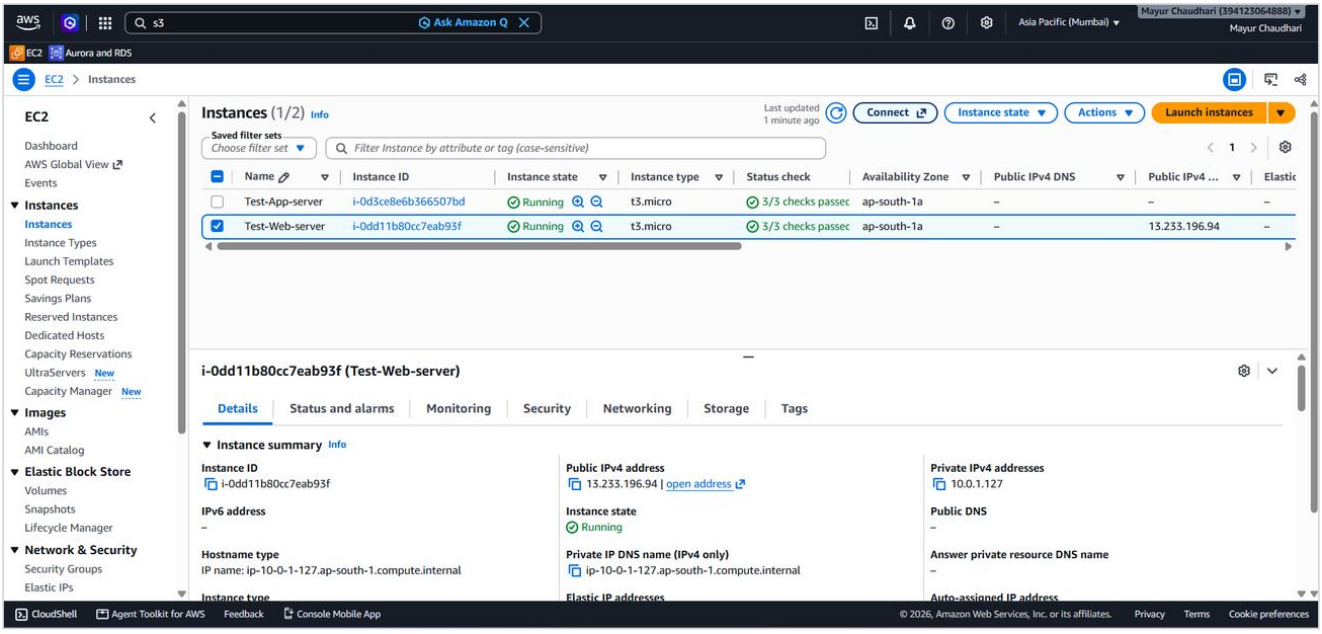
Step 10.5 nginx.conf: paste the internal load balancer DNS in proxy_pass.



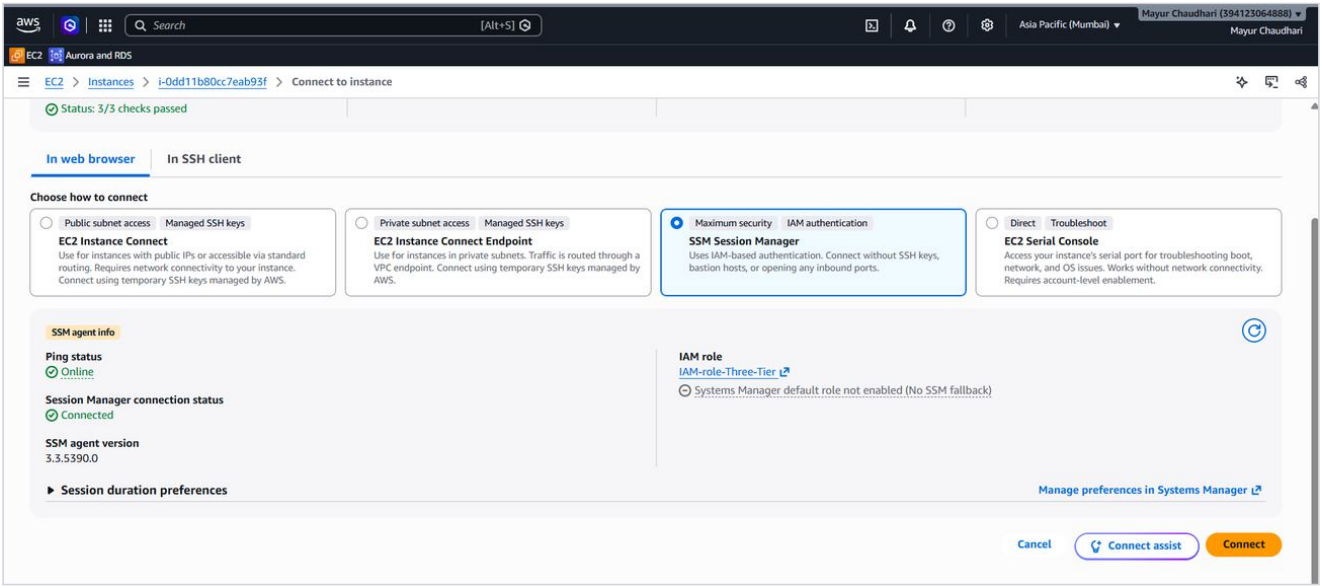
Step 10.6 nginx.conf ready to upload to S3.



Step 10.7 Test web server running.



Step 10.8 Connect with Session Manager.



Step 10.9 Copy the web-tier code from S3.

```
sh-5.2$ sudo -su ec2-user
[ec2-user@ip-10-0-1-127 bin]$ cd /home/ec2-user
[ec2-user@ip-10-0-1-127 ~]$ sudo aws s3 cp s3://three-tier-app-bucket-28-10-26/application-code/web-tier/ /home/ec2-user/web-tier/ --recursive
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/README.md to web-tier/README.md
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/package.json to web-tier/package.json
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/public/robots.txt to web-tier/public/robots.txt
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/App.test.js to web-tier/src/App.test.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/components/Burger/Burger.js to web-tier/src/components/Burger/Burger.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/public/index.html to web-tier/public/index.html
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/App.css to web-tier/src/App.css
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/components/Burger/index.js to web-tier/src/components/Burger/index.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/components/Burger/Burger.styled.js to web-tier/src/components/Burger/Burger.styled.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/App.js to web-tier/src/App.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/components/Home/Home.js to web-tier/src/components/Home/Home.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/components/Menu/index.js to web-tier/src/components/Menu/index.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/components/Menu/Menu.js to web-tier/src/components/Menu/Menu.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/components/DatabaseDemo/DatabaseDemo.css to web-tier/src/components/DatabaseDemo/DatabaseDemo.css
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/components/index.js to web-tier/src/components/index.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/components/DatabaseDemo/DatabaseDemo.js to web-tier/src/components/DatabaseDemo/DatabaseDemo.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/hooks.js to web-tier/src/hooks.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/components/Menu/Menu.styled.js to web-tier/src/components/Menu/Menu.styled.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/assets/3TierArch.png to web-tier/src/assets/3TierArch.png
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/theme.js to web-tier/src/theme.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/reportWebVitals.js to web-tier/src/reportWebVitals.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/setupTests.js to web-tier/src/setupTests.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/global.js to web-tier/src/global.js
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/index.css to web-tier/src/index.css
download: s3://three-tier-app-bucket-28-10-26/application-code/web-tier/src/index.js to web-tier/src/index.js
[ec2-user@ip-10-0-1-127 ~]$ sudo chown -R ec2-user:ec2-user /home/ec2-user/web-tier
[ec2-user@ip-10-0-1-127 ~]$ sudo chmod -R 755 /home/ec2-user/web-tier
```

Step 10.10 Install Node.js with nvm.

```
[ec2-user@ip-10-0-1-127 ~]$ curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.7/install.sh | bash
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  16555 100  16555    0     0 930.7k      0                              0
=> Downloading nvm as script to '/home/ec2-user/.nvm'

=> Appending nvm source string to /home/ec2-user/.bashrc
=> Appending bash_completion source string to /home/ec2-user/.bashrc
=> Close and reopen your terminal to start using nvm or run the following to use it now:

export NVM_DIR="$HOME/.nvm"
[ -s "$NVM_DIR/nvm.sh" ] && \. "$NVM_DIR/nvm.sh"  # This loads nvm
[ -s "$NVM_DIR/bash_completion" ] && \. "$NVM_DIR/bash_completion"  # This loads nvm bash_completion
[ec2-user@ip-10-0-1-127 ~]$ source ~/.bashrc
[ec2-user@ip-10-0-1-127 ~]$ nvm install 16
Downloading and installing node v16.20.2...
Downloading https://nodejs.org/dist/v16.20.2/node-v16.20.2-linux-x64.tar.xz...
######################################################################################################################################################################################################
Computing checksum with sha256sum
Checksums matched!
Now using node v16.20.2 (npm v8.19.4)
Creating default alias: default -> 16 (-> v16.20.2)
[ec2-user@ip-10-0-1-127 ~]$ nvm use 16
Now using node v16.20.2 (npm v8.19.4)
[ec2-user@ip-10-0-1-127 ~]$ cd /home/ec2-user/web-tier
```

Step 10.11 npm install.

```
npm WARN deprecated rimraf@3.0.2: Rimraf versions prior to v4 are no longer supported
npm WARN deprecated @humanwhocodes/object-schema@2.0.3: Use @eslint/object-schema instead
npm WARN deprecated @humanwhocodes/config-array@0.13.0: Use @eslint/config-array instead
npm WARN deprecated uuid@8.3.2: uuid@10 and below is no longer supported.  For ESM codebases, update to uuid@latest.  For CommonJS codebases, use uuid@11 (but be aware this version will likely be d
d in 2028).
npm WARN deprecated @babel/plugin-proposal-numeric-separator@7.18.6: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-t
m-numeric-separator instead.
npm WARN deprecated @babel/plugin-proposal-private-methods@7.18.6: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-tra
private-methods instead.
npm WARN deprecated @babel/plugin-proposal-class-properties@7.18.6: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-t
-class-properties instead.
npm WARN deprecated @babel/plugin-proposal-nullish-coalescing-operator@7.18.6: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babe
n-transform-nullish-coalescing-operator instead.
npm WARN deprecated @babel/plugin-proposal-optional-chaining@7.21.0: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-t
m-optional-chaining instead.
npm WARN deprecated @babel/plugin-proposal-private-property-in-object@7.21.11: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babe
n-transform-private-property-in-object instead.
npm WARN deprecated svgo@1.3.2: This SVGO version is no longer supported. Upgrade to v2.x.x.
npm WARN deprecated eslint@8.57.1: This version is no longer supported. Please see https://eslint.org/version-support for other options.

added 1545 packages, and audited 1546 packages in 1m

281 packages are looking for funding
  run `npm fund` for details

84 vulnerabilities (16 low, 5 moderate, 13 high)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
npm notice
npm notice New major version of npm available! 8.19.4 -> 12.2.1
npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.2.1
npm notice Run npm install -g npm@12.2.1 to update!
npm notice
[ec2-user@ip-10-0-1-127 web-tier]$
```

Step 10.12 npm run build.

```
[ec2-user@ip-10-0-1-127 web-tier]$ npm run build

> aws-3tier-web-layer@0.1.0 build
> react-scripts build

Creating an optimized production build...
Compiled successfully.

File sizes after gzip:

  74.73 kB  build/static/js/main.73346a18.js
  1.78 kB   build/static/js/453.203979ea.chunk.js
  493 B     build/static/css/main.b20b6ac4.css

The project was built assuming it is hosted at ./.
You can control this with the homepage field in your package.json.

The build folder is ready to be deployed.

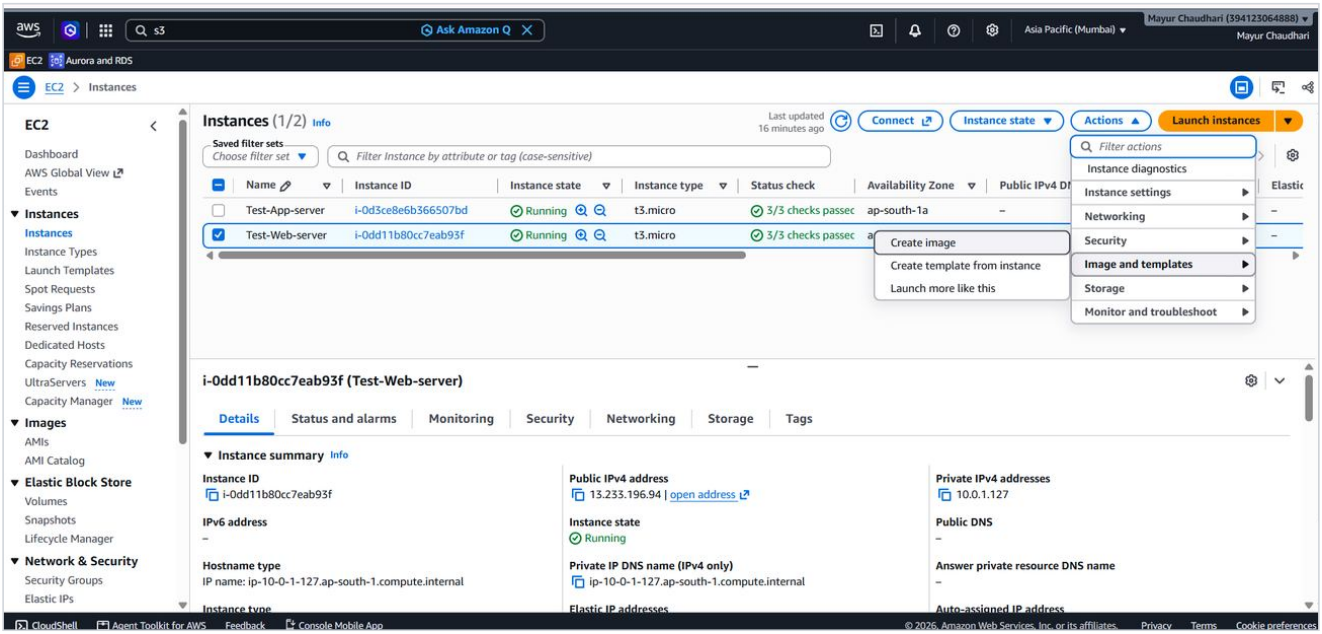
Find out more about deployment here:

  https://cra.link/deployment

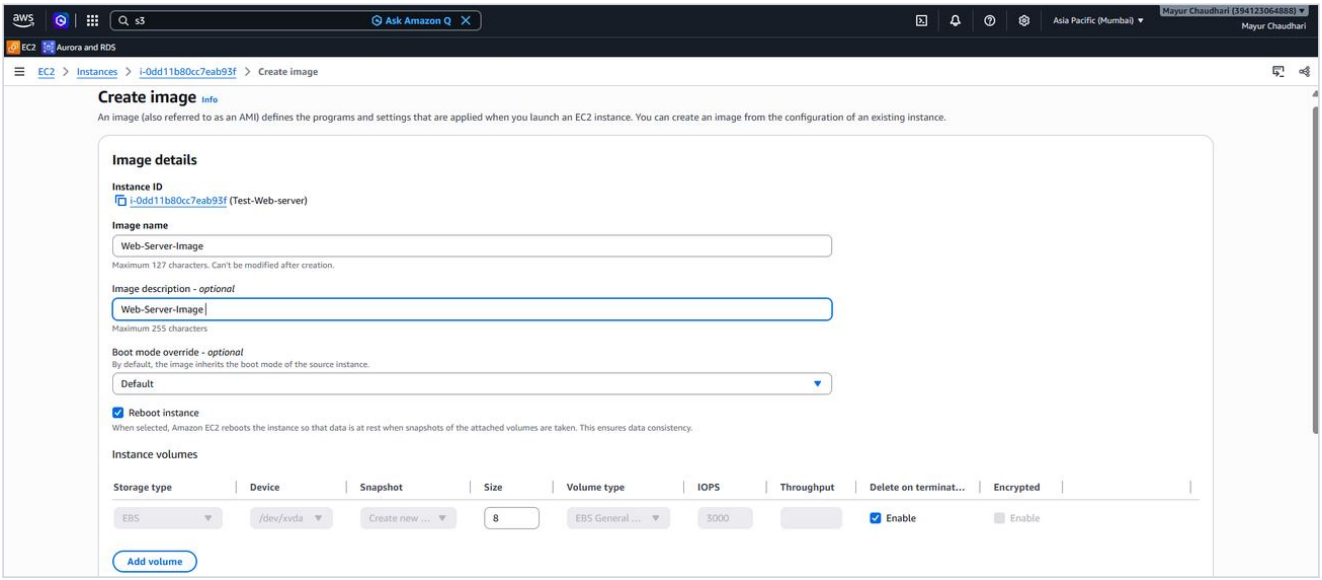
[ec2-user@ip-10-0-1-127 web-tier]$
```

Step 11 - Create the Web-Server AMI

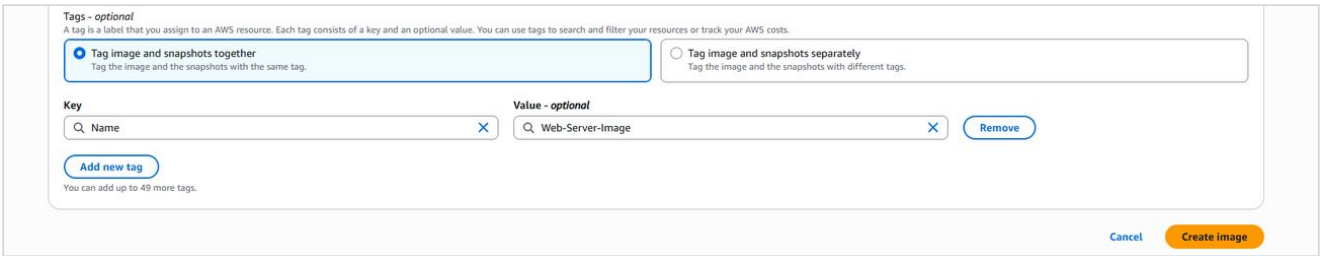
Step 11.1 Test web server > Actions > Image and templates > Create image.



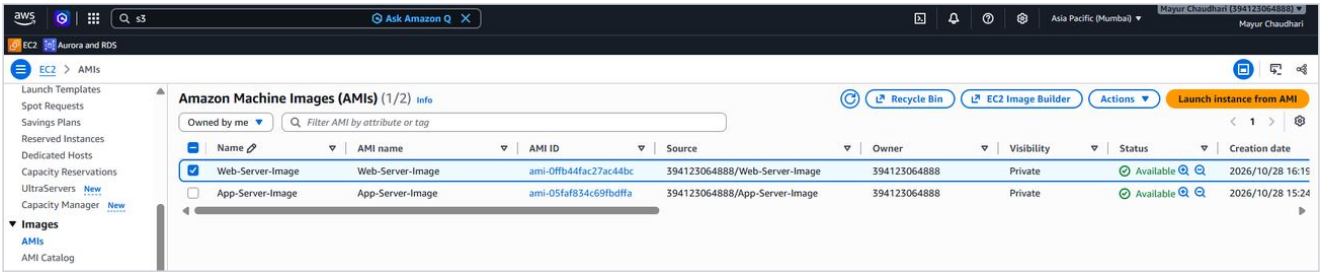
Step 11.2 Create image form.



Step 11.3

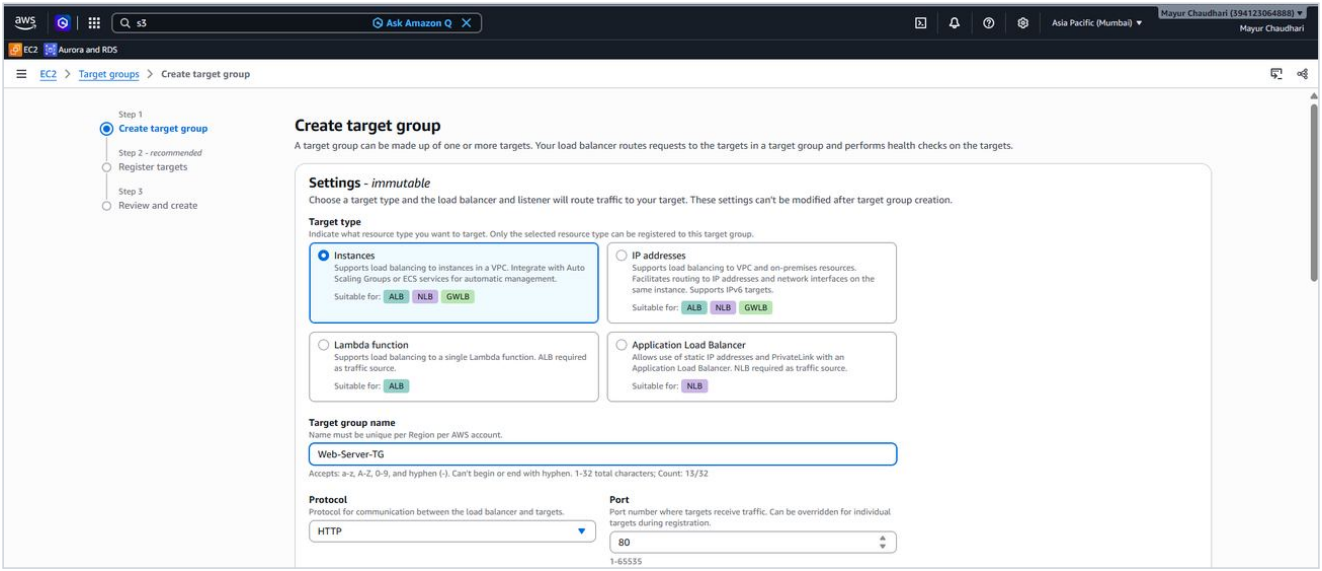


Step 11.4 Web server AMI created.

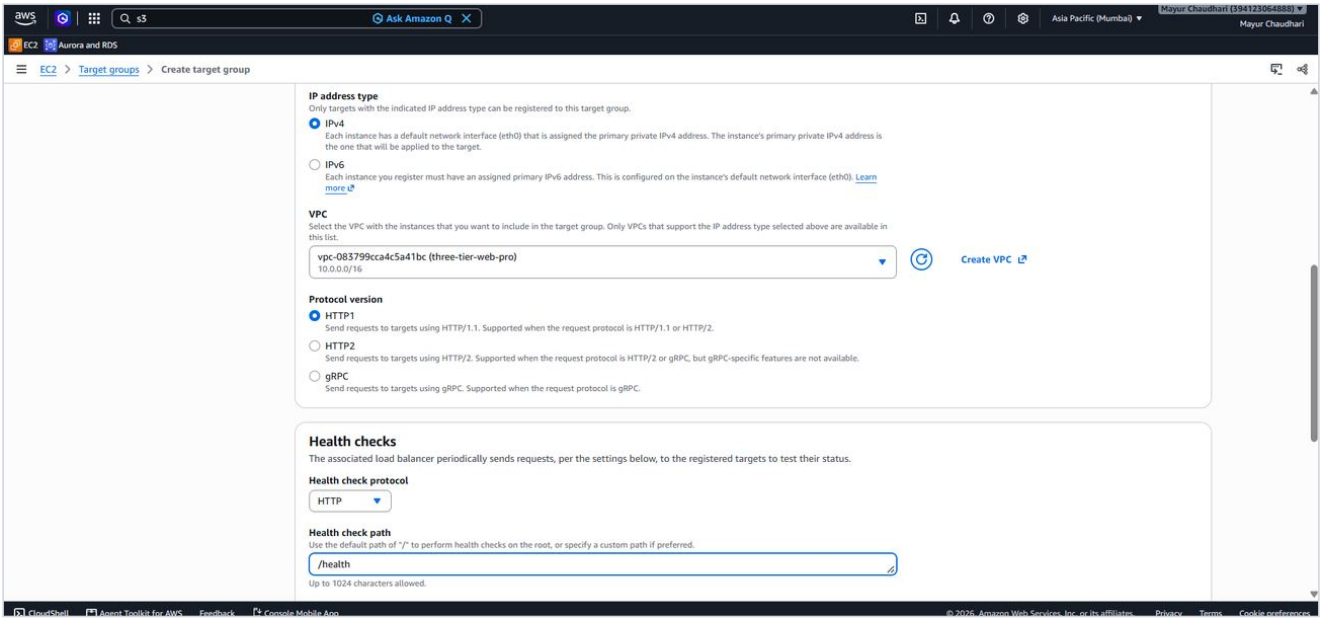


Step 12 - Web tier: target group, launch template, external load balancer

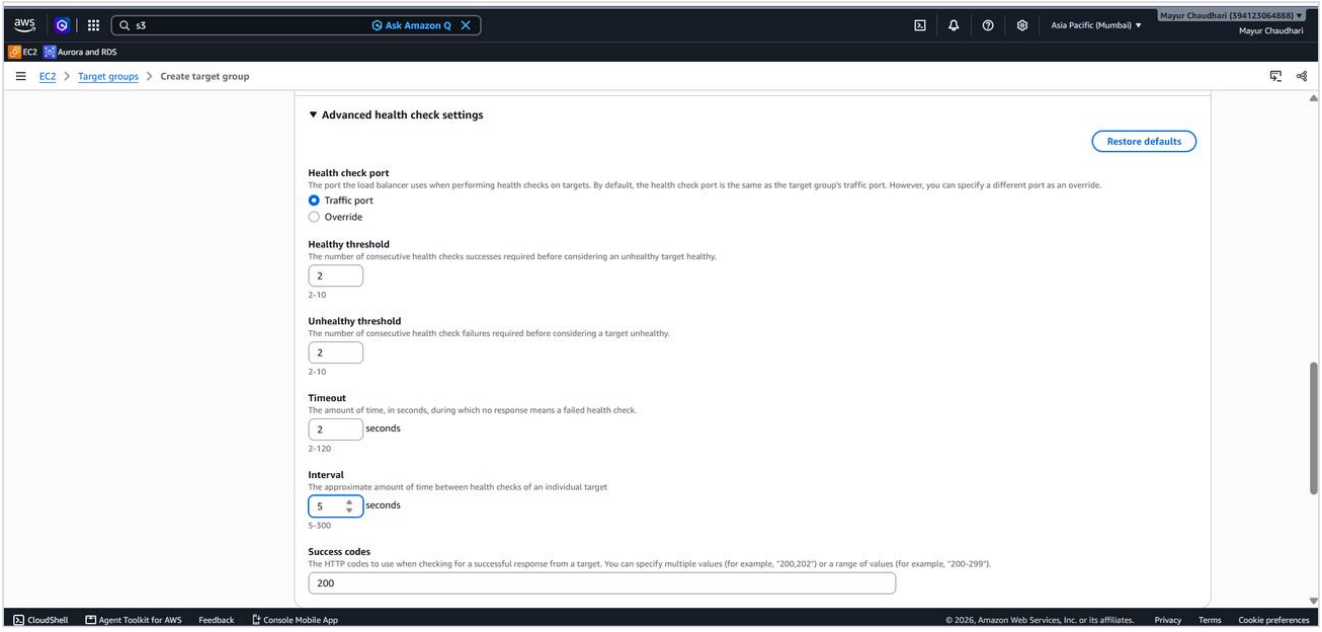
Step 12.1 Create target group for the web tier (port 80).



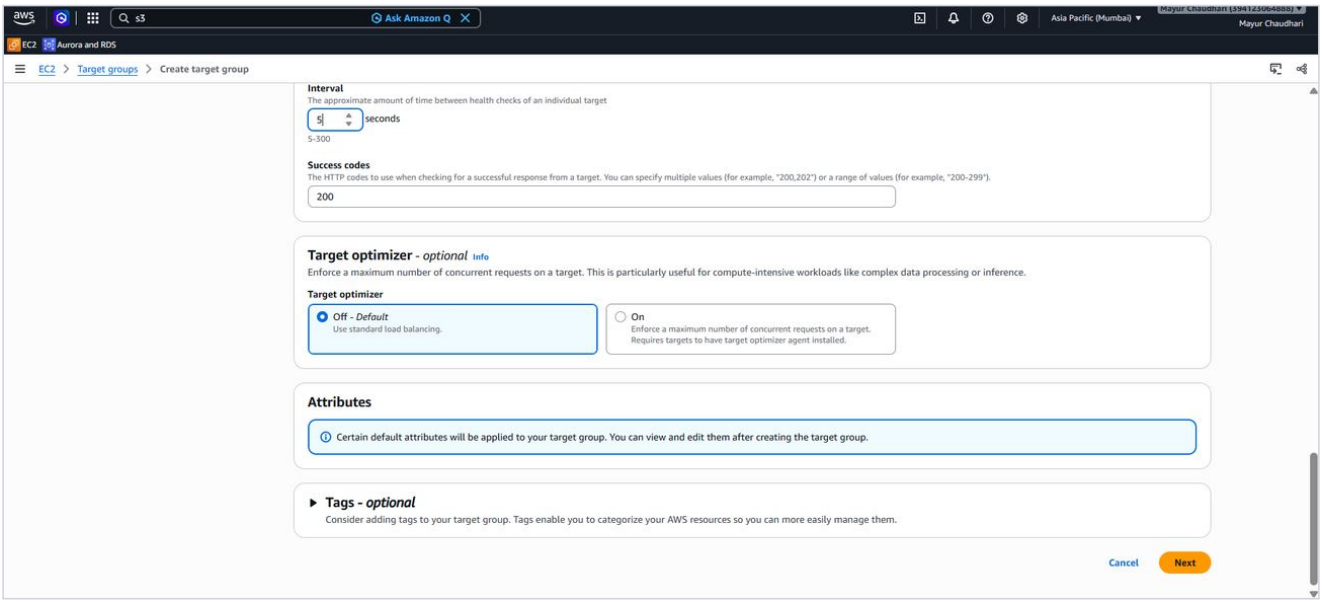
Step 12.2



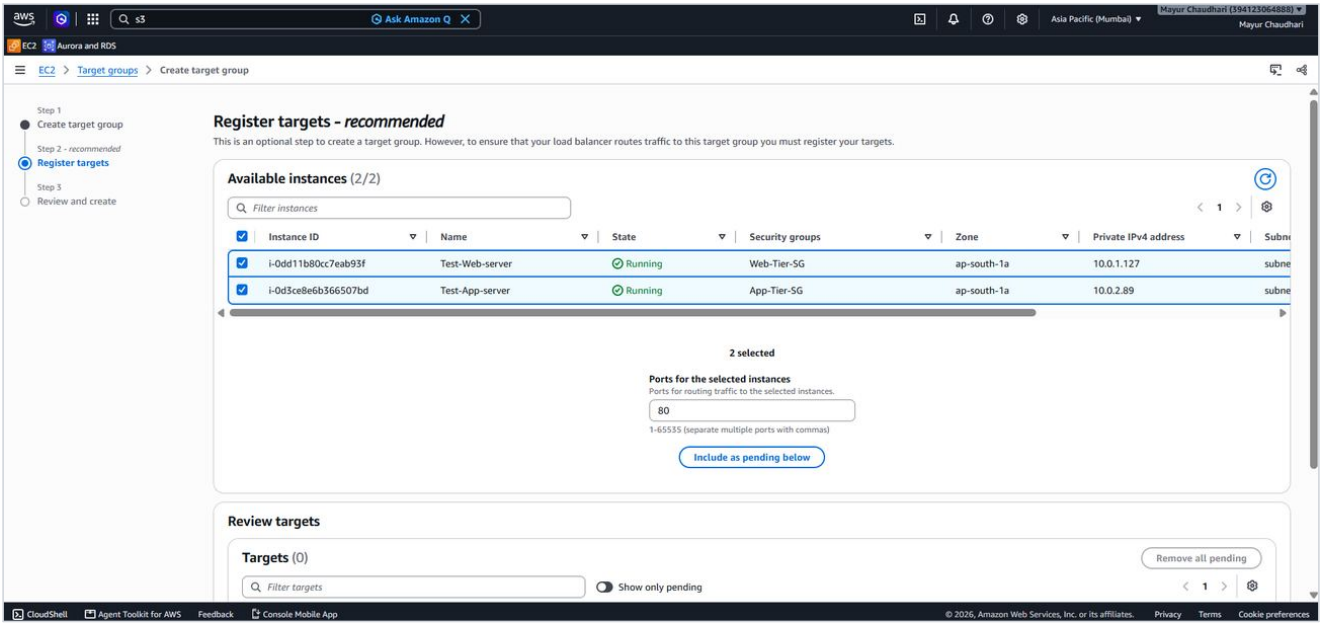
Step 12.3



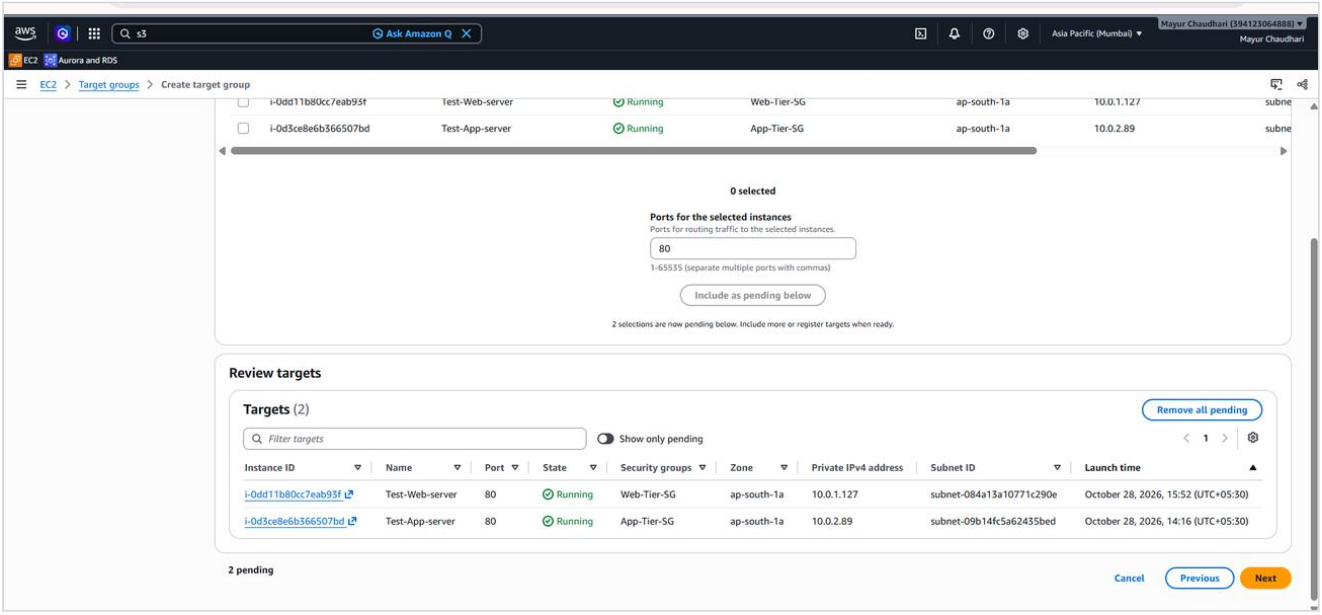
Step 12.4



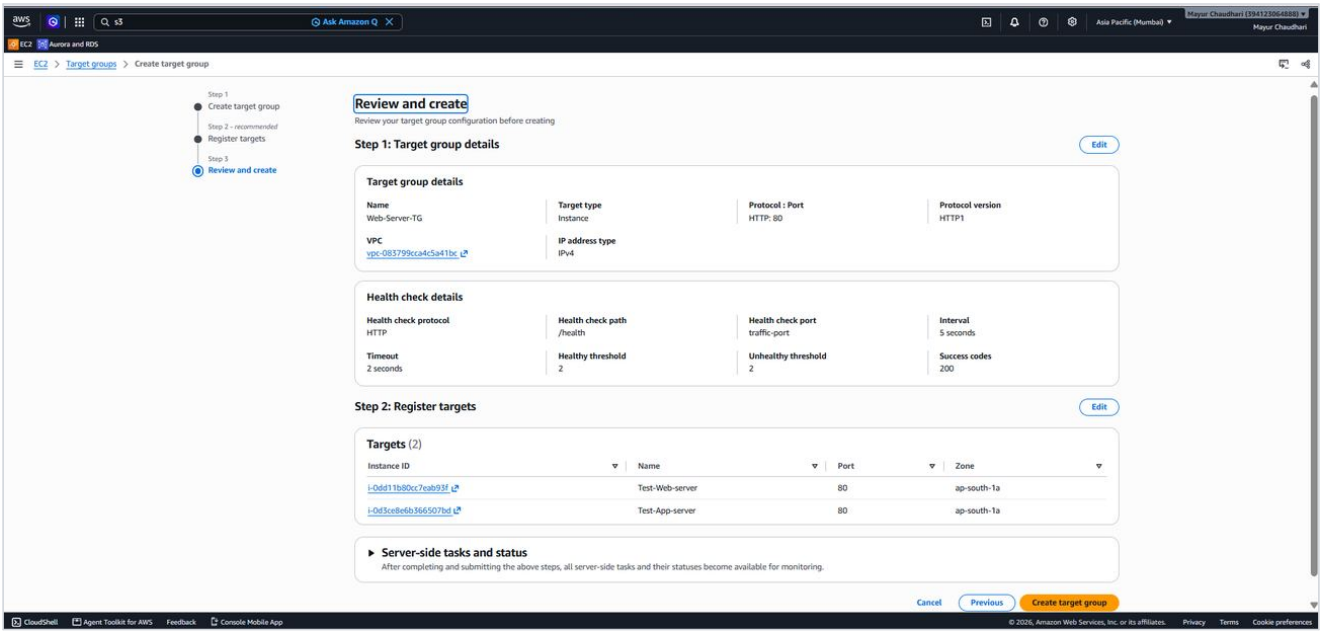
Step 12.5



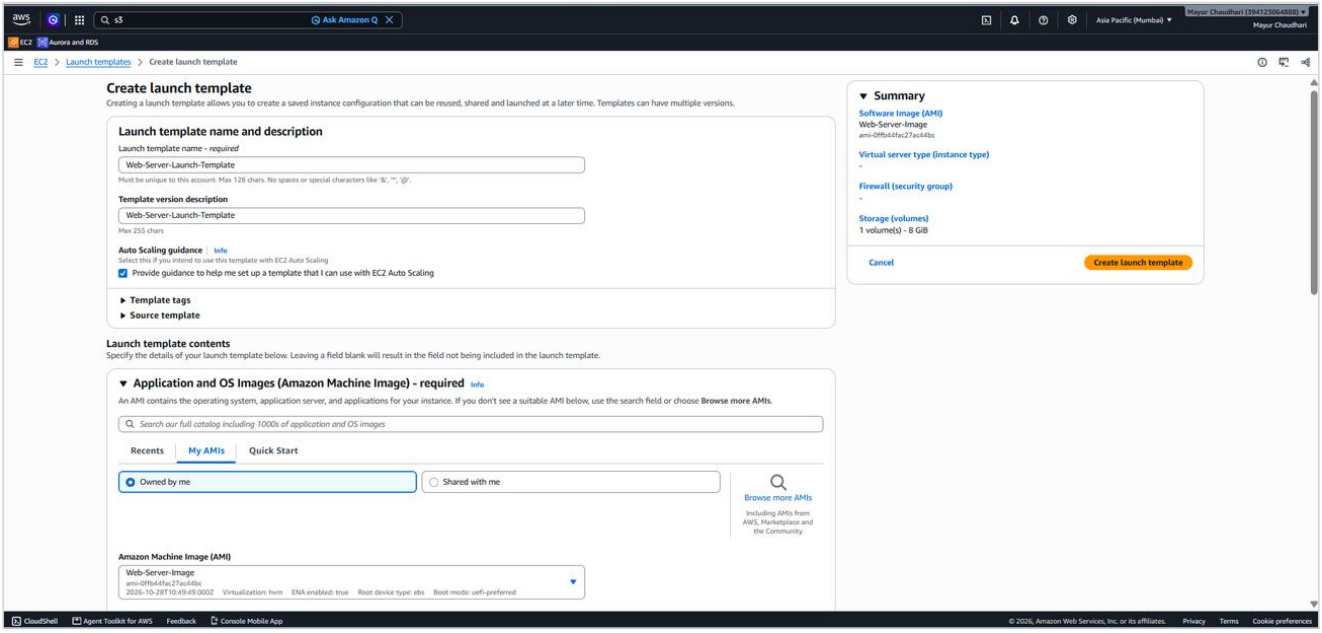
Step 12.6



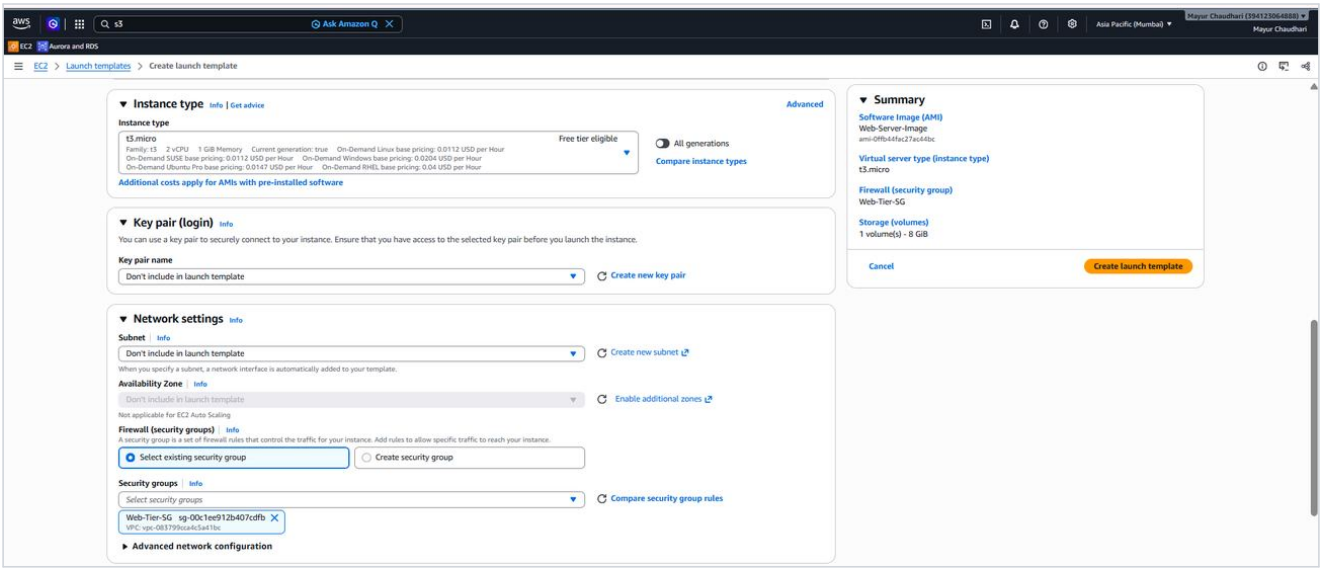
Step 12.7 Review and create the target group.



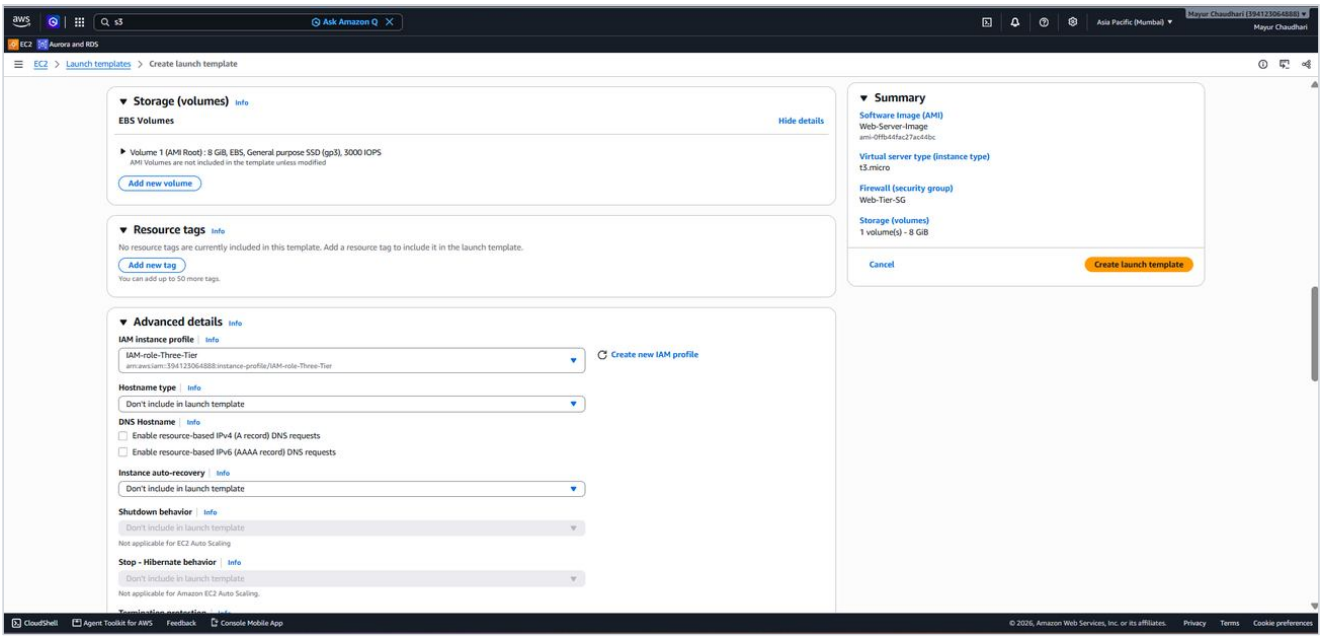
Step 12.8 Create launch template: choose the web AMI.



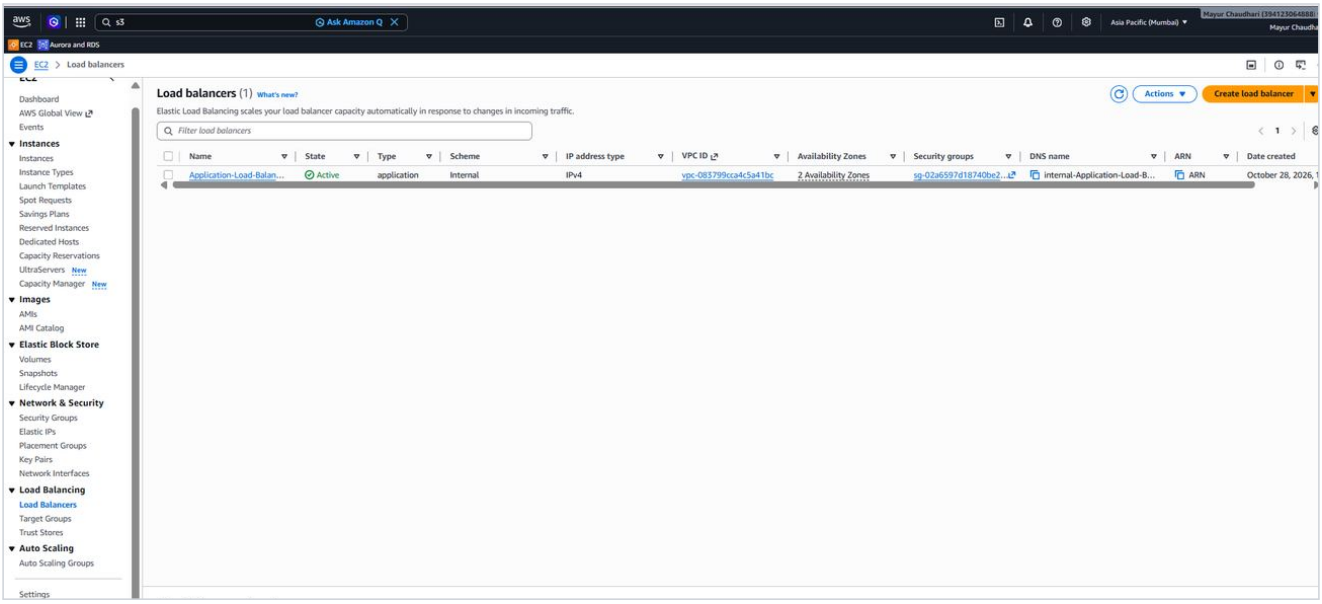
Step 12.9



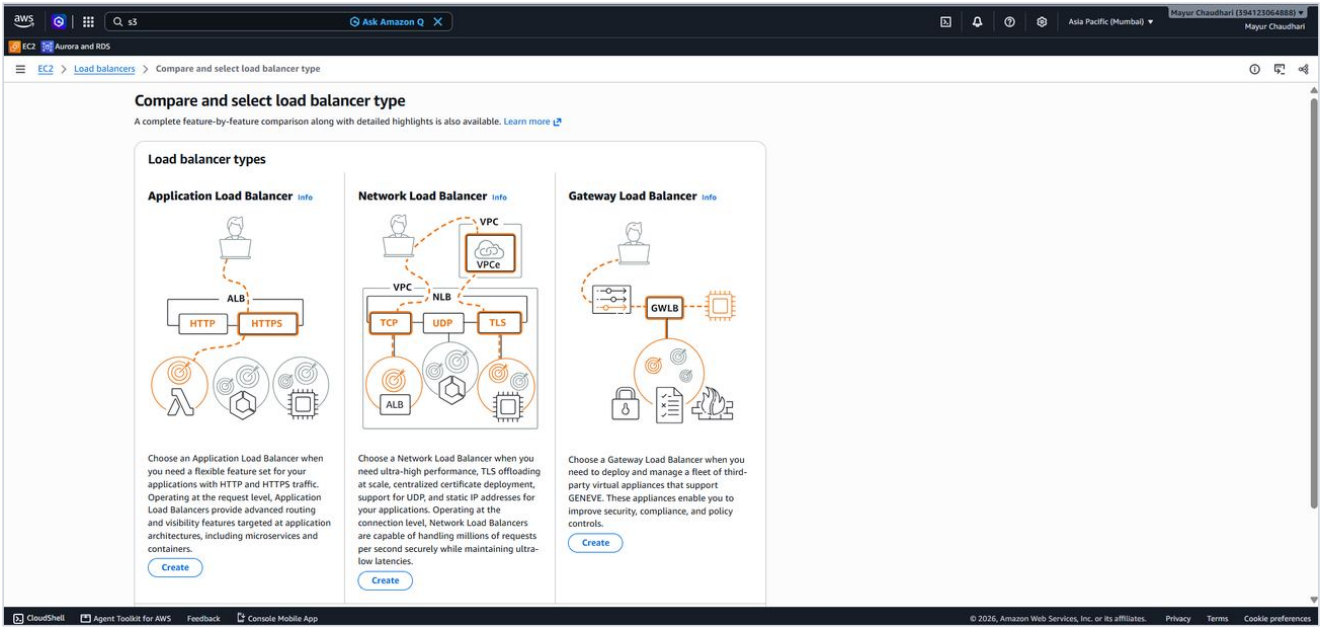
Step 12.10



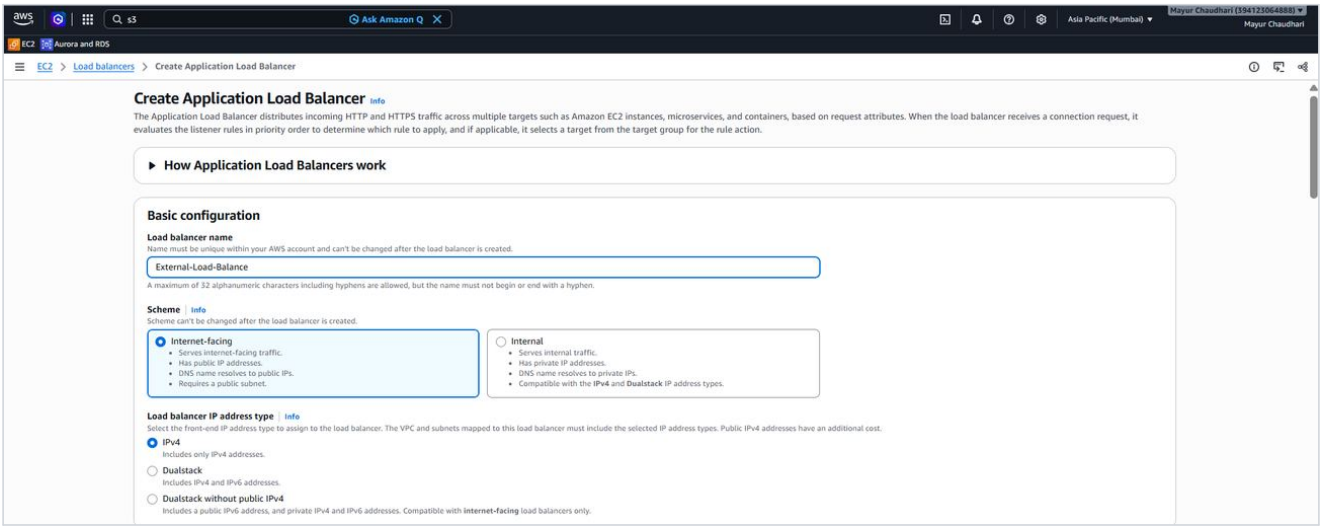
Step 12.11 Load balancers page.



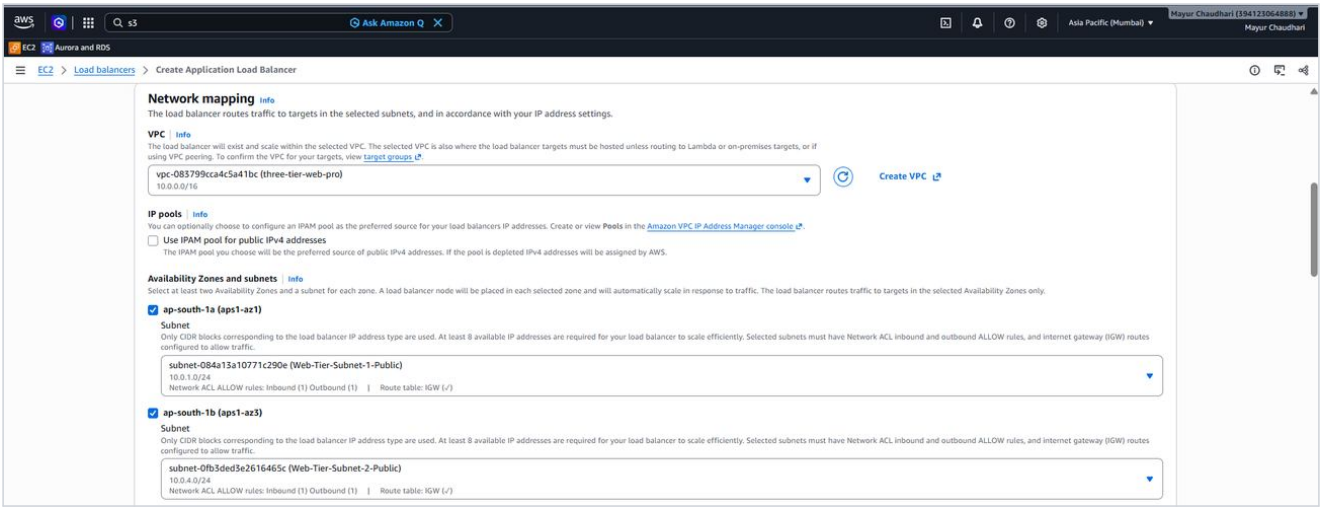
Step 12.12 Choose Application Load Balancer.



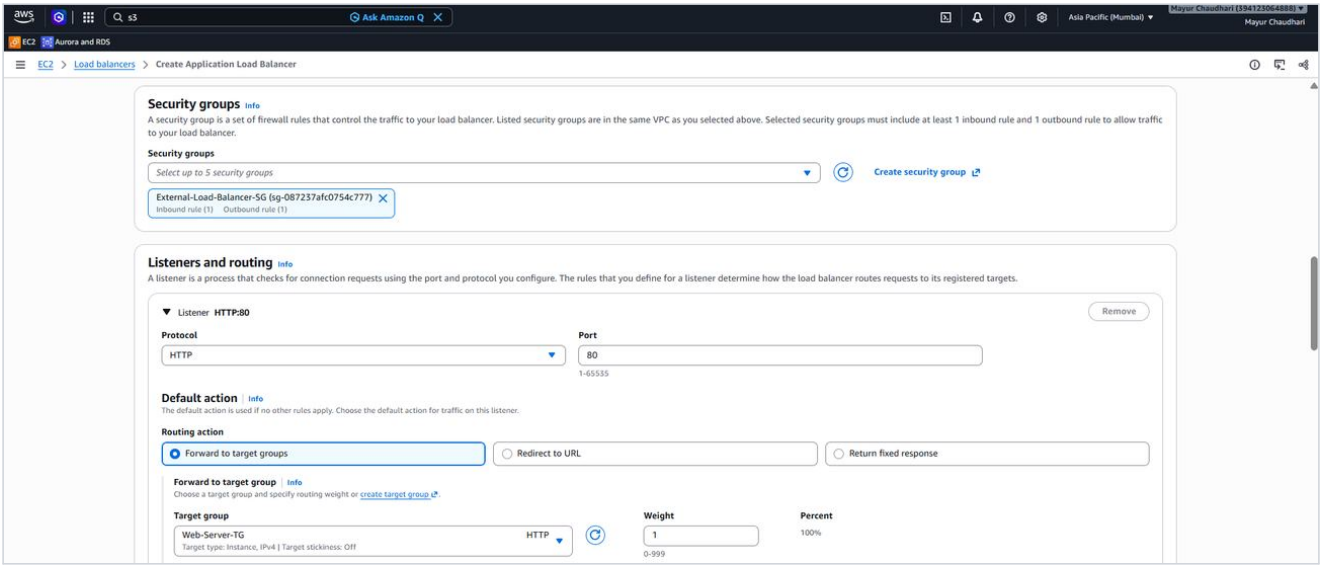
Step 12.13 Create the external (internet-facing) load balancer.



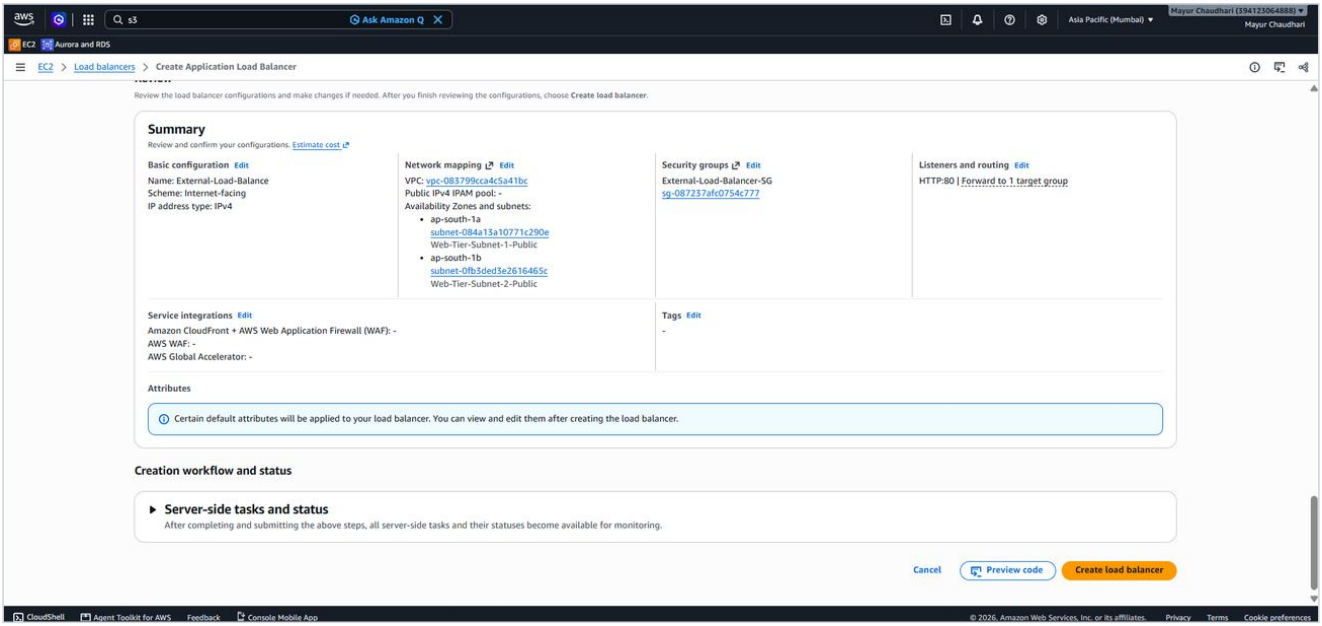
Step 12.14



Step 12.15

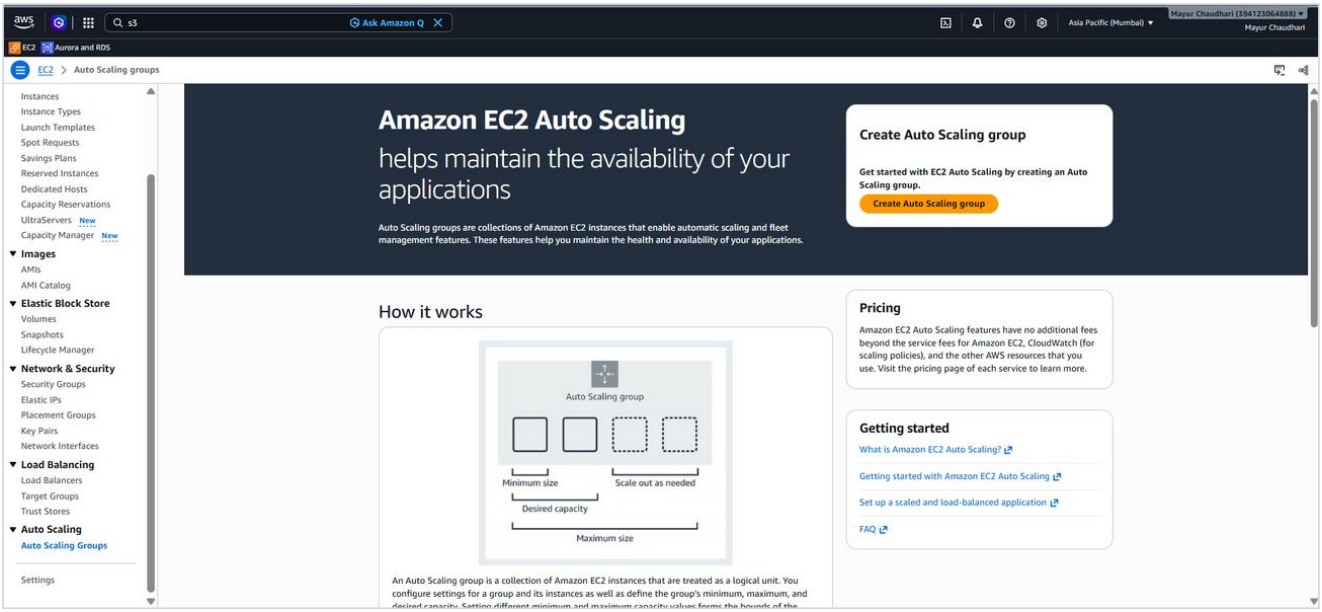


Step 12.16 Review and create the external load balancer.

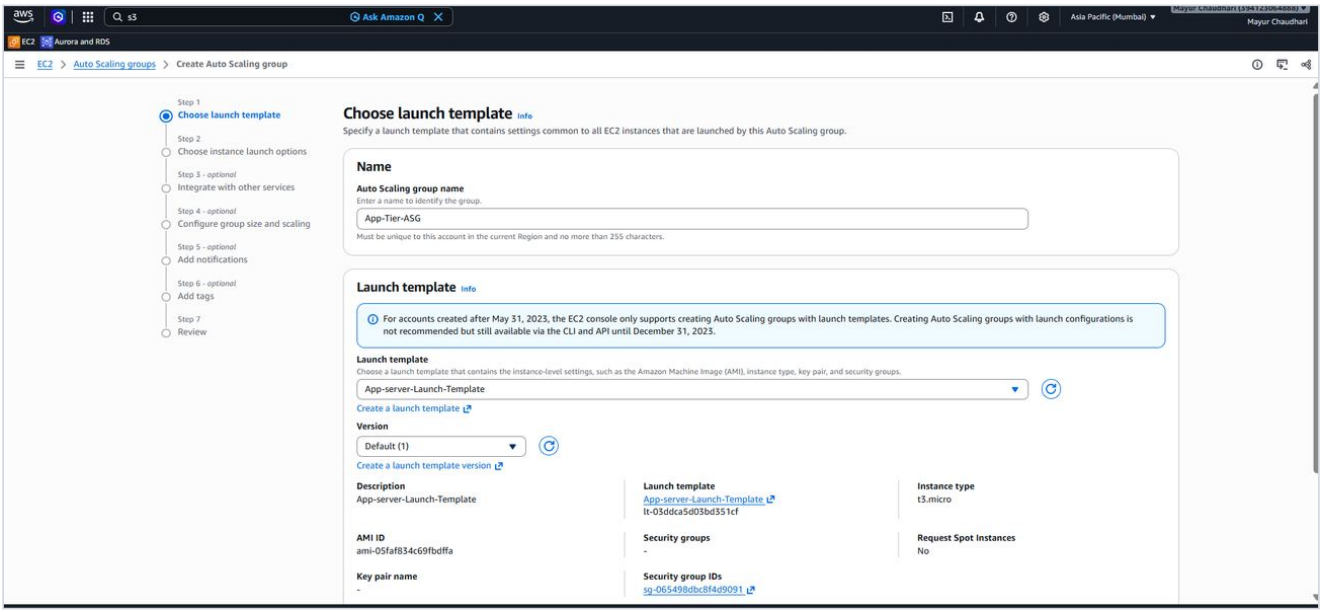


Step 13 - Auto Scaling groups

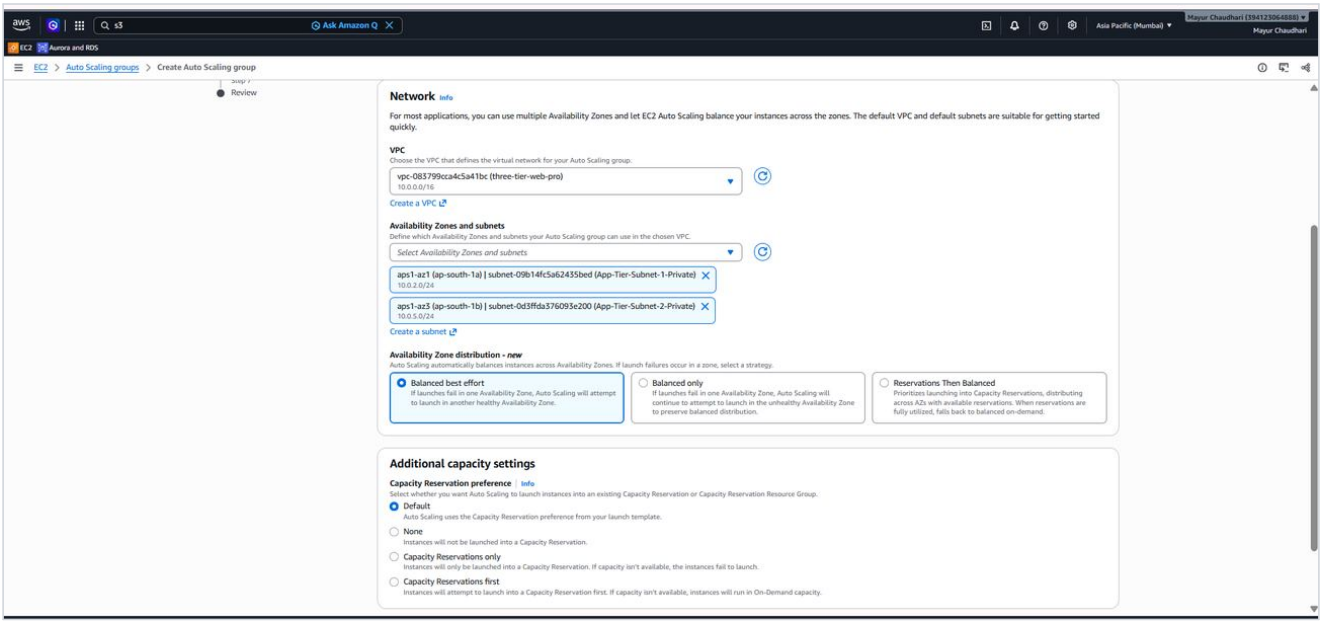
Step 13.1 EC2 Auto Scaling landing page.



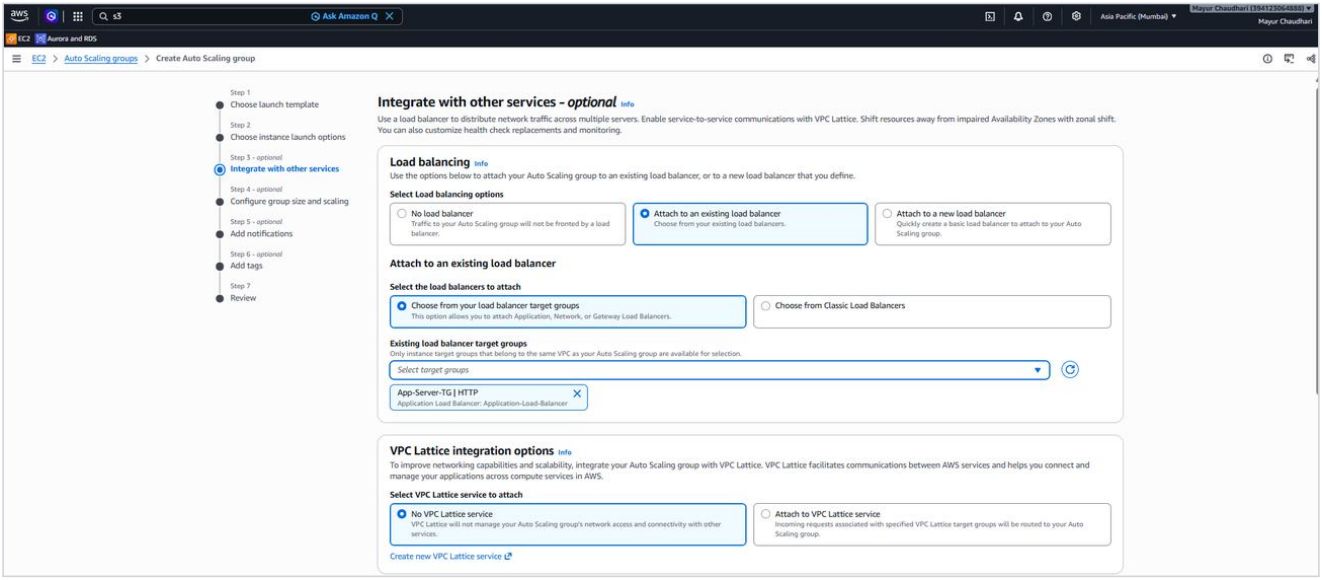
Step 13.2 Create Auto Scaling group: choose launch template.



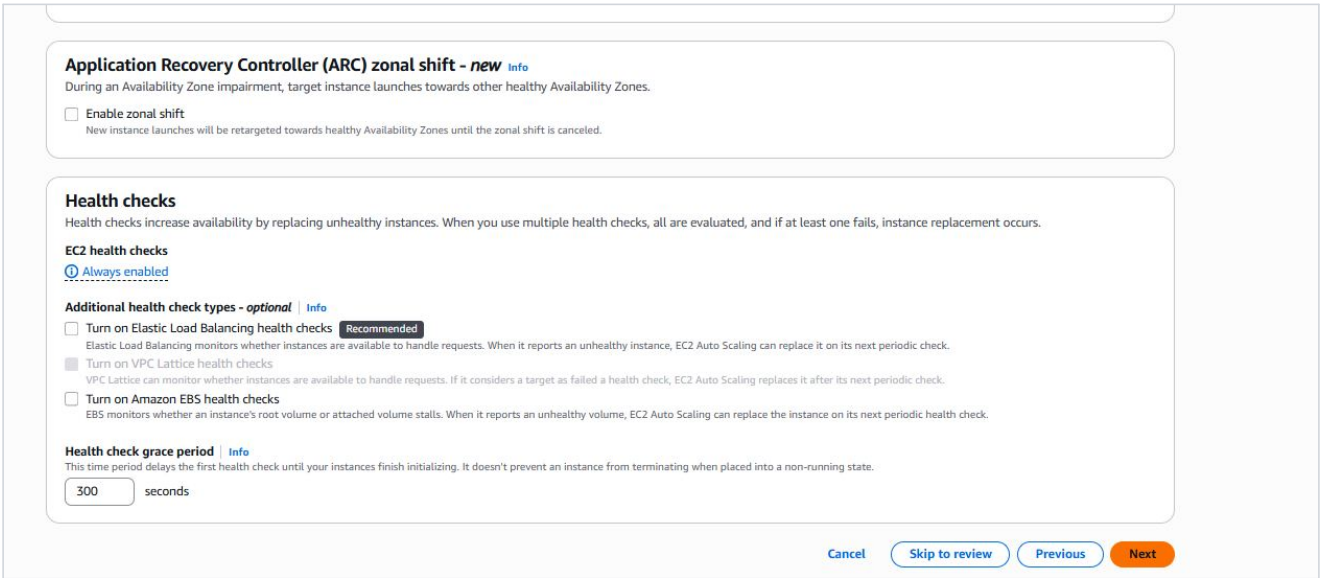
Step 13.3 Choose network: VPC and subnets.



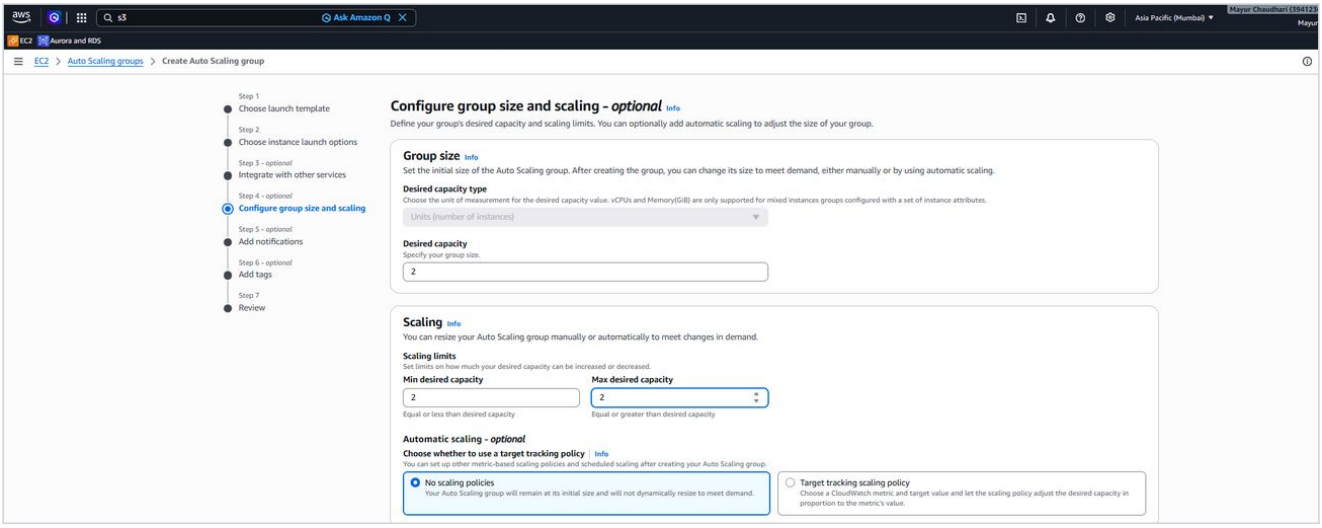
Step 13.4 Attach to the load balancer target group.



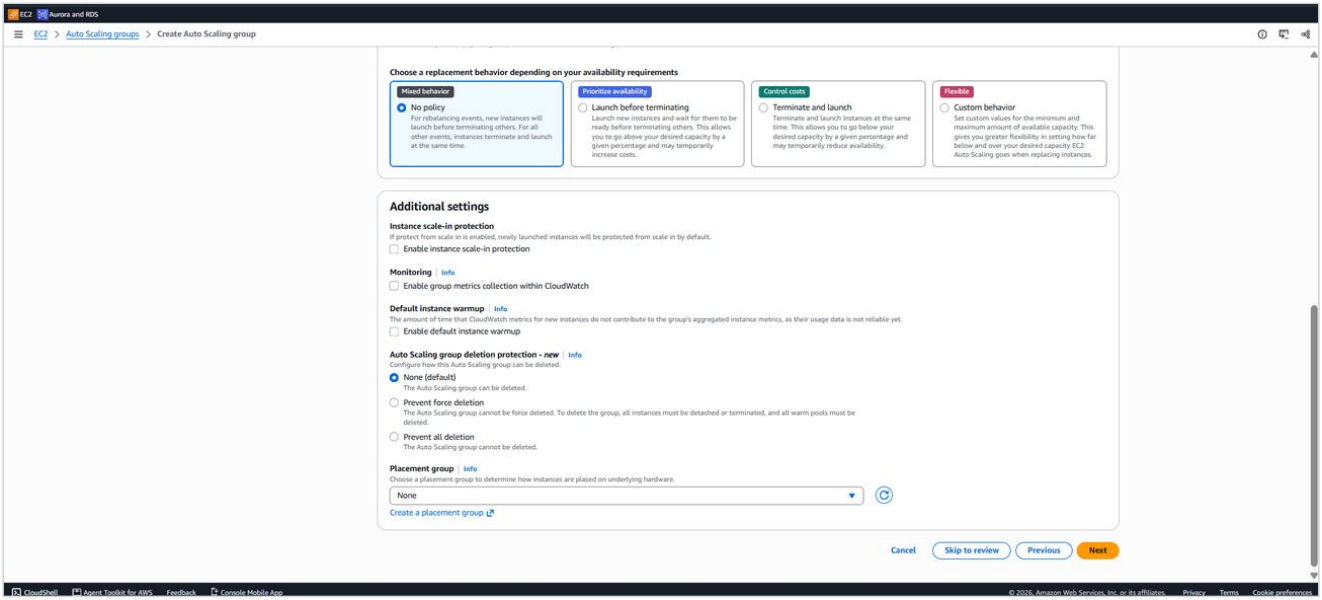
Step 13.5



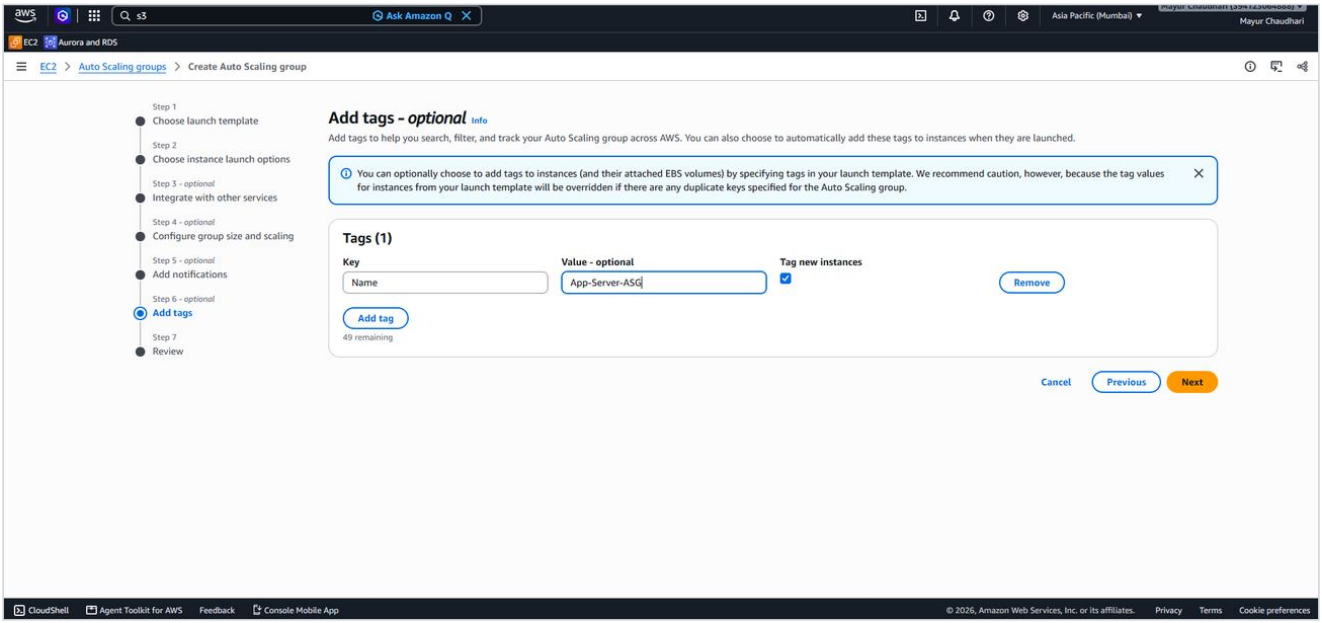
Step 13.6 Set group size and scaling.



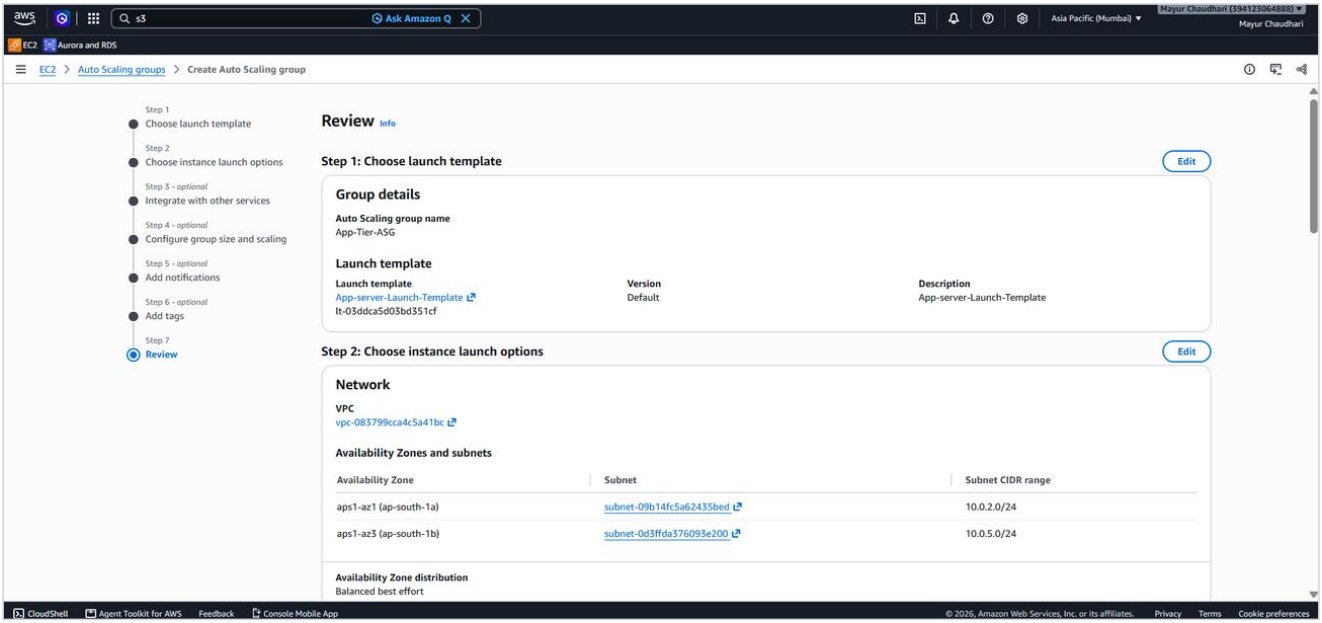
Step 13.7



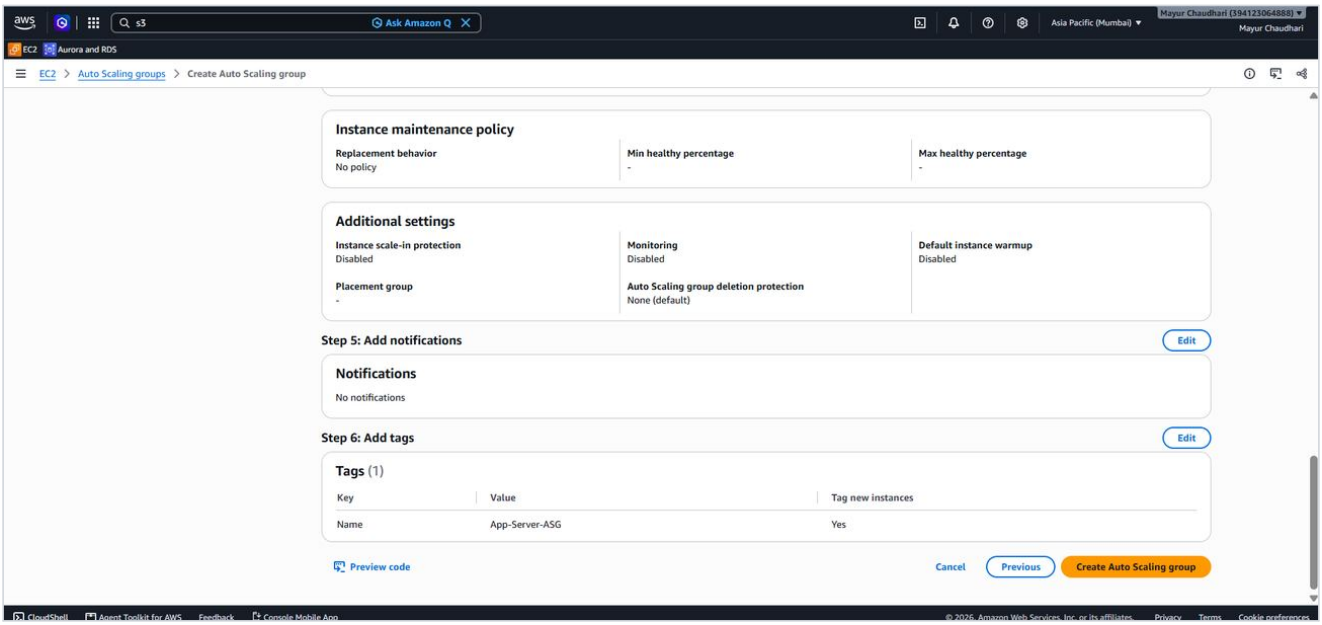
Step 13.8



Step 13.9 Review the Auto Scaling group.



Step 13.10 Auto Scaling group created.



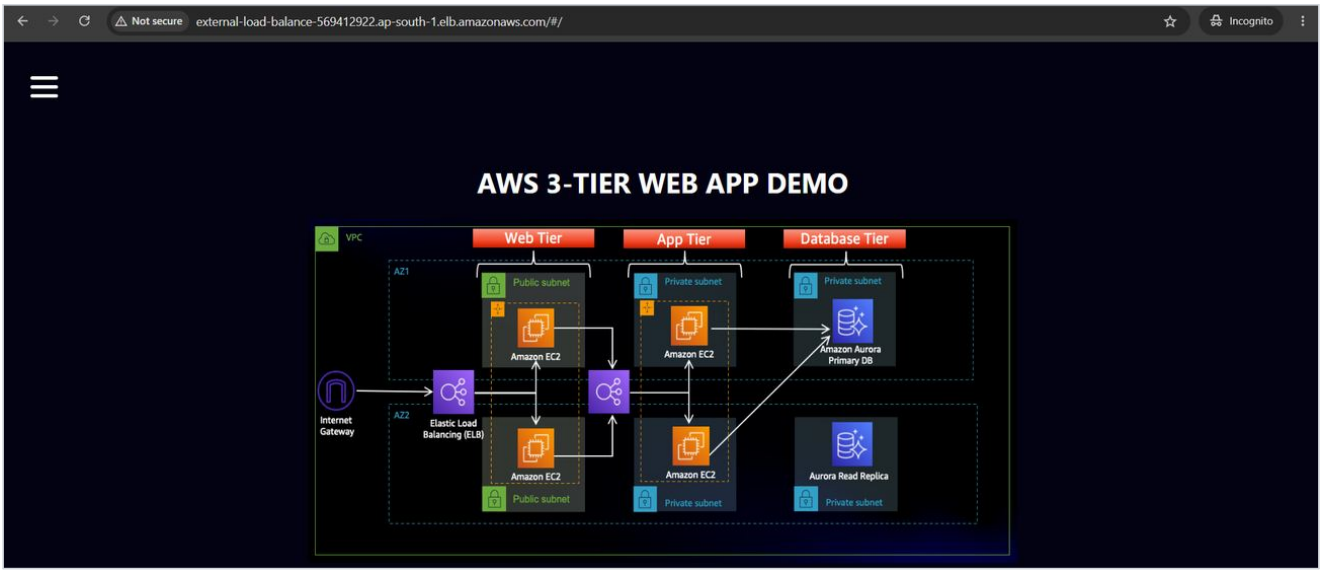
8. The journey of one click

#	What happens	Where
1	User opens the External LB DNS name in the browser	Internet
2	Request enters through the Internet Gateway to the External Load Balancer	Public
3	ALB picks a healthy Nginx server (AZ-a or AZ-b)	Web tier
4	Nginx returns the web page; for `/api` calls it forwards to the Internal LB	Web tier
5	Internal LB picks a healthy Node.js server on port 4000	App tier
6	Node.js runs the logic and queries Aurora on port 3306 (read/write on the writer)	App to DB
7	Answer travels back along the same path to the browser	All

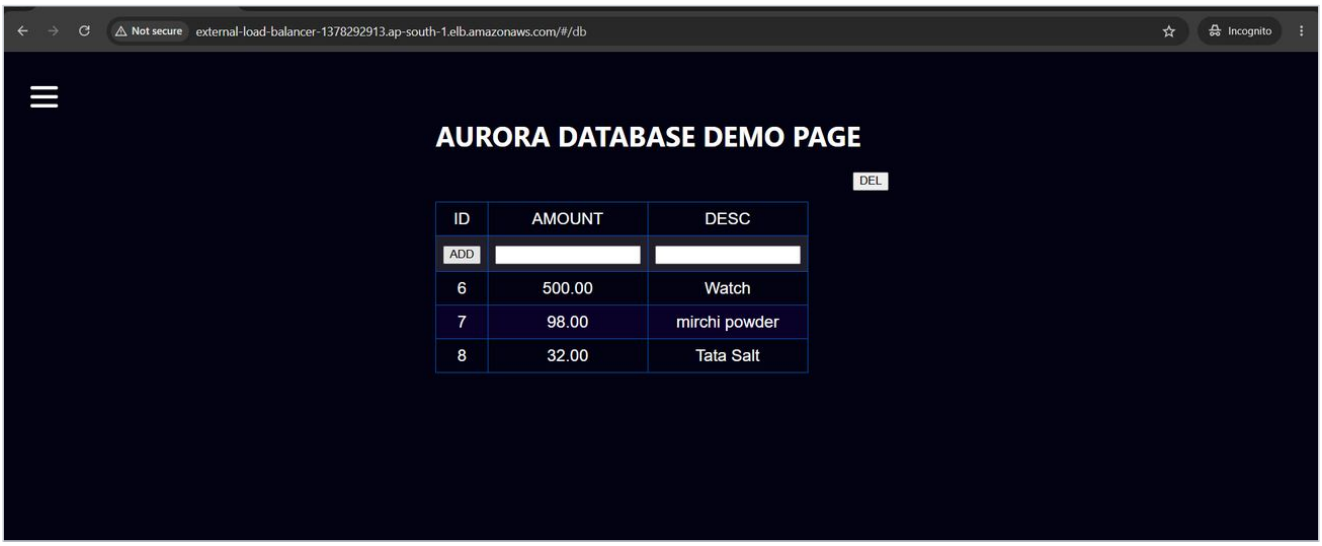
How to test it works
Open the External LB DNS in a browser (use Incognito if it does not load). Open the **DB DEMO** tab: you should see the sample row (groceries, 400). Add and delete items to prove the full path web > app > database works. Tip: using the raw LB DNS is ugly; with a domain name and **Route 53** you get a friendly URL.

Step 14 - Open the application

Step 14.1 Application opens using the External Load Balancer DNS name.



Step 14.2 DB DEMO page shows data read from the database.



9. Clean up (so you are not charged)

Delete in this order, otherwise AWS says "resource in use":

1. Auto Scaling Groups	**9.** NAT Gateways
2. Load balancer listeners, then Load Balancers	**10.** Elastic IPs
3. Target Groups	**11.** Route tables (remove subnet associations first)
4. Launch Templates	**12.** Internet Gateway
5. AMIs	**13.** VPC

6. Snapshots	**14.** S3 bucket
7. EC2 instances	**15.** IAM role
8. Aurora DB (turn off delete protection first)	

Biggest hidden costs if forgotten: **NAT Gateways** (hourly + data), **Aurora**, **load balancers**, and unattached **Elastic IPs**.

10. Production level: what is different?

The original project is an excellent **learning** build: the architecture (3 tiers, 2 AZs, private subnets, SG chaining) is already the same pattern used in real companies. What changes for production is **how it is built, secured, watched and recovered**. Below, each demo shortcut and its production fix.

10.1 Demo shortcut vs production practice

Area	In the demo	In production
Secrets	DB user/password typed into `DbConfig.js` and stored in S3 (and shown in the doc)	**AWS Secrets Manager** (or SSM Parameter Store). App reads the secret at start using its IAM role. Never in code, S3 or Git. Rotate regularly. If a password was ever shared in a document, **change it**.
DB user	Master user *admin* used by the app	App gets its own limited MySQL user (only the needed tables/rights). Master user kept for admins only.
HTTPS	Plain HTTP port 80 on the LB DNS name	Domain in **Route 53**, free certificate from **ACM**, HTTPS listener 443 on the ALB, redirect 80 to 443. Optional **AWS WAF** + **CloudFront** in front.
Building servers	Click in console, SSH-like commands typed by hand, `curl | bash` installs	**Infrastructure as Code** (Terraform / CloudFormation / CDK) for the network and services. Server images built by **EC2 Image Builder** or Packer in a pipeline, not by hand.
Deploying code	Edit file, upload to S3, re-copy on a test server, make new AMI	**CI/CD** (GitHub Actions / CodePipeline): build, test, then rolling or blue-green deploy through the ASG (instance refresh). Version every release.
Runtime versions	Node.js 16 (end-of-life), `npm audit fix` by hand	Supported Node LTS, pinned dependencies, automated vulnerability scans (Dependabot / Inspector).
Database	Single Aurora instance (the doc mentions Multi-AZ standby idea)	**Writer + at least one reader in another AZ** so failover is automatic. Encryption at rest (KMS), automated backups 7-35 days, deletion protection ON, final snapshot always. Consider **RDS Proxy** for connection pooling.
Server access	Session Manager (good)	Keep Session Manager, keep port 22 closed, log sessions to CloudWatch/S3. Enforce **IMDSv2** on EC2.
IAM	S3 ReadOnly on *all* buckets	Least privilege: allow only that one bucket/prefix. Separate roles for web and app tier. MFA on human users; no root usage.
Scaling	Desired = 2 servers, fixed	Min 2 / max N across both AZs with a **target-tracking policy** (e.g. keep CPU at 50% or requests-per-target steady).
Health checks	Basic check	Real `/health` endpoint that tests DB connectivity; ASG uses the **ELB health check** to replace sick servers.

10.2 The production-grade architecture (what to add)

Same three tiers and same two AZs. These pieces are added around them:

Layer	Add this	Why
Edge	Route 53 + ACM + (CloudFront) + AWS WAF + Shield	HTTPS, caching, block bad bots and attacks

Layer	Add this	Why
Network	**VPC endpoints** for S3 and SSM (gateway/interface); Flow Logs on; consider Network ACLs	Private traffic to AWS without NAT, cheaper and safer; network audit trail
Compute	ASG across 2-3 AZs, instance refresh, latest hardened AMI, patching via SSM Patch Manager	Zero-downtime updates and security patches
Database	Multi-AZ Aurora with reader, encrypted, backups, optional cross-region replica	Survive AZ loss; recover from mistakes
Secrets / keys	Secrets Manager + KMS	No passwords in code; encrypted data
Observability	CloudWatch Logs, metrics, dashboards, alarms; SNS/PagerDuty alerts	You know about a problem before users do
Audit	CloudTrail + AWS Config + GuardDuty + Security Hub	Who changed what; detect threats and drift
Governance	Separate AWS accounts for dev / test / prod; tags on everything; Budgets and cost alerts	Safe changes, cost control, ownership

10.3 Monitoring: alarms you should have on day one

What to watch	Alarm when	Likely action
ALB 5xx errors / unhealthy hosts	Any sustained errors or 0 healthy targets	Check app logs; ASG should replace servers
ALB target response time	p95 above your SLA (e.g. 1 s)	Scale out; look for slow queries
EC2 CPU / memory	CPU above 70% for 10 min	Scaling policy should react; raise max
Aurora CPU, connections, free memory	Connections near limit; CPU above 80%	Add reader, RDS Proxy, tune queries
Aurora replica lag / failover event	Any failover	Investigate cause
NAT Gateway bytes / errors	Sudden spikes	Find the chatty server; add VPC endpoints
Billing	Forecast above budget	Review resources, remove idle ones

10.4 Backup and disaster recovery in simple terms

- **HA (high availability)** = survive one AZ failing. Your design already does this with 2 AZs.
- **DR (disaster recovery)** = survive a whole region failing. Needs a second region: copy snapshots/AMIs, optional Aurora Global Database, Route 53 failover.
- Decide two numbers with the business: **RPO** (how much data loss is OK) and **RTO** (how long can we be down). The design follows from them.
- **Test restores** regularly. A backup you never restored is only a hope.

11. Production go-live checklist

Tick every box before real users arrive.

Security
☐ No secrets in code, S3 or documents; stored in Secrets Manager
☐ HTTPS only (80 redirects to 443); valid ACM certificate
☐ DB not public; only App-Tier-SG can reach port 3306
☐ No SSH (port 22) open; Session Manager only
☐ IAM least privilege; MFA on all users; root account locked away
☐ WAF enabled; IMDSv2 required; EBS and RDS encrypted

Reliability
☐ Every tier in at least 2 AZs; one NAT Gateway per AZ
☐ ASG min 2 with scaling policies and ELB health checks
☐ Aurora writer + reader, backups on, deletion protection on
☐ Restore from backup tested once

Operations
☐ All infrastructure in Terraform/CloudFormation, in Git
☐ CI/CD pipeline with rollback
☐ Alarms + dashboards + on-call alerts
☐ Centralised logs (CloudWatch) with retention set
☐ Runbook: what to do if web, app or DB fails

Cost and governance
☐ Tags on every resource (env, owner, project)
☐ Budget alert set; idle test servers/AMIs/snapshots removed
☐ Separate dev / prod accounts
☐ Savings Plans or Reserved capacity considered for steady load

12. Quick revision (remember only this)

- **3 tiers** = Web (Nginx, public) + App (Node.js, private) + DB (Aurora, private).
- **Traffic path:** Internet > IGW > External ALB > Nginx > Internal ALB > Node.js > Aurora.
- **Security chain:** each security group only trusts the one before it.
- **Scaling recipe:** Test server > AMI > Launch Template > Target Group + ALB > Auto Scaling Group.
- **NAT** = private servers go out; **IGW** = public servers go in and out.
- **Production** = same design + secrets manager + HTTPS + IaC + CI/CD + monitoring + Multi-AZ DB with backups + least privilege.

Source: "Deploy a Three Tier Web Application on AWS" by Subodh Kumar (GitHub: SubodhK143/three-tier-app-on-aws). Production recommendations follow AWS Well-Architected Framework principles (security, reliability, operational excellence, cost).

THANK YOU!

for going through this project documentation

★ ★ ★

Throughout this project, I interacted with the following components on AWS.

1. Amazon Elastic Compute Cloud (EC2)
2. Amazon Simple Storage Service (S3)
3. Virtual Private Cloud (VPC)
4. Auto Scaling Group (ASG)
5. Application Load Balancer (ALB)
6. Identity and Access Management (IAM)
7. Amazon Relational Database Service (RDS)
8. Amazon Machine Image (AMI)
9. Launch Templates
10. Target Groups
11. Internet Gateway (IGW)
12. NAT Gateway
13. Route Table (172.16.0.0, 172.16.1.0, 172.16.2.0)
14. Security Groups

A big thank you to everyone who supported and guided me throughout this journey.
This project has been a valuable learning experience in building **HIGHLY AVAILABLE, SECURE, SCALABLE** and **FAULT TOLERANT** three-tier web application on AWS.



Prepared by

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AWS & DevOps Enthusiast

- Passionate about Cloud Computing and DevOps
- Exploring | Learning | Building | Deploying
- Always learning new technologies and building better solutions

★ **KEEP LEARNING. KEEP BUILDING. KEEP GROWING!** ★

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